SOCIAL NETWORKS

NPTEL Online Course | Prof. Sudarshan Iyengar, IIT Ropar

Week 10 - Lecture Transcripts

Lectures 126-142 | 17 lectures | 107 pages (source transcript pages 1060-1165)

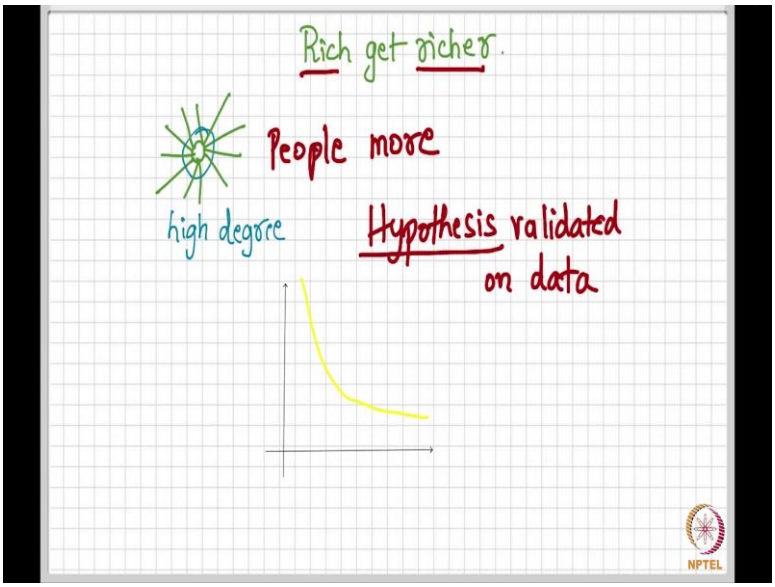
INDEX

LECTURE	TOPIC	PAGE IN THIS FILE
L126	Rich Get Richer - A Possible Reason	2
L127	Rich Get Richer - The Long Tail	11
L128	Epidemics - An Introduction	19
L129	Introduction to Epidemics (contd.)	27
L130	Simple Branching Process for Modeling Epidemics	30
L131	Simple Branching Process for Modeling Epidemics (contd.)	38
L132	Basic Reproductive Number	43
L133	Modeling Epidemics on Complex Networks	49
L134	SIR and SIS Spreading Models	54
L135	Comparison Between SIR and SIS Spreading Models	68
L136	Basic Reproductive Number Revisited for Complex Networks	74
L137	Percolation Model	79
L138	Analysis of Basic Reproductive Number in Branching Model (Problem Statement)	85
L139	Analyzing Basic Reproductive Number 2	89
L140	Analyzing Basic Reproductive Number 3	96
L141	Analyzing Basic Reproductive Number 4	100
L142	Analyzing Basic Reproductive Number 5	104

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 126
Rich Get Richer - A Possible Reason

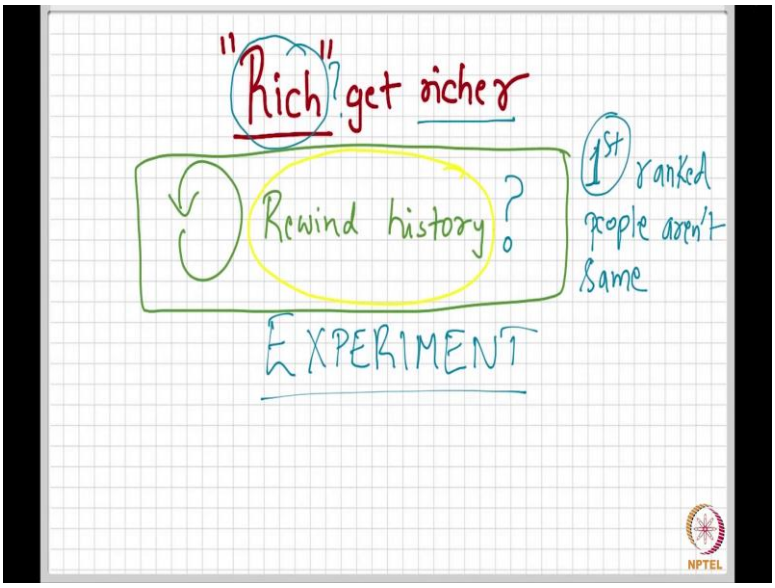
(Refer Slide Time: 00:10)



We saw in the previous lectures; the paradigm called the Rich Get Richer. It was a rich get richer. So, what was happening there? People with high degree they can do attract people more people who have high degree. If they have high degree, then they have more propensity to attract more people towards them. So, a person who has more friends will attract more friends more friends. Since, he has more friends he will attract more friends and so on and so forth. This is a sort of a hypothesis; this is a hypothesis validated on several data; validated on data. That does not make this hypothesis true, but it is highly suspected that this phenomena is indeed happening.

What phenomena? The phenomena that the rich get richer because of which we observe the power law happening right, the power law as in the drop in the curve if you remember the drop in the curve correct, fine. So, now, we will try to ask this question; this question slightly unrelated from networks, but related of course, and quite an intriguing question.

(Refer Slide Time: 01:44)



When we say rich get richer right, how did this people become rich in the first place to begin with right? I mean you say that people who have more friends attract more friends. How it they start getting this so called more friends, right?

So, let us take an example. If I were to rewind history, do you think Alice in Wonderland still would be a very popular story ever written? Do you think JK Rowling's Harry Potter would still be a bestseller right? There are many such authors who wrote many novels and stories which did not make it to the best seller status just the way Harry Potter did. If I were to re rewind history that is my question, if I have to rewind it do you think successful people will continue to be successful? Well, how do you even rewind history, how do you even mimic such a experiment, how do we try to sort of re do a stuff and then see if popular people continue to be popular?

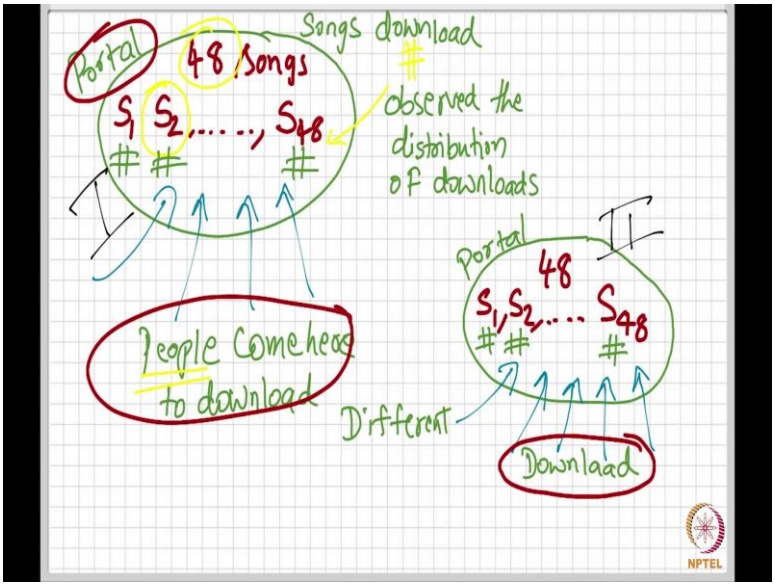
If history where to remind itself and then play once again, do you think Britney Spears would still be the best? Do you think Obama will again become the president of the United States? With this is very tough to answer right. So, probably, one should think of an experiment which sort of gives the tries to attempt an answer to this question. So, what kind of experiment can one do; is my question. What experiment can you think of?

So, I hope my question is clear to you all. The question is people who are rich tend to become richer right. Who are this rich people? Everybody start a fresh with 0 money right. So, who are these people who become famous? JK Rowling before her first work

was not known to people right and she got the people got to know of her through her Harry Potter's work, the philosophers stone which is the part 1 of Harry Potter.

And then she became really famous and whatever she wrote sold. Of course, the work was good too but then still it gave her an edge because she was successful with her first attempt right. So, is this true in general that if you are good you will make it or is it because by luck you make it the first time and you become rich and you become richer because you were rich and so on. There was a nice experiment conducted which I am going to describe right now.

(Refer Slide Time: 04:49)



People took some music songs for download, they took 48 songs ok; song 1, song 2 up to 48 songs. And a bunch of people were asked to come to this portal, this is a portal basically, this is a portal. Let me write that down. This is a portal for songs download, portal for songs for the downloading of the songs ok; songs down download people come here to download songs.

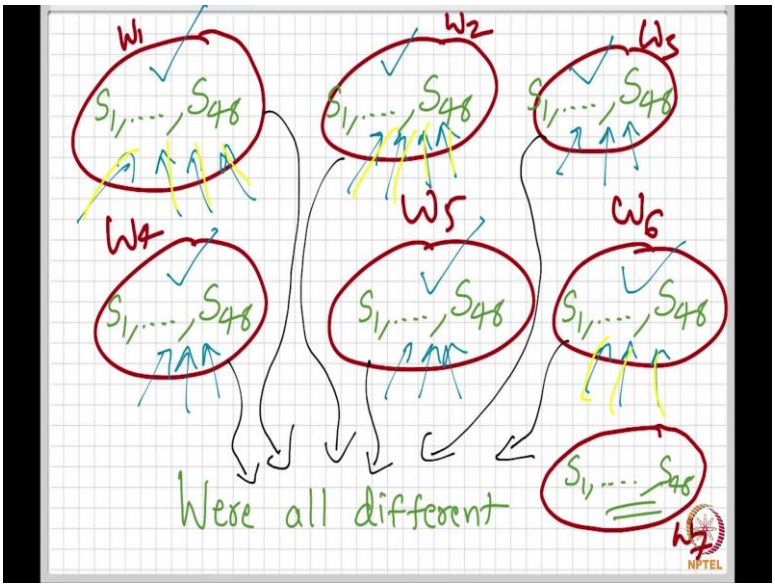
Now, with time people observe that song 2 was being downloaded a whole lot. People do get to see what song is downloaded how many times then the number of downloads is visible and people did see this and eventually it was observed that songs that were downloaded more often became more and more popular ok.

People observed the distribution of downloads, they observed the distribution of downloads. Some of them were really popular and some of them were not very popular ok. Now, here goes our question. If I were to take another world ok, a parallel world as I call it and then take the same 48 songs as I took here and display them for downloading on a different portal. Here was a portal ok, this is a different portal another portal ok. And then again I ask I let some new bunch of people to come here random bunch of people to see and then they download the songs as usual as in the previous case and now I see the distribution of how much how many times songs 1 was downloaded, how many the number of times song 2 was downloaded, the number of times song 48 was download and so on, all the songs.

I have made note of this as well right in the previous case ok. So, now, what do you expect? Let me be clear. The people here are different they are different from people here to download the songs.

This is one world this one is world number 1 this one is world number 2 ok.

(Refer Slide Time: 08:019)



So, if I want to create many such words let us say world 1, world 2, world 3, let say world 4, 5 and 6; I have six different worlds and then all these worlds I have the same bunch of songs; song 1 to song 48 I had 48 songs ok, same thing everywhere alright. So, let me just copy paste and I have the same thing here and then the same thing here in w3 and w4 I have the same bunch of songs and assume I have this 5 what is call the parallel

universe, where different bunch of people are allowed to come and download the songs. Here people are different; here people are different and so on.

So, please note this people do not know about other portals. There are six portals and different people assigned to different places. So, people were assigned here are different from people of same here and people were same here and so on. So, it was observed that the distribution; the distribution of w1 and the distribution of w2 and that of w3 and that of w4 and that of w5 and that of w6 were all different.

What do I mean by that? By that I mean winner here the most downloaded song here was different from most downloaded song here was different from what it was here different from here different from here and different from here.

Now, pause for a minute and think what is happening? You have 6 different worlds where different people were assigned the task of downloading music and rating the music and they it was observed that the number of downloads of a song was different in different portals. Now, why is this? What just happened here?

(Refer Slide Time: 10:39)



The reason here is precisely what we were discussing before, the rich getting rich richer phenomenon; rich get richer phenomena. The fact that you displayed the downloading information or let us say rating information right, you displayed the download information the download information not downloading information download

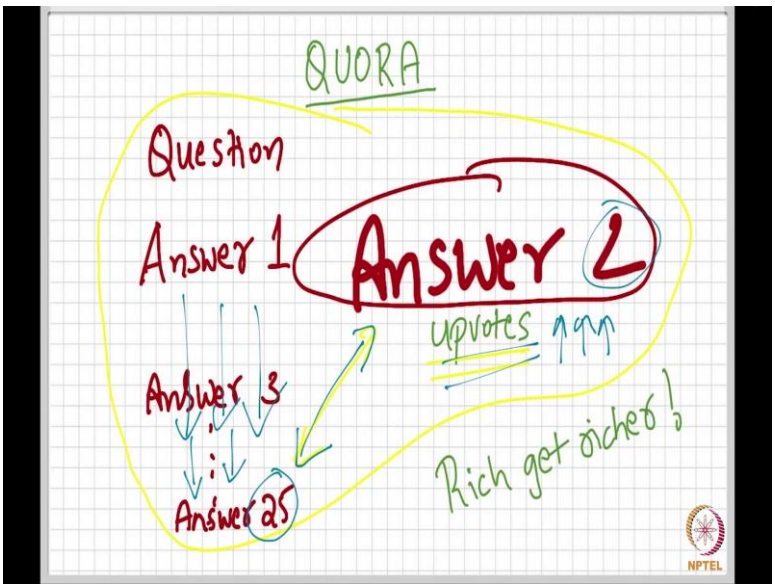
information and people saw that and sort of became biased right. So, highly downloaded songs it was displayed there of course, this are the highly downloaded songs biased new people new people to download these songs and hence they downloaded and that added to the download count once again so on and so forth ok.

So, there was an there was an experiment which involved another world with the same songs where no information of the download was display and yet again the distribution was very different here ok. So, this was the world let say 7th world alright ok. So, what does it signified?

Signifies that the fact that something is good results in that something becoming further good right. Good gets good becomes good even further rather the right world is better. Good gets better and better gets let say more better sounds like a wrong English grammar, but there is there is no other way for [laughing] me to explain this; good gets better, better becomes more better and more better becomes more and more better.

Why? That is because if something is very good. People tend to get attracted to it and they added to the popularity of the stuff right. So, this is called the rich get richer phenomena. This is observed in almost all the places. I will give one instance where you can relate this phenomenon very well and that is your Quora portal.

(Refer Slide Time: 13:02)



If you did not know Quora is a portal where people ask questions. There is a question and there is an answer. There is not just an answer; there is an answer 1, answer 2, answer 3 and so on. So, what happens in Quora is just in case answer let say answer 2 becomes really really popular ok; answer 2 becomes really popular; you know what happened then ok? The sort of dominates this question thread, answer 2 becomes popular and it gets more what is called upvotes and more people see answer 2 and they will upvote it all the more.

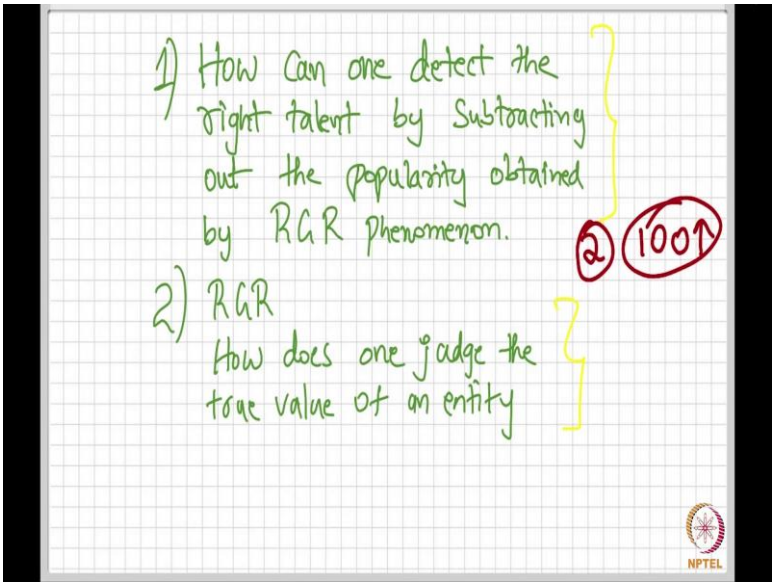
And just in case a new answer comes, let us say answer 25, which is much better than this answer; nobody at all even if it is much better than answer 2. Assume answer 25 is way better than answer 2, nobody will see it. It will just be flushed down somewhere because Quora displays answers based on upvotes right.

At least so far, the main logic behind displaying the answers first is based on the sorted order of their upvotes. So, more the upvotes, more people will see it and more will they upvote the answer and answer become popular and no answer that is below answer 2 or lower will ever be displayed. So, do you see the rich get richer phenomena happening here.

Now, one can ask several questions here. So, getting back to a previous question let me go back slide where we were discussing JK Rowling's popularity etcetera. So, if I write to rewind history right, it is most probably true that people the first ranked people are not the same. First rank does not first few rank people. Although people who do really well will continue to do really well, people who do not do well will continue to do if continue not to do very well, but the first rank, but the top few ranks will actually shuffle.

By this I mean, Obama may not become a president if history were to rewind, JK Rowling may not become the bestselling author if history were to rewind and our experiment sort of gives a nice ideas to why the our hypothesis is true that the person who starts of afresh might become rich out of luck to begin with and then rich get richer and he or she becomes richer and richer and richer and attracts more um popularity and becomes all the more richer ok.

(Refer Slide Time: 16:40)



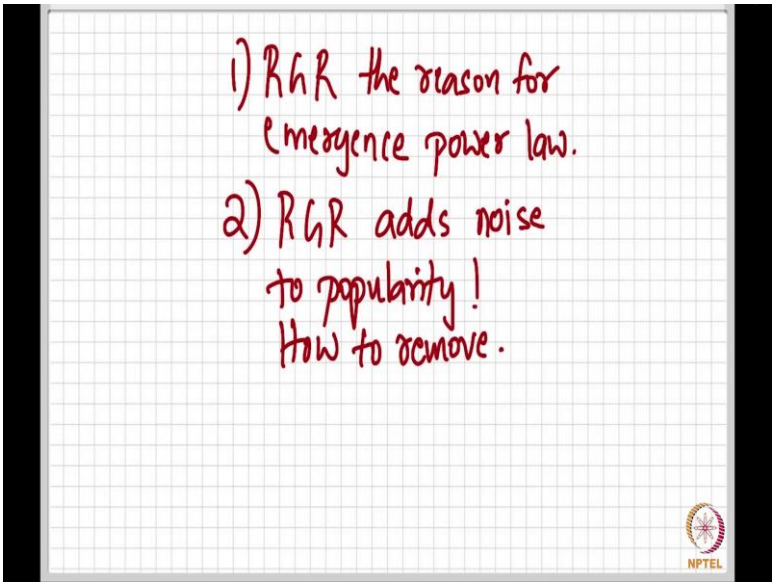
A good and nice question to ask is how do you how can one detect the right talent by subtracting out subtracting out the popularity obtained by this rich getting richer phenomenon. So, how can one do this? This is this is one nice question that one can ask this is one thing.

And second thing is if something has had this something has exhibited this rich getter getting rich phenomena how does one judge the true value of an entity which is grown because of rich getting richer phenomena right. So, both sound sort of similar, but there is certain difference between them. The first one being something that is over already how would you know who is the right person by subtracting out the popularity, the second one is you should not just rank them, but you should understand that true value.

For example, in Quora upvotes if the worlds are beyond 100 should you really take it seriously right? A questions answers is beyond 100, but was, but as an answer 2, second answer to the question; one should probably subtract out a few votes here to sort of normalize and to understand the exact popularity of the answer right.

So, it this allows still open questions in the domain where people have not really understood. Firstly, we do not understand if rich get getting richer phenomena is the reason why power law emerges, first one right.

(Refer Slide Time: 18:58)



Is rich getting richer the reason? The reason for the emergence of power law this we have not understood it well it is just to hypothesis. Secondly, rich getting richer phenomena adds noise to popularity. How to remove this noise right? How does one go about removing it, how to remove this noise? So, that the true color, the true picture, the true popularity of a person is revealed.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 127
Rich Get Richer – The Long Tail

We have seen power law on networks. There is a different kind of power law that is in play in nature right and we are going to study more about it in this chapter in this particular lecture.

(Refer Slide Time: 00:22)



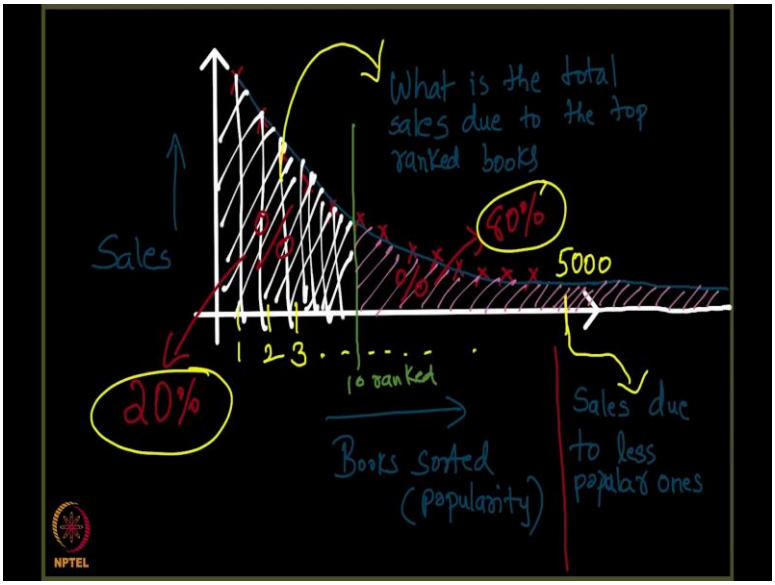
So, let me write more power law. So, this is mostly outside the realms of graphs and networks. So, let us go to a bookstore. So, assume you go to a bookstores and you browse through some books right, go to a bookstore and you see a book 1 and there are several books in the shelf as you know.

I am not good at drawing; let me make some attempts to write books. So, here is a bookstore with a lot of books alright, books. There are different types of books, different genres ok. Now, there is a huge collection generally, as you would have observed there is generally huge collection of books and a question is why would a bookstore have such huge collections right.

There firstly, there are different genres of books as a huge collection. Why would anyone have something like this right? If I will were to run a bookstores, I would simply have the best sellers right. I would simply have the best sellers. So, roughly let me assume that 10 percent of the books that are there in the bookstores are best sellers, best sellers. So, why not just keep the 10 percent of it and discard the remaining 90 percent; it only makes sense you see right.

If I were the book shop owner, I would simply keep the best sellers and discard the rest why then do bookstores any big bookstore that you visit they have a whole lot of books, why cannot they just keep the best sellers? Now, what is what is the obvious answer to this question? The obvious answer to this question is that we should probably ask this question in a different way.

(Refer Slide Time: 02:52)



I should plot this graph ok, let me just make it clear to you people slightly involved.

I will see the most popular the first most popular book and the second most popular, the third most popular book, so on and so forth. Assume, I have some 5000 such books. So, what is my x axis? My x axis is going to be popularity rank of the book. These are these are books sorted according to popularity, according to popularity fine. What is on the y axis? It is the sales of this books, the sales of this books. Obviously, the most first rank book will obviously, have very high sales, second rank book will have slightly less sales, third rank that is what you mean by ranking you say, correct.

Evidently there will be this power law like distribution correct, correct. So, on let me join these points and as you can see it will be a curve like this. Now, this looks like the same old theory of power law, but power law is slightly different as you know, it is about the distribution of the node degree correct. So, but the curve here resembles that. Now, let us get back to our question, let me go to the previous slide and let me help you remind let me see the question that I asked.

Why not have only the best sellers and discard the not so best sellers ok? I want to discard this and only retain this. Why because at my profit if my profit is mainly based on a bestseller, why then have this the remaining stuff 90 percent of the books are not best sellers. Let me just discard them. Now, what is so, wrong about this hypothesis of my bookstore business that I am going to start very soon of only the best sellers?

Well, that is where we need science rather smart accountancy or smart economics understanding of economics the situation. What we should do is think for a minute, what one should look at it is what is the amount of sales of the top few. Let say let me let me just take the first 10 which is let us say here; let us say first 5 or first 10 whatever that is. So, let us say first 10 ranked books; top 10 rank, the first 10 books I will see what the sales of this is.

A very small a very whatever I mean, it does not take a lot of time for you think. What is the answer for this. Let me just write that down. So, that it seems to our minds. My question is what is the total profit that I might possibly make profit due to the top ranked books. What do I mean by this? I just mean what exactly I mean it is thought will tell you all I am trying to say here is what is this plus, this plus, this plus, this plus, this plus, this plus this. So, basically it is the area below this curve right.

I mean whatever is the area that will denote what will that denote that will denote the total profit total sales rather not necessarily profit I think I should remove the word profit here, it is not profit it is the sales. What is the total sales due to the top ranked books? It is going to be this curve correct and definitely this is going to tell me what is the sales due to the top ranked books and the remaining stuff which is this will tell me what is my sales, what is my sales so on; this will go on as you know right.

This will extend, this does not stop here it just extends to all 5000 books. What does this denote? This denotes this denotes what? Sales, sales due to a line here; sales due to due

to less popular books, less popular books. So, what I am trying to say here? All I am trying to say here is I am looking at my sales due to my popular products versus sales because of my niche product. If these sales is very high; now please note these are the sales because of best sellers. Just because they are best sellers and just because they sell in big numbers does not mean that they make most of my complete sales.

So, a good question for you all to ask which I am sure this is running in your mind already is what percentage of my sales is this and what percentage is this. To get your clear understanding I think you people should rather not see the curve this way which is slightly misleading you must know that the curve here the blue line extends like this, so does the white line here, the y x axis and then the area below the curve also extends like this.

So, good question to for one to ask here is what percentage is this ok, what percentage is this and what percentage is this right. Just in case, this was only 20 percent; you know what I am going to where I am getting at right now. This will be 80 percent. Now, you know very well that I cannot have my bookstores with just the popular products, I need the niche products. So, niche means, the less popular ones.

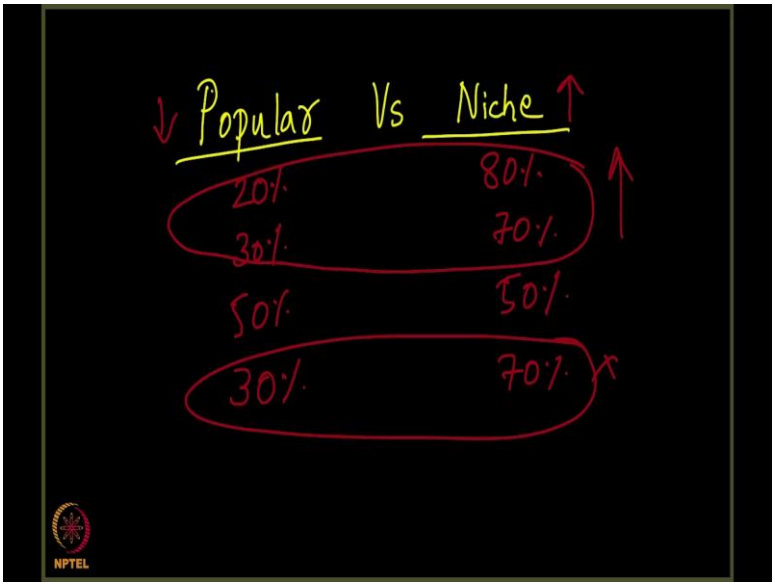
So, what are all here? As you can see let us say the past best sellers might be here, the best seller will be a best seller for a few months and then it will come here right or may be less popular books. Science books unfortunately are less popular because not many people read it. So, more scientific books are all here or technical books all here, novels and all the masala stuff is here, although you see that this may not be a big chunk of your sales, it can just be 20 percent and this can just be 80 percent.

So, the point here is you may want to concentrate on this more than this although your sales is mainly because of the bestselling book. So, you think if you write a curve like this and then see you will realize that it may not be. So, one thing that you almost observed is this. Do you think a book here continues to stay here? It will probably come somewhere here very soon right. Every best seller has it is shelf life after while it will stop being a best seller and it will become a niche product; it will become a classic right.

So, it is very interesting to see how this curve continuous to stay like this where more popular stuff keeps coming here and less popular stuff with time goes to this part of the

curve. And what is more interesting is what exactly is the percentage? 20 percent, 80 percent is what I am saying it could be different.

(Refer Slide Time: 11:36)

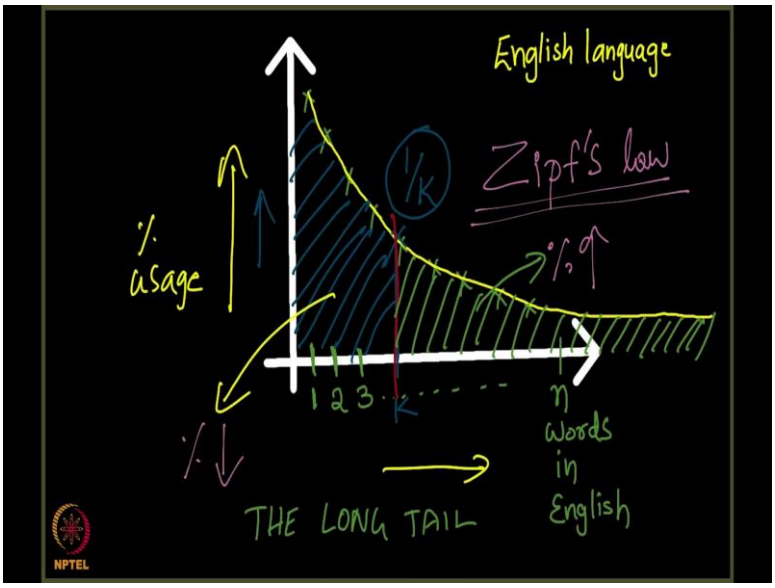


So, in your business you may have to see what is the popular products, popular versus the niche ones. You should have a good understanding of this ratio. It could be 80 percent; I am sorry just a minute whatever I said 20 percent and 80 percent or this could be 30 percent and 70 percent or it could be 50 percent and 50 percent it is always observed that something like this never happens, something like this never happens.

It is slightly on the higher side; the niche is on the higher side, popular is slightly on the lower side right. So, this speaks in volumes on why when you go to a bookstore you have all sorts of products. You cannot just keep the best selling once right. This is generally true no matter what business you are thinking of.

Let us say the kind of udio videos that people watch on YouTube, YouTube cannot afford to keep only the most watched video saying that people who watch this videos less rather I mean videos which are being watched less should just be removed right. So, the big chunk of what people watch is probably indeed the niche ones is probably the niche ones and less so of the popular once ok.

(Refer Slide Time: 13:22)



So, this has been there historically, in sciences people have observed this property in many many works of science.

One such thing is a very interesting fact which is let me try plotting and tell you what is this. If you look at the word frequency, let us not get lost. I just started off with this bookstore example and then I showed you this phenomena that happens and then I told you that popular sales, sales of popular products or sort of less in ratio as compared to the niche products and I also observed we also observed that a product that is there as a popular product might become a niche product.

It is interesting to see how this curve is preserved throughout in sales right ok. So, this is not this very similar to what happens in general with nodes as well. A node that is popular may not continue to be popular with time. It might become less popular and it might go this side right. What kind of questions do you have in mind? In fact, one can ask several research questions on this topic. So, I just right to me over email in case you want some cool questions on this topic. There are several nice very nice questions that one can attempt that the world does not know rather ok.

So, as trying to plot something very interesting about English language; about your English language, ok. If you take up the most popular word in English which is the most frequently used word in English call it the first ranked word and the second most

frequently used word in English call it the second ranked word and the third most frequently used English used word in English call it third rank and so go on.

And write down all the words let us say there are n words in English, you write n units on x axis English and then you see what is the number of times what is the proportion of times that you use this word. The first ranked word how many times, what percentage is it is usage. For example, in a document, if you think the word t h e the is most frequently used I will ask what percentage of the document is it frequently used. So, the plot looks again similar to this is and then if you join this points you will get a power law like that is not a good way to write a curve ok.

And you will again get the curve like this and this curve is known to be known to follow this function 1 over k. So, if this is K, this will be 1 over K and of course, I have used scale y axis. So, this might appear to be like a this distance seems the same as this distance, but do not mind my drawing the fact is this turns out to be 1 by K, if you look at the distribution of words in English language based on their ranking sorted based on their ranking and this is popularly called the Zipf's law. You can take a look at it on the internet.

It is a very popular law found several decades ago and this again says that the amount of frequently used words in English, same old theory back again if you see. That are there are there is this part of the curve which talks about the frequently used words in English, these are the words that are very frequently used rather popular words and this side is the sort of less popular words, but if you see this tail goes on like this ok; this is a big percentage this a big percentage.

I mean the area here is way bigger compared to the area here; this is a smaller percentage. Again, this is observed in words to know. So, coming for coming getting back to where we started from, it is not just true in books bookstores like business. It is also true in English language know that is surprising. It is observed in many places right and this is called the fat tail phenomena rather the long tail not the fat it is called the long tail. What is the long tail mean here? I am sure you all of guessed what I am trying to say here.

By this I mean this part of the curve this part of the curve is it goes on like this. It is start from here; it goes on like this given that this tail is longer do not have this part of the

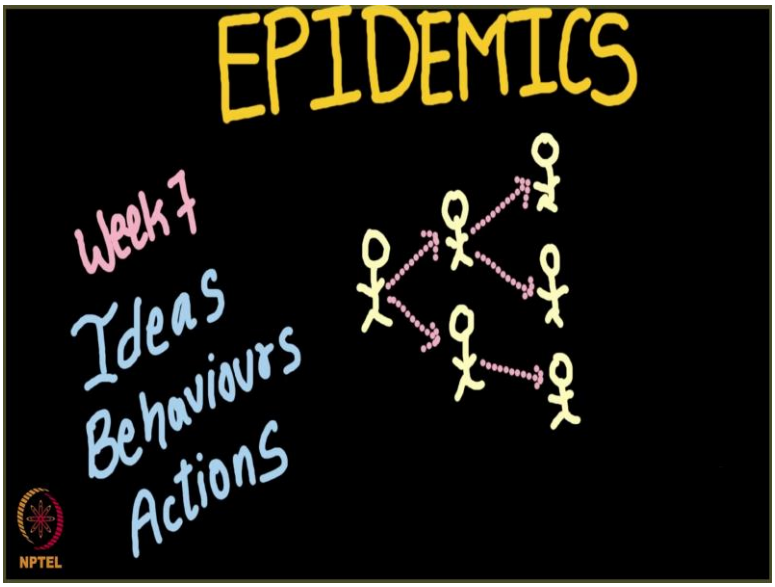
curve looks resembles the tail of an animal that is probably the reason why they call it the long tail.

Since, it is long, it acts to a whole lot of the area as compared to the popular part of the curve right. So, that is why this is called the long tail phenomena. What is long tail phenomena? Long tail phenomena is nothing but the fact that; so, the popular products are less in number although they are popular and less popular products are more in number although they are less popular the sort of dominate the space.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

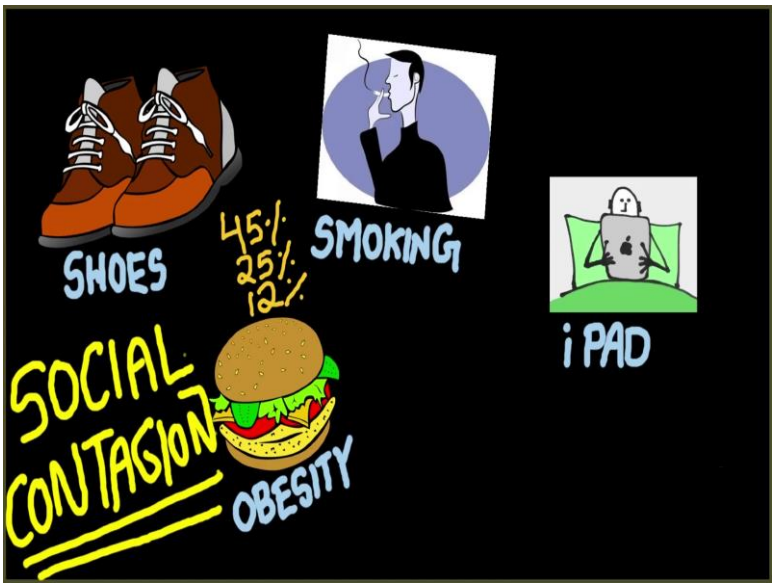
Rich Get Richer Phenomenon – 2
Lecture - 128
Epidemics- An Introduction

(Refer Slide Time: 00:08)



This week we are going to start of with a new chapter and the name of the chapter is Epidemics. So, before starting off with this chapter, I would like to I would like to all of you to recall what we studied in week 7 the cascading behavior in networks. So, we have looked that how ideas, behaviors and actions spread on a network. We have looked at people who are connected with the help of a network and then will looked at how these ideas behaviors and actions spread between these people and we have talked about a number of ideas and behaviors.

(Refer Slide Time: 00:47)



So, let me recall some of the examples. We talked about the sports shoes example where one of my friend who is in a healthcare industry and he had come up with his own brand new technology of sports shoes and he wanted me to advertise his sports shoes, which I advertised to some people and some people means some of my friends and they advertised it to some of their friends and so on, there was this cascade of sports shoes on these networks.

And then we have looked that smoking as a behavior is contagious. One of my friend could be smoking and when I look at this and when I look at my friend I could feel like smoking is cool and I might also start smoking which leads to a cascade of smoking on the network.

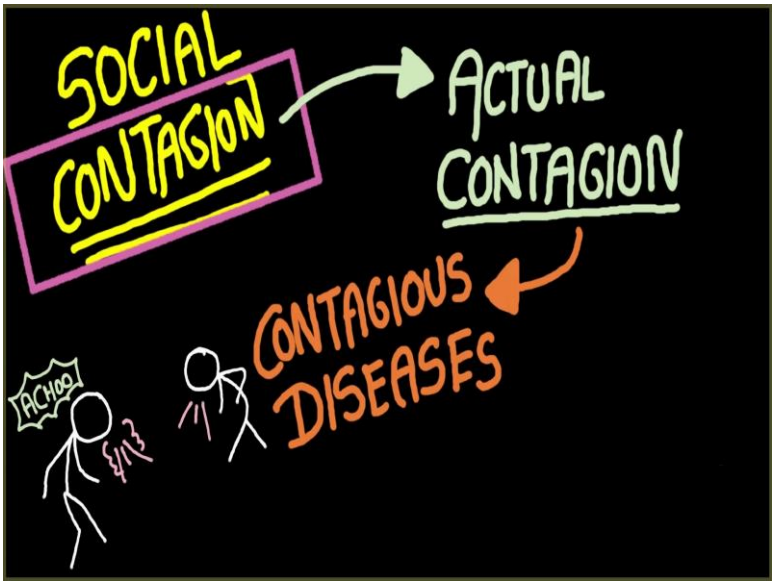
And then obesity; obesity is contagious. We have also studied this in the preliminary video. It does not appear much like obesity is contagious. Obesity seems to be quite an individualistic behavior, but actually not. There is a very interesting piece of research which says that if your friend is a obese, your chances of becoming obese increases by 45 percent. If your friends friend is obese still your chances of becoming obese increases by 25 percent; if your friend is obese still your chances of becoming obese increases by 12 percent and its only when you reach to a friend that there is no longer of a correlation between that guys body weight and your body weight.

All in all obesity is also a behavior which spread on a network similar to obesity depression and happiness are also the behavior which spread on a network and yet another product which can spread on a network is iPad. I basically wanted to recap that there are these ideas, behaviors and actions which can spread on a social networks and we have studied all of these in a week 7 which is cascading behavior in networks and most of these behaviors which are spreading are known as social contagion social contagion.

Why social contagion? Contagion is a word which is generally used for the spreading of a contagious disease. Let us say flu, measles, chickenpox, etcetera.

But here, contagion refers to something its spreads from person to person and what is spreading here is a social behavior, social norm; hence, the term social contagion.

(Refer Slide Time: 03:32)



So, we have talked in detail about social contagion in week 7. What we are going to do in this chapter is we are removing this word social and we are going to top just about contagion that is an actual contagion, a real contagion. And what is an actual and real contagion is nothing, but a contagious diseases; a contagious disease like flu. So, in this chapter we are going to see how we can model the spreading of contagious diseases on a networks.

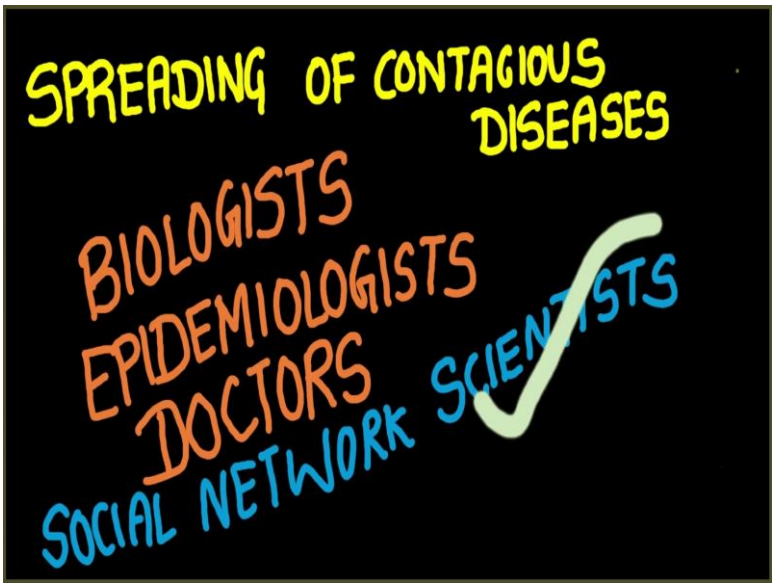
(Refer Slide Time: 04:02)



Now, one question is why is it even important, why the modeling of contagious diseases is important? And the answer have actually flashed it on the screen and the answer is it helps us in fighting epidemics. Whenever a contagion is spreading on a network and it spreads way too much; way too much means, it infects a lot of people almost everybody in the world, we say that it has become an epidemic.

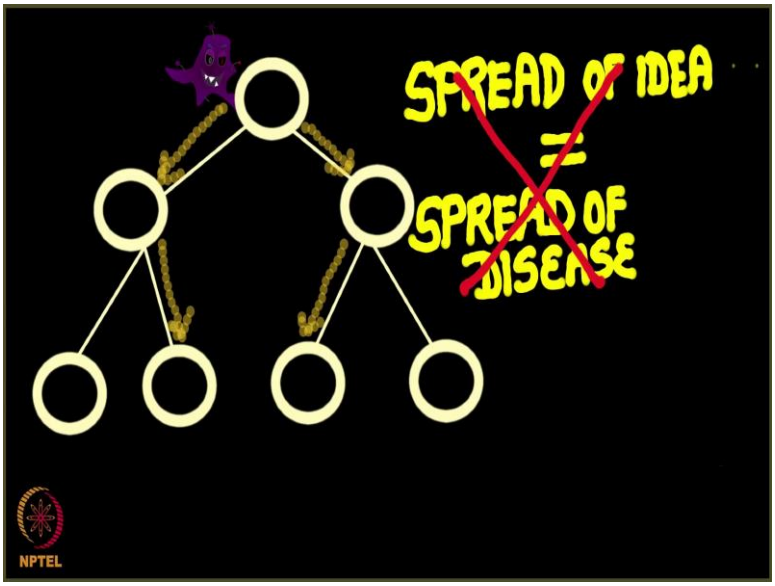
So, these contagious diseases can become epidemics overtime and such epidemics the need to be controlled, we cannot afford to have epidemics. World has suffered from dangerous epidemic from time to time and all of us know about various deadly epidemics. For example, Ebola; the most recent one, Swine flu and then Black death and there is a large number you can search on Google or Wikipedia. The world has suffered from a lot of epidemics throughout the history.

(Refer Slide Time: 05:12)



So, we are going to see in this chapter how can we model this is spreading of contagious diseases. But, now again a question should not it be biologists who should be studying how this contagious diseases spread or should not it be epidemiologists or rather medical doctors. Though social network scientists have much to offer there what will social network scientist have to what they will be having to say about the spreading of contagious diseases, can be even work in this direction? And the answer is a big yes. Why the answer is a big yes is actually very simple.

(Refer Slide Time: 05:53)



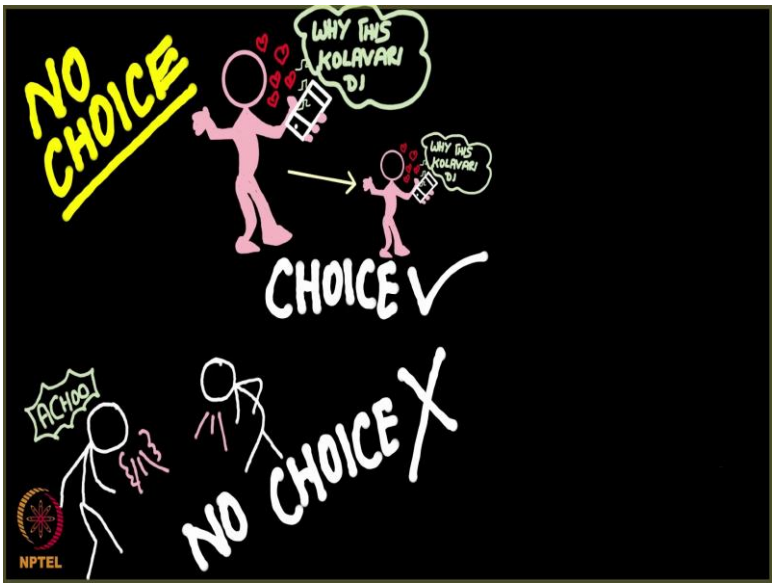
The answer is like anything this contagious diseases they spread on a network. So, as you can see, here I have a network and here is this one node and this node can infect this node which can infect this node and so on.

So, our world can be modeled like this network and our contagious disease is actually spreading on this network which is shown here. So, let us see a contagious disease starts spreading from here. So, this person gets infected the first and then it makes its way through this network. But wait now, does not do you notice something; is not it exactly like this spread of an idea? An idea also spreads the similar way.

There is a node from where a new idea starts and then it looks at its neighbors and some of its neighbors adopts this idea and similarly, this idea travels or rather diffuses through this network. So, it seems like spreading of a diseases and spreading of an idea they are the same things; looks like spreading of an idea is same as the spread of a disease.

But the answer is no, they are two different processes. Why two different processes?

(Refer Slide Time: 07:12)

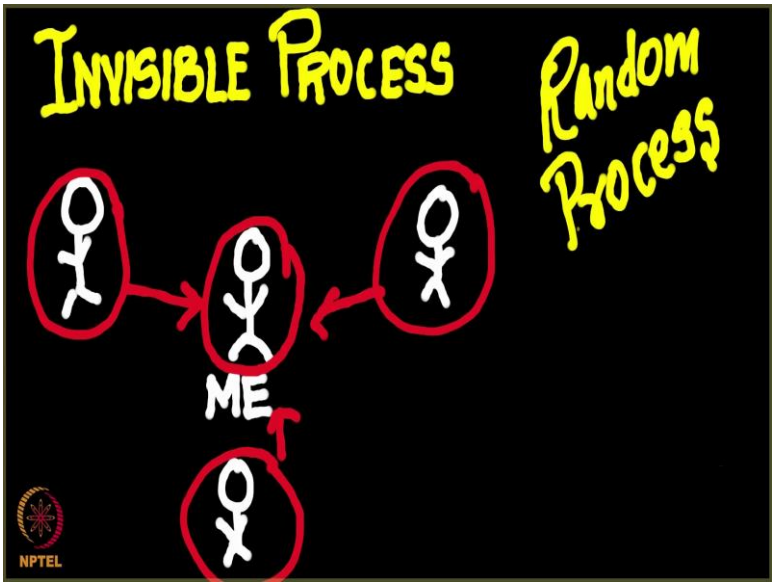


The first reason is no choice. When we talk about an idea which is spreading; let us say there is a song and my friend hear this song why this Kolavari Di and he likes it very much. And next day he tells me about it, I also listen to the song like it very much, I adopt the song do you see there is a choice hear my friend told me about this song I like this song and I adopted this song.

So, well there is a choice when we talk about the spreading of an idea. But, let us I am talking about the spreading of a disease. This time instead of a song my friend has come to me with a flu. So, he has this flu and he is sneezes on my face and now flu virus come and its entering my body. Can I first of all I notes know this virus is entering in my body? But still, let us say I know this. So, can I say that I do not like this flu virus, I am not going to take it?

While in the case of a song, you have the choice to say this is a bad song, I do not like it, but you are you cannot, you cannot do the do this in the case of a disease. So, there is no choice. If the disease has to infective, it will infective there is no way out. So, you have less of a control over the spreading of a disease as compared to the spreading of an idea.

(Refer Slide Time: 08:40)



This second difference is the invisibility of the process. What is invisibility of the process? I will tell you a small anecdote; this was when I was in my fourth standard. So, here is me and I am in my fourth standard and there are three people in my family; my mom, my dad and my brother and all of these three people they suffered from eye flu. So, eye flu was spreading that time and all of these three family members of mine they suffered from this eye flu.

And now, I can catch flu from any of these three right and the very next day I also catch eye flu. So, I have got this eye flu here. But I do not know which is the person out of these three who has infected me with it eye flu; till today, I do not know. So, the

spreading of a disease is quite in invisible process as compared to the spreading of an idea where you might have a very clear clue of which is the person from whom you have adopted the idea from, but it is very difficult to say which is the person from, whom you have adopted a disease.

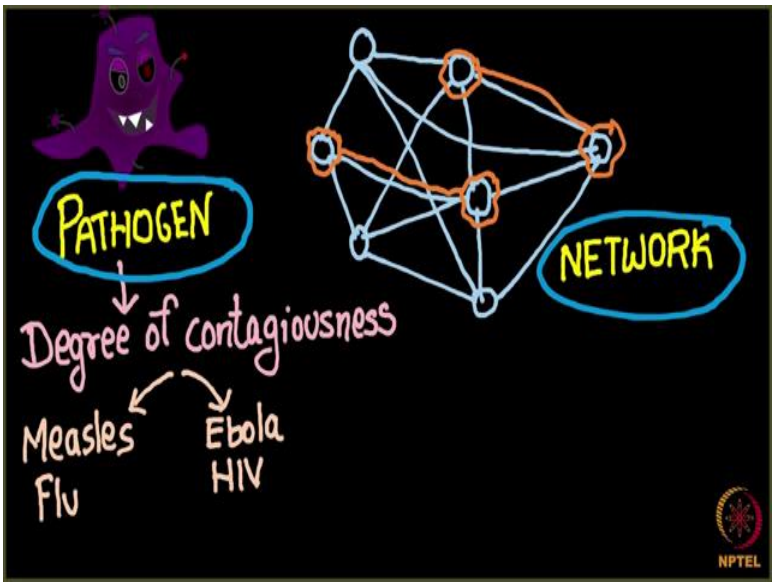
Hence, we need different model for the spreading of an idea, which we have looked that before and now we need a different model when we want to talk about the spreading of disease because the spreading of a disease is more of a random process as compared to the spreading of an idea and will explore it further.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 129
Introduction to epidemics (continued...)

So, now, before starting with the question or before looking into how an epidemic spread lets us look at 2 things, 2 important things. So, I would like you to ask this question from yourself which 2 factors do you think are the most important which 2 factors are required when we want to model a disease spreading on a network? Assume there is this classroom and there are some people, some students in this classroom and one of the student catches a flu there and this flu is spreading through the network. So, which 2 things will you require if you want to model the spreading of this flu? And here goes the answer.

(Refer Slide Time: 00:50)



So, the first one is the pathogen itself about the flu itself. So, what is spreading on this network is important. For example, it is degree of contagiousness matter, how contagious this flu is. Taking an analogy with these spreading of an idea as we have seen, something like a piece of code spreads less quickly as compared to a juicy piece of gossip.

Similarly, in the case of diseases, there are certain diseases which is spread more quickly as compared to other diseases. If you talk about diseases like measles and flu, they spread quite quickly as compared to the diseases like Ebola and HIV.

So, first of all what is important is the pathogen. We need to know how contagious this pathogen is and the second which is the obvious for a network scientist of course, is the network, yes we need the network on which this pathogen is spreading. Does the network really matter? Yes it matters. So, if you look at the network which is shown to in this figure, you can see that there are quite a bless number of edges in this network.

The network is less dense rather we call that this network is (Refer Time: 02:06) So, you put any disease over there it will slowly move to this network; rather it can die away quickly, but if your network is say something like this and I put a lot of edges there between these people, so you put any disease on this network and this disease will quickly spread on this network because, your network is dense.

So, yes one thing is the density of the network or let us say the sparsity of the network it affects how your contagion is going to spread on the network rather, there is one thing very interesting to note here is do you see that this pathogen has something to do with the network? So, let us say I want to model these spreading of flu and I can be having a network like this and the network is going to be quite dense and actually network is dense for the spreading of something like a flu because, even if you come in because flu spreads even if you coming close proximity with someone, so even if you just stand and talk to a person for 5 minutes you can catch flu right.

And even people do not need to come in close contact together, let us say that you have worked on a piece of code on your keyboard and then you go away and then your sister comes and she also works on the same keyboard, even this thing can make a flu to spread. So, this common flu, common cold flu, common cold virus, it can spread even through keyboard.

So, the network is going to be very dense in the case of such a contagion, if the contagion is a flu, but let us say I do not want to model the spreading of a flu, I want to model spreading of let us say HIV. So, in the modeling of a HIV will the same contact network work? And the answer is no, in the case of HIV since it is spreads with the help

of sexual contact, there will be quiet less number of edges which will be which will be counted in this contact network.

So, the network in the case of HIV is going to be very sparse. So, whether your network is dense or your network is spars also depends upon what kind of a pathogen we are talking about. If this is the pathogen like flu, then the network is going to be quite dense and if the pathogen is something like let us say HIV then this network is going to be very sparse.

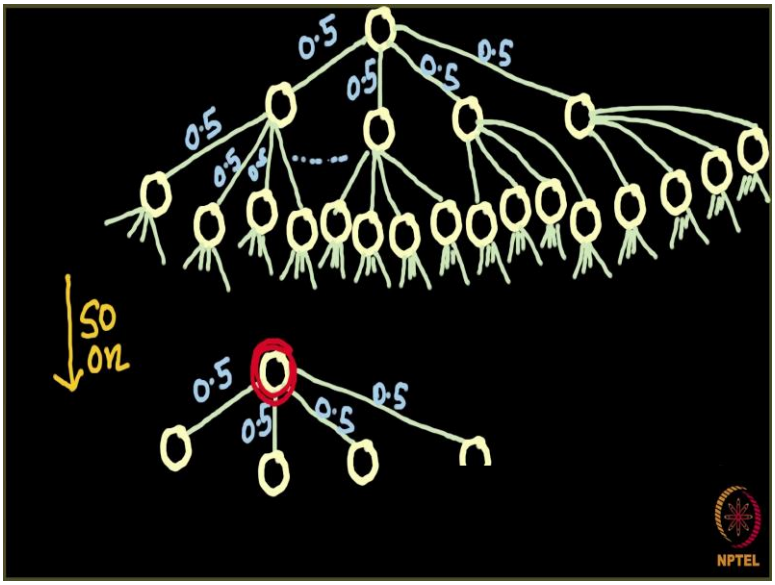
So, here while your blue I just repeat your contact network for the spreading of flu orange edges or the orange nodes here, this is the contact network for the spreading of HIV which is quite sparse. So, these are the two things which are most required for modeling the spreading of a disease; the first one is the pathogen how contagious it is and second one is obviously, the network structure, so will be using just both of these 2 things to model the spread of diseases on our social networks.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 130
Simple Branching Process for Modelling Epidemics

So, now we have covered all the prerequisites, so the prerequisites were start. So till now we have covered all the prerequisites which we were required for understanding the modelling of epidemics and we have looked at the two most important factors there. The first one is the pathogen and second one is the network. So we are now all set to go ahead and model this spreading of a disease. So let us say how we can go about it.

(Refer Slide Time: 00:35)



So, let us say here is a node, here is a person and now this person over here it has let us say four friends here as shown in this figure; so there is this person and this person has four friends here.

And then these four friends each of these has each of these have again four friends each. So you see we are taking a very simple example for modelling; so we will start from a very simple example and we will move towards a complex network. So we are taking the most simple network that is which is in the form of a tree; obviously, we know that social networks are not in the form of tree; because there will be triads on this network

and these nodes will be connected to each other. But only for the sake of simplicity for the time being we are taking the network to be a tree network, so one node having four children each of these four children again having four children.

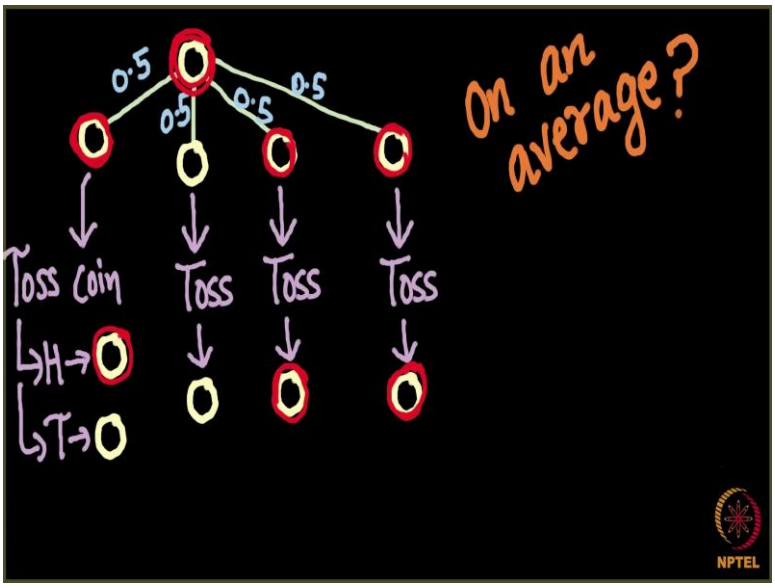
And then which of these again having four children each. So this is the kind of network on which we will be modelling a disease. So this is the network here now let us say this if you see this edge over there so I write a number on this 0.5. So as we have discussed previously there were two factors one was the network and another was the pathogen so the network we have seen here is in the form of a tree over here and second thing is the contagiousness of the pathogen. So we model the contagiousness of the pathogen in terms of probability.

For example, I have given this edge a probability of 0.5; what does it mean? It means that if this node here is infected then there is a 50 percent chance that this node will be able to infect this node. So with the 50 percent probability 1 by 2 probability the infection will transmit across this edge and if this node is infected it is this particular child will be infected. So I will say that the probability of infection across this edge is 0.5 and again for the sake of simplicity, I say that the probability of infection across all the edges is 0.5.

So, again if this node is infected it infects this node with the probability of 0.5; similarly, across this edge similarly across this edge and similarly across all of these edges. So every edge has a probability 0.5 of infection; what can we say from here? So let me just take the first level of this tree so I redraw this network here I have this node which is the same node here. So I take this node here and then this node has four children and it can infect each of these it infects each of this these children with the probability of 0.5.

Now if this node is infected so I show it with the red circle this node is infected. How many children do you think it will infect here? So each of its child it infects with the probability of 0.5 and when it infects these how many children do you think will become infected.

(Refer Slide Time: 03:53)



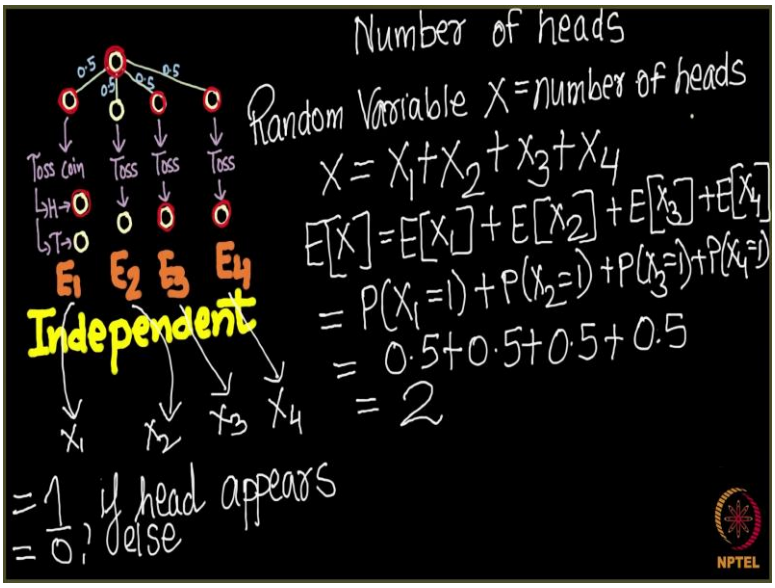
So, I redraw this network here just for the sake of simplicity and my question is how many of its children will it infect ok. So if we see here this node has a 0.5 probability of infecting this node what does that mean.

That means that if I toss a fair coin here and unbiased coin here I toss a coin I get head with the probability of 0.5 right. So if head turns up I will mark its the particular child to be infecting and if there is a tail I will say that this child is not infected. So how do I model this? If the probability of infection is 0.5, I will take a fair die toss it I get a head it means this node is infected else node.

And I do this for all of its children, so I do it for second child and let us say I get a tail and if I get a tail this node remains uninfected. And then I do it for the third child and let us say I get a head and I infect this and I do it for the fourth child let us say I again get a head.

So, I infect this fourth child also so three children are infected, so this happens for this particular case what do you so this three guys got infected over here. So what do you think happens on an average, on an expectation so what is the expected number of its children which will get infected here.

(Refer Slide Time: 05:26)



So, let us quickly see that; so I again redraw it so here I have four cases; in this case I am tossing the coin here, I am tossing the coin here, here, here. So four of these are different experiments, which are independent of each other so I have four experiments here.

I toss one coin for this, one coin for this, one coin for this and one coin for this and all these coin tosses are independent of each other. So I say that these all experiments are independent one has nothing to do with another; all of these are independent. And now our aim is to find the number of heads expected number of heads rather so if I the number of heads I get is equal to the number of nodes infected; because if I get a head I infect that particular node so our aim is to find the expected number of heads.

So whenever we have to find the expected number of a variable we modulating the form of a random variable. So I take a random variable X here and the value of this random variable X is equal to the number of heads I am going to get. And how do I find the value X over here it is with the help of these four things so what I do is, I take four more random variables. One random variable for each experiment so for experiment 1 I have a random variable X1 rather we call it indicator random variable. So indicator random variable is a random variable, which takes the value only 1 or 0.

And then for E2 I have another indicator random variable, for E3 I have 1 for E4 I have 1. Indicator random variable is something very simple what is happening in experiment 1 and either getting a head or I am not getting a head. So I say that the value of X1 is 1 if I

get a head and 0 if I do not get a head. So this is what do I mean by an indicator random variable it is 1 for success and 0 for failure. So I have X1 which is 1 if head appears, 0 if tail appears similarly X2 is 1 if head appears 0 if tail appears and same for X4 and X4.

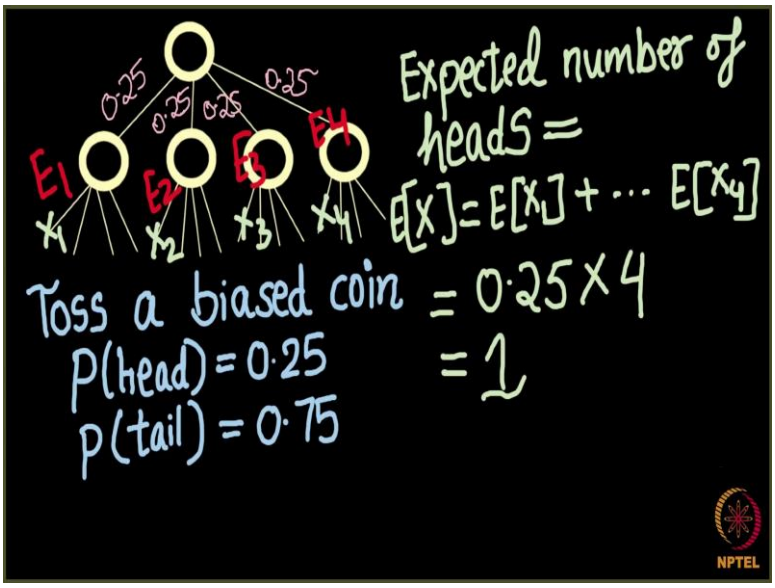
So, now can you write this X in terms of X1, X2, X3 and X4 so what is my random variable X; it is the number of heads. So I know that X1 is what the number of heads I get from experiment 1 right. If it is a success, I get 1 head if it is a failure I get 0. Similarly, X2 is a number of heads I get from experiment 2 which again can be either 1 or 0 similarly I have X3 and then I have X4. So what do I can say is X equals to X1 plus X2 plus X3 plus X4. So wherever I get a head out of these experiments I can add that add that value and it becomes equal to X ok.

What is our aim? Our aim is to find the expected value of X so I will take an expectation on both the sides; left hand side and right hand side and I get expected value of X is expected value of whatever is there in the right hand side and I can actually distribute this expectation. So I can write expected value of this complete term X1 plus X2 plus X3 plus X4 as something like this. So this is called linearity of expectations I will not go in much detail so I get expected value of X is this and then again here is a you can take it as a black box. So just understand this if I am talking about the expected value of an indicator random variable.

So, here X1, X2, X3, X4 are indicator random variable either 0 or 1 so if I talk about the expected value of an indicator random variable it is nothing, but the probability that this variable is 1. So, I can write it like this expected value of X is probability that X1 equals to 1 plus probability that X2 equals to 1 plus probability that X is equals to 1 and so on. And then we know what is the probability that X1 is 1; when is X1 1? X1 is 1 when I get a head in the first coin toss and the probability of that happening is nothing, but 0.5 and same is for X2, X3 and X4.

So, overall I get the value 0.5 into 4 which is 2. So I have done it rigorously you even if you did not understand it is perfectly it comes to (Refer Time: 10:31) also that. If let us say I toss a coin four time so on an average how many heads will you get and intuitively your mind tells you that the answer is 2; 0.5 into 4 so even if you did not understand this calculation of expectations it is perfectly fine ok.

(Refer Slide Time: 10:51)



Now, let us take another example; again I have a node here and then this node has four neighbours and then each of them has four neighbours and then let us say the probability of infection across every edge this time is 0.25 instead of 5. So in the previous case we have tossed a coin so the probability was 0.5 we tossed a coin we looked at whether it is a head or it is a tail and accordingly we did what we have to do. Now what I do in this case can I toss a coin in this case; I do not know so coin gives me head with probability 0.5 tail with probability 0.5; what do I do in this case?

So what we can do in this case is we toss a biased coin and my biased coin is such that it gives me head with the probability of 0.25 and gives me tail with the probability of 0.75. So biased coin is something like it is not fair coin, fair coin gives you head and tail with equal probability. So a biased coin can be something like if you remember the movie the movie Sholay; so there was a coin which has which had head on both the sides so that was a biased coin in that case probability of head was 1 probability of tail was 0.

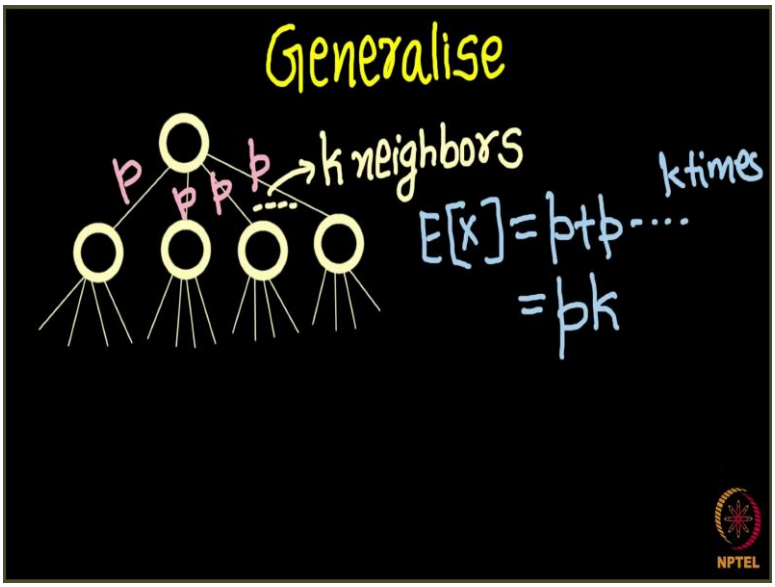
No matter whatever you do you are going to get a head every time; that was a biased coin. Here I am using another form of biased coin which does not give you head always. It gives you head 25 percent of the times and gives you tail 75 percent of the times. And again what I am going to do is if I then again I will have these four experiments here for each experiment, I am going to toss a coin and then if I get an head I infect the particular

node, if I get a tail I do not infect the particular node. And here what do you think is going to be the expected number of heads.

So, it is again very simple its almost the same what we have done previously so again we take four indicator random variables here X1 X2 X3 and X4 and we know that expected value of X is expected value of X1 + X2 + X3 + X4 which can be written like this. And we have looked at what is the expected value of a indicator random variable is nothing, but the probability of 1 which is probability of head which is 0.25 in this case. So I am going quiet I am going quiet quickly because it is similar to what we have done previously you can pause the video here and see what is actually happening.

So, we get here 0.25 into four which is nothing, but 1 so the expected number of heads here is 1 which means that if this guy here infects each of these nodes with the probability 0.5 on an expectation on an average it will infect one of its children ok.

(Refer Slide Time: 13:53)



Now, what we are going to do is we are going to generalize what we have done till now. We have looked at two examples and now we are going to generalise whatever we have done.

And it is very easy how do you generalise it I take a network here and now I say instead of having let us say 4 children so instead of every node having four children, I say that every node is having k children or k naught neighbours. Every node is having k children

so the node here it will have four children and then the node here it will have its own four children node here four children and so on I am very sorry not four it will be k.

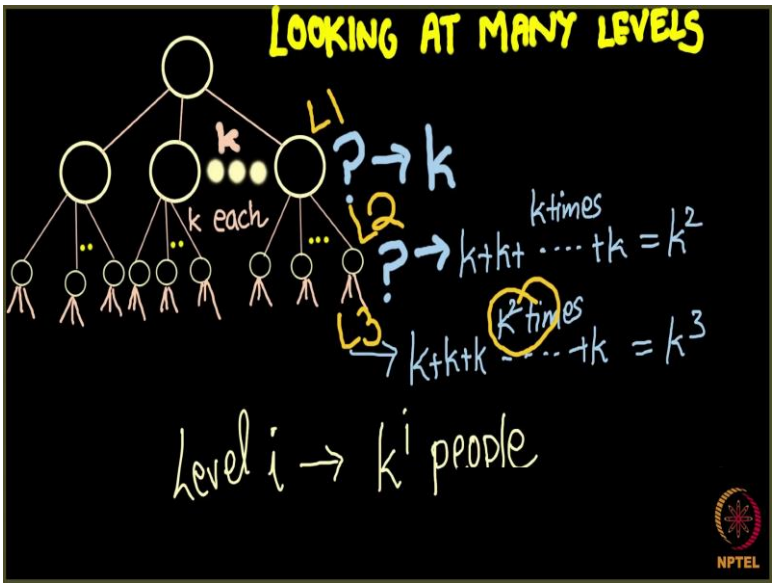
So, this very sorry this node here will be having k children this node here will be having k children this node here will be having k children and so on. And let us say the probability of infection is p so we have taken 0.5 in the first example 0.25 in the second example here we generalise it and take this probability to be p. Now what do you think will be the expected number of infected people, so it will be simply be so you can write each of these.

So how many experiments are we doing here? We are doing k experiments here because there are k children and what is the probability of success in every experiment is p. So the expected value of X that is expected value of you know infected people here is going to turn out to be p + p + p k times which is going to be equal to p * k.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 131
Simple Branching Process for Modeling Epidemics (Continued)

(Refer Slide Time: 00:09)



Now, what we are going to do is, we are going to look at more levels, many levels. How do we, what do I mean by looking at many levels? So till now we have looked at if there was one person who is infected and this person has some children how the disease is going to spread in this case according to what is the number of children this person has and how is the pathogen, what is the contagiousness of the pathogen we have looked at that in terms of probability.

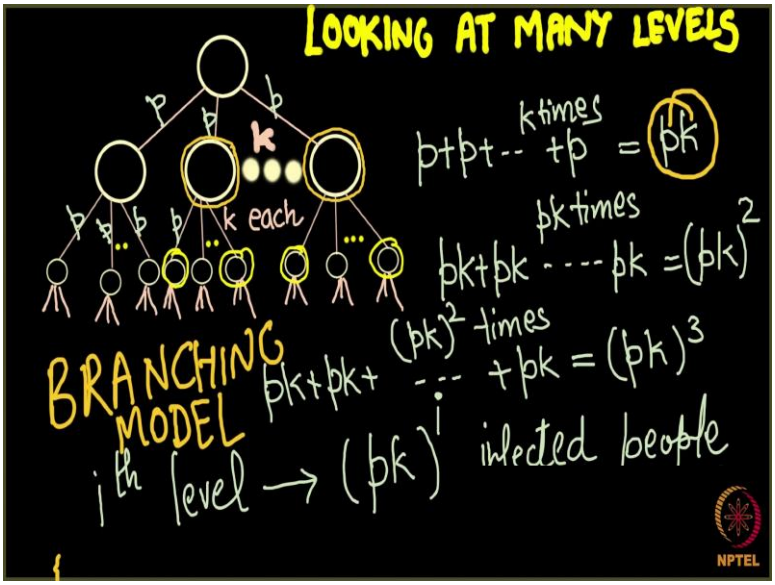
Now, what I am going to do is, we are going to look at this network in terms of many level. So this is the first node here and let us say this node here has got k children k children and each of these k children is again having some k children each of these children has these k children and so on. Each of these k children are again having some k children; what do you think? So let us call this particular level to be level 1 so this is your level 1.

So, what do you think are the number of people present at level 1 and we know it very clearly this is k because our node here it is having k people? So the number of people at this level are going to be k people now what do you think is the number of people at the second level. So this is our level 2 so what is the number of people here at level 2 so a little bit of maths tells us what is the number of people. So in the previous level we were having k people and each of these k people have k children.

So, how many people here we have; k + k + k k times which is equal to k^2 similarly let us look at level 3. So how many people will be the here at level 3 so at level 2 we were having here at level 2 we were having k square people and each of these k^2 people now are having k children. Hence at level 3 we have how many people k + k + k k^2 times because each of these k^2 people are having k children each.

So, we get an answer k^3 so on and so forth so if I ask you the question what is the number of people at level i; so what it will be at level 1 we had k people k^1 at level 2 we had k^2 people; at level 3 we had k^3. Similarly, at level i we are going to have k^i people. So, we have looked that how the different levels how do the number of people increase; what we want to do next is.

(Refer Slide Time: 03:06)



Let us look at how the contingence spreads over here and again looking at many levels and here we have a node. So this node is here having k children each of these k children is again having k children and each of those is again having k children.

The probability of infection here is p; I know that there are k people at this level, but what are we interested in now is out of these k people on an expected on an average how many people are infected here. And we know the answer right so if there is a person here and he has k neighbors each of the edge each of the edge has a probability p of infecting. So we have looked at it previously that there are going to be p + p + p k times, that is on an average p * k people are going to be infected at this level.

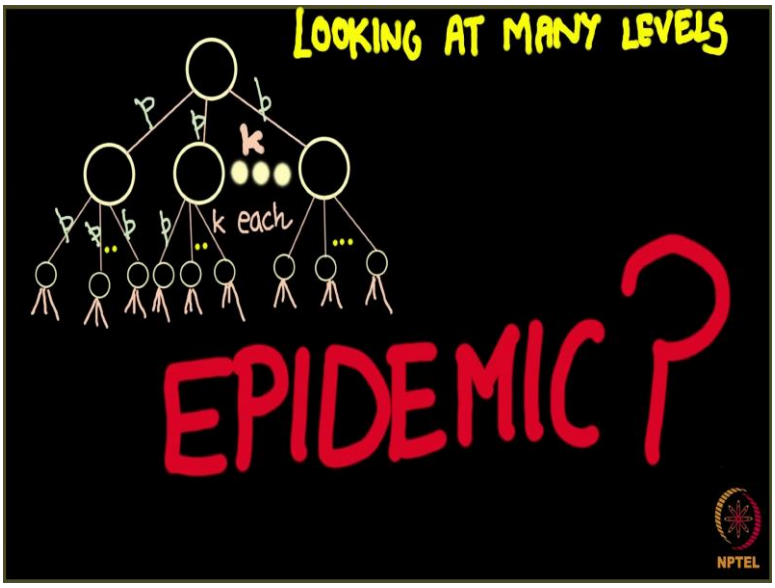
What about the next level how many people are now going to be infected the next level; can you pause this video for 2 minutes and then try to figure it out by your own and then you can resume the video. Now here I am having p * k number of infected people let us say these are those infected people; so, one is here, one is here, so here are p * k number of infected people. What happens next? This person here how many people is it going to infect in the next level again the question is the same.

One node is infected here. And this node is having k neighbors each of the neighbors gets infected with probability p as before on an average it creates p * k number of infected people. Similarly, this node here creates p * k number of infected people and how many such nodes are there p * k such nodes are there. So how many people get infected at this level is p * k + p * k + p * k times which is p * k^2. What happens at the third level is again similar.

So, let us say these are those these people are those pk^2 number of people who are infected. So now, on the next level how many people will be infected? So each of these guys over here will infect on an average p * k number of people so in the next iteration p k + p k + p k and there are p^2 people so we add it p * k^2 times and we get p * k^3. So in the third level p * k^3 number of people gets infected.

Similarly, what happens at the ith level is p * k^i people get infected. I think it is clear now so in this tree kind of a network at ith level we have how many people we have k^i people. And then if we talk about the number of infected people there are p * k^i number of infected people. This is all about the basic epidemic modeling and this model which we have discussed about this is known as the branching model. So, this model it is known as the branching model.

(Refer Slide Time: 06:59)



We have looked that this example; always keep this example in your mind so one person having k children each of these having k children and then we looked at the branching model here we found out the expected number of people infected at the ith iteration is p * k^i. What I said in the beginning, why are we studying these why are we studying this contagion, how does a contagion spread? And the answer was to understand whether something goes epidemic or not.

When will I say something has become epidemic on this network shown over here I go I keep going down in this network keep going down in this network and when I go lot of down let us say through some infinite level and I see a person infected here. I can say that this disease has become an epidemic. If it was not to become an epidemic it would have died somewhere here, but this disease starts from here infect these people, infect some of the people at this level we keeps we keep going down and it even infect some of these people at too much bottom or let us say at the infinite level.

Then I say that this disease has become an epidemic now from all the information I have given you that there are some probability p and number of neighbors k p * k^i people infected at the ith level. Can you tell me whether this disease is going to be an epidemic or not? So based on the values of p and k can you comment about whether contagion is going to be epidemic or not. Is it even possible to say it and the answer is yes and in next lecture we see how do we do that? How just by looking at the value of p the probability

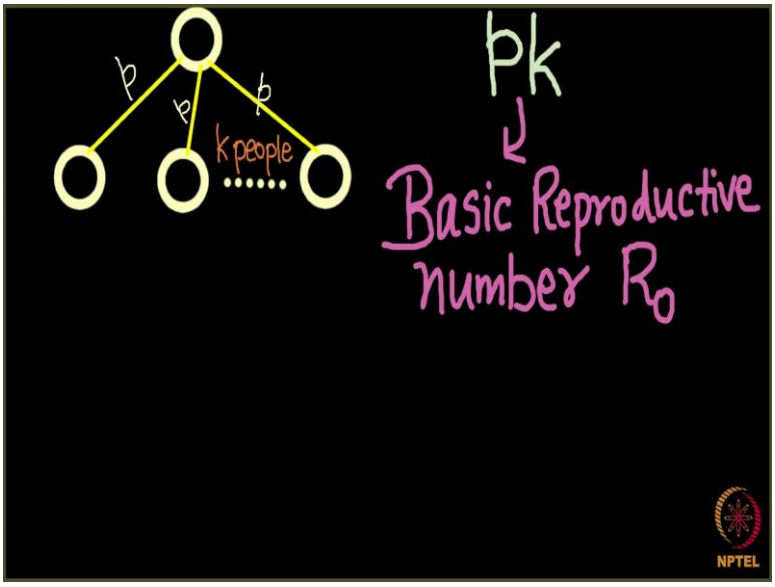
and k, the number of children every node has, we can comment about whether this contagion is going to become an epidemic or not.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture – 132
Basic Reproductive Number

In the last lecture, we said that just based on the probability p that is the contagiousness of the pathogen and number 2 by just looking at the number of children every node has, we can comment on whether a disease is going to become an epidemic or not. How do we do that?

(Refer Slide Time: 00:26)

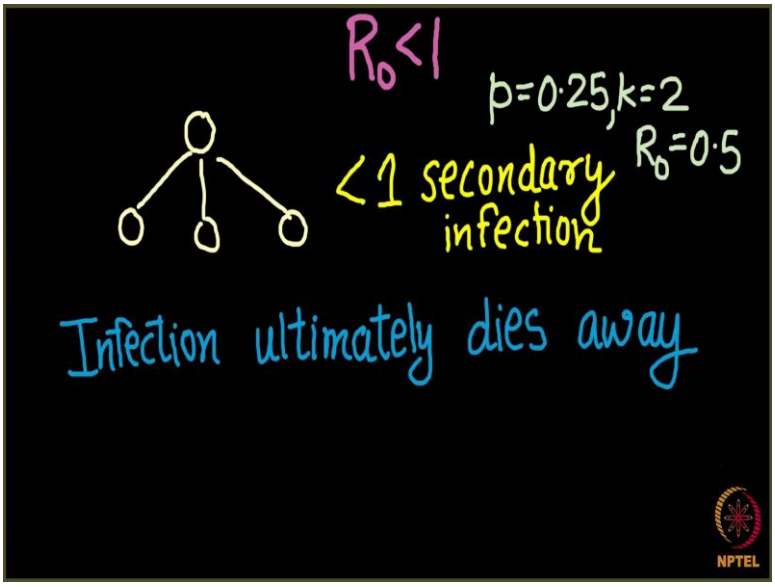


Let us look at that. Here is a node this node has some k people, or this node has some k children. Each of these children gets infected with the probability p. This is what we have studied till now. We looked at this number p * k here, what was this number p * k? This number p * k was the expected number of people who are going to be infected here.

If you can see what is this expected number of people who are going to be infected here. You can see if this node is infected p * k tells us the number of secondary infections produced by one person; primary infection here and it creates some number of secondary infections and that number the expected value of that that number is p * k. This number p * k is important. It is called the basic reproductive number denoted by R0 and it is very

very important, because this number is going to reveal a lot of secrets. This number itself is going to tell us whether a disease is going to become an epidemic or not.

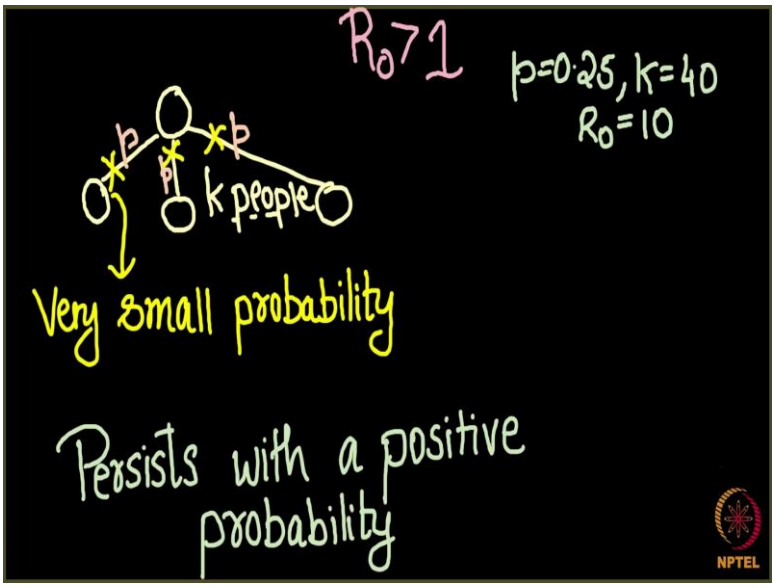
(Refer Slide Time: 01:40)



And I tell you here that if this number R0 < 1. Let us see what will happen. So, this number R0 < 1. Recall what is R0? R0 is nothing, but p * k that is the number of secondary infections. If R0 < 1, what happens? There are less than 1 secondary infections. If you look at this node over here, it creates less than 1 secondary infection what does it mean? It is unable to infect even one person here. You see that disease dies over here itself and let us say by hook or crook, the disease somehow survives here. But then in the next iteration, it has a very high probability of dying out. First of all, it dies away here only, even if here it somehow survives is able to infect one node. In the next iteration, it will not be able to infect even one node.

If R0 < 1, what is going to happen is let us take this example first. Let us say p is 0.25 that is the probability of infection is 0.25 and every node is having two neighbors, then a basic reproductive number R0 is going to be 0.5. That is this node here on an expectation, it is going to infect half of its child, half of its children. Half of its children says that it is unable to infect even one person here and the disease dies away in this case. What happens? So, when R0 < 1, your infection ultimately dies away which means that this contagion is not going to become an epidemic.

(Refer Slide Time: 03:25)



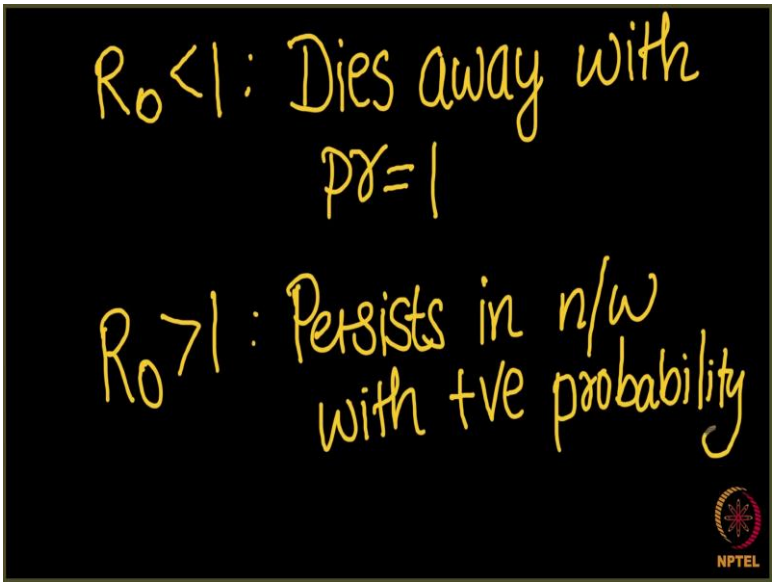
What happens in the other case, R0 > 1? If R0 > 1, we again take the same network k people probability of infection p across every edge and let us see, let us say p is 0.25, k is 40. There are 40 people and each person gets infected with the probability of 0.25. The basic reproductive number is 10. So, you would see what will happen here. This node here will infect 10 people here and then those 10 people will infect almost 100 people and those 100 will infects some 1000 people.

So, you see how this disease is going to become an epidemic. So, in this case when a very sorry so, in this case when your R0 > 1, your disease becomes an epidemic. There is a slight difference between both of which is very important. Previously here when your R0 < 1, we said that infection ultimately dies away, and it dies away with probability 1. It is for sure that this infection will die away. I am not going to tell you now, exactly how, I will be covering that in the advance part of this lecture, but you take my words in this case infection will ultimately die away for sure that is with probability 1 infection will vanish from this network.

But in this case here what is going to happen is, your infection yes it persist in the network it is an epidemic, with the high probability it is an epidemic, but still here is a chance that it will die away. So, in this case when R0 > 1, your infection your contagion is an epidemic, but not with the 100 percent certainty. There is still a very small probability that it can die away. How it can die away is let us say at the first level itself.

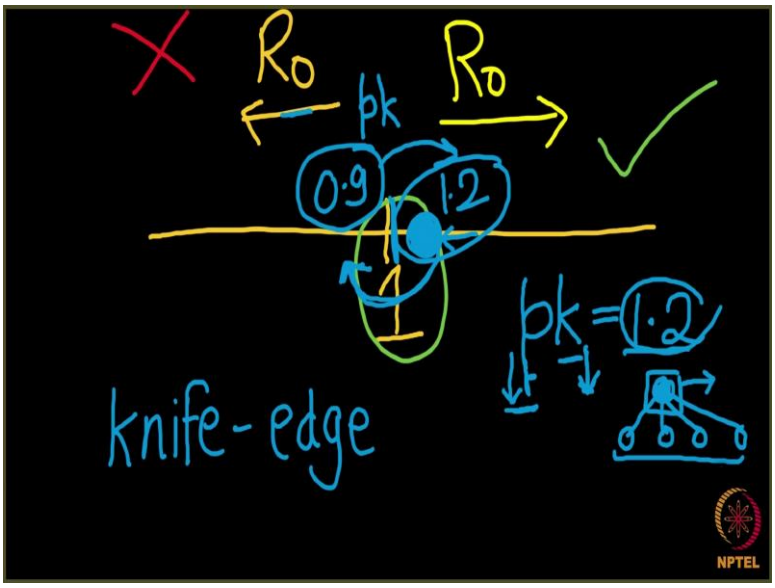
All of these links fail and the probability of happening that is very small. There are 40 links and out of this 40 links each of these links fail, very less probability. But still it is positive; there is some positive probability that all of these connections fail. And if all of these edges fail will very sorry; if all of these edges fail in transmitting a disease, it means that your infection is died away. So, when R0 > 1 with some positive probability which is high or infection persists become an epidemic, but with a very small probability it can die away also. So, persists with a positive probability.

(Refer Slide Time: 06:25)



While write this statement here for you if your R0 < 1, then your disease your contagion dies away certainly, dies away with probability equals to 1. But if your R0 it is greater than 1, then it persists in the network. It becomes an epidemic; your infection persists in the network with positive probability. I do not say that this probability is 1, but with some positive probability your infection persists in the network. So, we have looked at the importance of the value of R0. If R0 < 1, the diseases for sure going to die away; if R0 > 1, the disease becomes an epidemic. Do you see something very interesting there? The importance of this value R0.

(Refer Slide Time: 07:26)



So, here let us see here is the value 1 and we say that if R0 it is less than 1, your infection dies away; infection no longer exists. But if your R0 it is greater than 1, R0 is on this side then your infection is going to become an epidemic. So, this value of 1 here it is becoming so very important. Why so very important? Let us say you have a disease, a contagion is spreading on a network and let us say the value of R0 > 1 right and let us say this value lies somewhere here.

So, here is your value of R0 let us say it is some 1.2 and it means that your disease is going to become an epidemic. Now you see what is happening. If somehow, I can do something here and bring this value a little bit down from 1 let us say from 1.2, I make this value to be 0.9. Do you see the disease suddenly dies away? So, this disease, which was an epidemic here, this disease was an epidemic here; I played little bit with its value of R0 and suddenly this disease will die from this network. Is not it so very interesting and so very important also? We have looked at R0, what was R0? It was p * k so, p * k is 1.2 here and I get you know play a little bit here.

I can maybe decrease the value of p a little bit or maybe k a little bit. How can I decrease p a little bit? Now what is p? p is the probability that the disease will attack you. So, I can decrease that probability. So, let us say a contagion is a spreading and epidemic disease is spreading, viral disease is spreading; I can put an advertisement on TV, I can ask people to take better health measures to have proper hygiene and this will you know

decrease my value of p. So, I can either decrease my value of p. Decreasing the value of k is also very easy. What was k? k was the number of people with this infected person was in contact with.

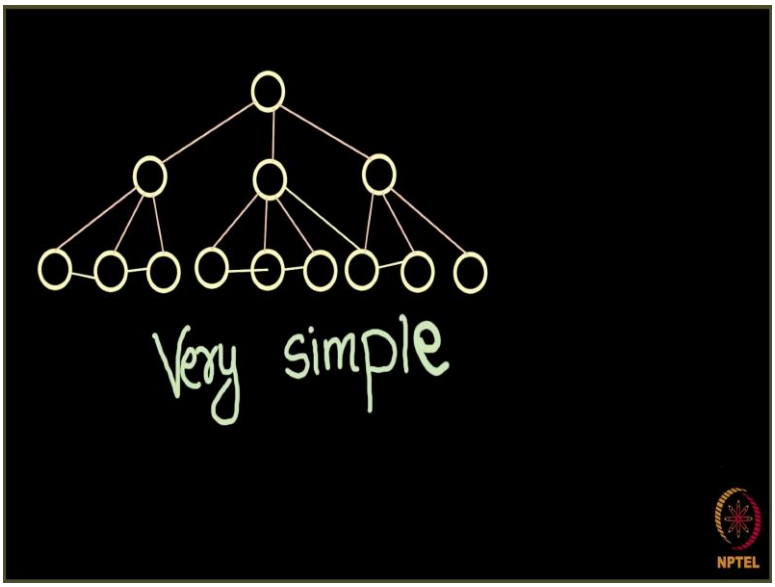
So, if we can decrease this number of people who are encountering this person. Basically, what I can do? I can for some time I can ask this person not to come out of his room that is called quarantining. So, I can isolate this person. So, as soon as I isolate this person, the value of k comes down. Either p comes down or k comes down, this value 1.2 even if it gets a little bit reduced; it will come on this side of the network and the disease will vanish. And the same thing happens here, here is a disease and the disease is not an epidemic but let us say the value of R0 is 0.9, a little bit increase in p or a little bit increase in k can now cause this disease to become an epidemic.

So, this particular property of this R0 is called knife-edge nice property. It is like the edge of an knife right, edge of an knife here. So, little bit on the side you come the disease will vanish from the network and little bit if it if there is an increase here a little bit, the disease will become an epidemic. So, that is the importance of this value, the basic reproductive number. It very well defines whether your disease is becoming an epidemic or not.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 133
Modeling epidemics on complex networks

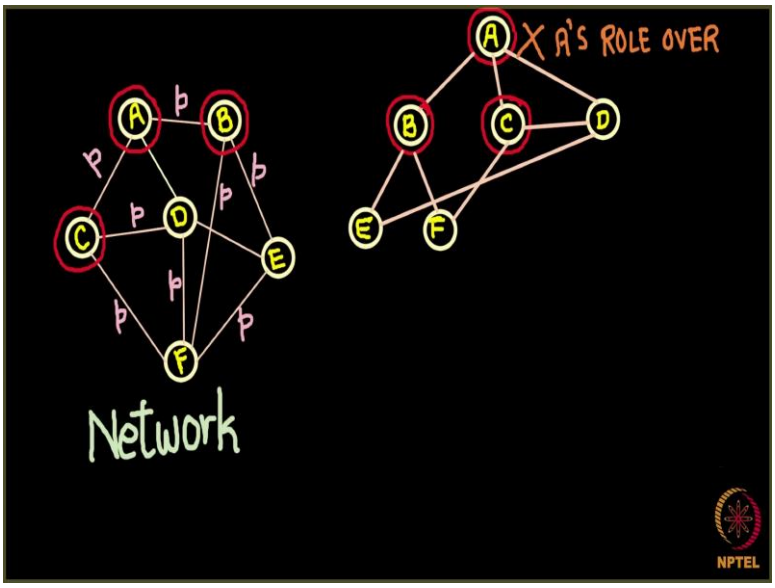
(Refer Slide Time: 00:11)



Till now we have looked at the simplest epidemic model, we will be considered the simplest network and the network was in the form of a tree something like this simple. Now, we know that a real world network do not look like this, a real world networks they look complicated and rather if you do remember the lectures from the previous weeks you will say that a networking never look like this, there should be some triadic closure here right.

So, if this guy over here is a friend with both of these there is a good chance, that both of these are friends, both of these are friends and maybe the this person here can be a friend with this one So, networks actually look very complicated something like this. How can we modify a most simple branching process what we did before for a real-world networks which are not trees? That we are going to see in this chapter.

(Refer Slide Time: 01:07)



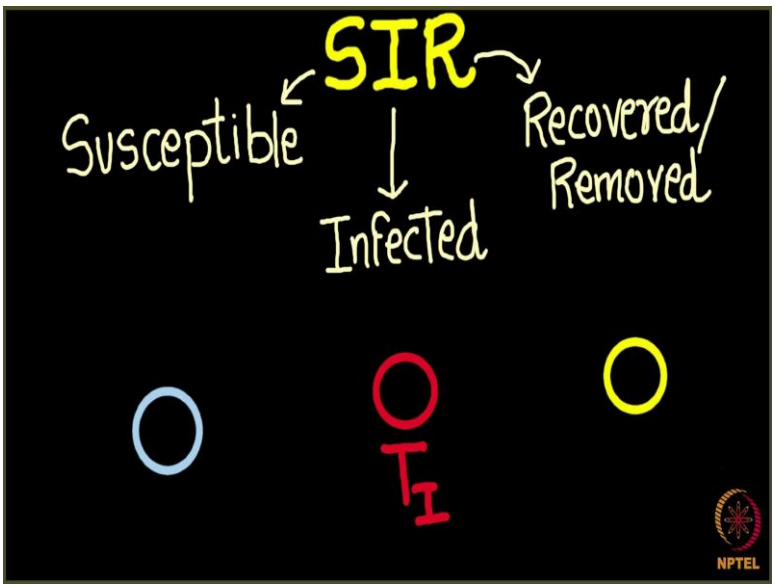
Let us say that we have a complicated network, not a tree. By complicate it what I mean is not a tree which looks something like this I have some nodes A, B, C, D, E, F here and every edge has a probability p of transmitting the infection as before. So, everything is same every edge is having a same probability. The only differences, in the layout of the network instead of the tree we have taken a proper network.

Let us say A is connected to D also here and the network look something like this. Now, I want to simulate the spreading of a contagion here how do we do that? Say initially A is infected and now A has three neighbors B, C and D and it tries to infect each of these again independently with the probability p. So, again here are three independent processes, we can flip coin for the edge A to C and if head comes, we infect C. Similarly, we can flip a coin for the edge A to D and if a head comes, we mark D as infect it and same thing we can repeat for B. So, maybe A is able to infect B and C.

How can I layout? Layout in the form of a tree, what is happening here? Let us see here is A, and A was having three neighbors B C and D and A was able to infect B and C. And now you see as role is over. I will discuss more about it. What I mean here in a branching model also at this step A's role was over A go to one chance to infect and A infected B and C. Now, it is B and C's turn now B and C will go ahead and see which of their neighbors they can infect; why so?

It should be so right because if A keeps getting a chance rather if every node keeps getting a chance to infect their neighbors at ever time period then this entire network will get infected. And that is not what happens in a real world also right you get infected from some disease and then you remain infected for some period of time. During this time you keep infecting other people and after that you might recover from the disease. So, we need a better model here let us make this picture clearer. So, what will be doing now is we will be looking at a very realistic diseases spreading model branching process was something very theoretical and very simple.

(Refer Slide Time: 03:52)



Now, we are moving little bit more towards what happens in the real world and we are going to study a brand-new model not for the tree networks, but for any kind of networks and the model is SIR epidemic model. Why is it called SIR epidemic model is because it tells you, it talks about three life cycles of an infection.

So, whenever you get infected from a disease, you mainly pass through three phases. The first one is the susceptible, susceptible is the one where you are happy living your life you are not infected from any disease, you are not infected from the disease. And the second stage is infected, so you come in contact with somebody who is infected and you get infected and the last one is recovered or removed.

Recovered or removed is something that needs to be taken very seriously here, something which is very important. What do we mean by the recovered and removed

here is what makes this model the, what makes this model spatial this SIR model it is used for two kinds of diseases, first kind of disease is something which is the terminal illness. What happens in the case of a terminal illness is patient ultimately dies away. You are susceptible and then you get infected and finally, you will be killed by the disease and when you get killed by the disease you will be removed from the network.

Or a second case; second case is the diseases something like that which can occur to you only once in your lifetime. Something like measles, you are susceptible and then you get infected from measles. Now what will happen when you get infected from measles is your body will develop a lifetime immunity against this disease and you are never going to catch this disease again that time, we say that you are recovered. You are never going to go from infected back to the susceptible this disease is going to happen to you only once in your lifetime. So, either you will gain a permanent immunity, or you will be killed by this disease that is our SIR epidemic model.

In our next lecture we will see how can we simulate this model on an actual network; we are going to do that the forgoing. So, we are having. So, a node can here exist in three states as we saw and exactly one state at a time you cannot be susceptible and infected at the same time. And infected and recovered both at the same time you can be only in one phase at one time.

If you are susceptible I am going to show you by a blue circle, if you are infected I am going to show you by an infected circle I will tell you what this T_I is and if you are recovered or removed I am going to show you by yellow circle. So, yellow circles can be ignored from a network because they have recovered and removed, they have nothing to do with the spreading of the disease now.

What is this T_I here? T_I here is again something very logical what is T_I? When you get infected there is some time period for which you remain infected it can be one day, two-day, one week, two weeks, during this time only you are going to infect other people. So, T_I define the time for which you are going to remain infected. So, maybe for the flu this time can be let us say three days. So, during these three days you will be infected with flu and you will keep infecting other people.

If you look at this figure here, where I said that here is A, and then A infected these two people B and C. And then we said that as role is over and these are B and C were going

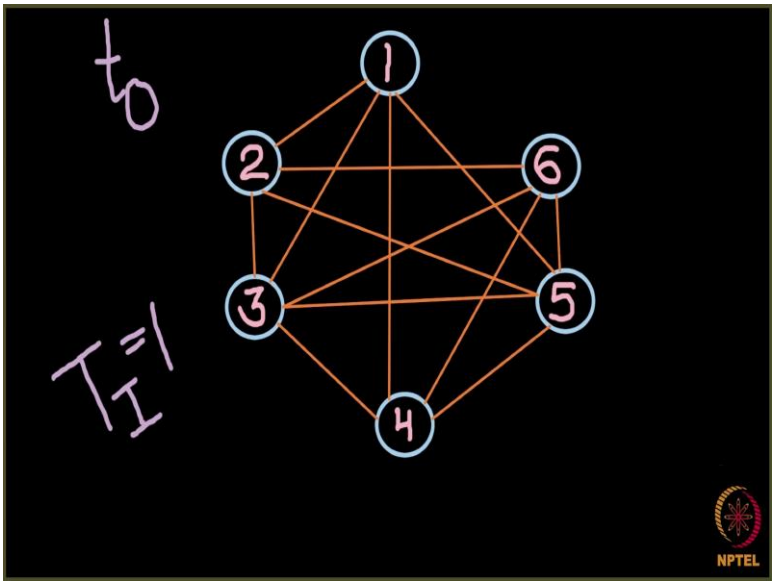
to further infect other people. Can you tell me what is the value of T_I here? So, T_I the time for which you are infecting here is 1 right because it was 2, after A infected B and C A will again get a chance to infect once more, we will look at it in detail just wanted to tell you the use of T_I. So, this is it this is an SIR model and in our next lecture we will see that how we can simulate this SIR model on a given network.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 134
SIR and SIS Spreading Models

So, we are now going to see how we can simulate the spreading of this SIR model we just discussed on a network. So, we will be modeling this is spreading in the form of number of iterations or number of days. So, you can imagine a situation where there is this measles is spreading on network. We will be looking at what happens at day 1, then what happens at day 2, what happens at day 3 and so on.

(Refer Slide Time: 00:32)

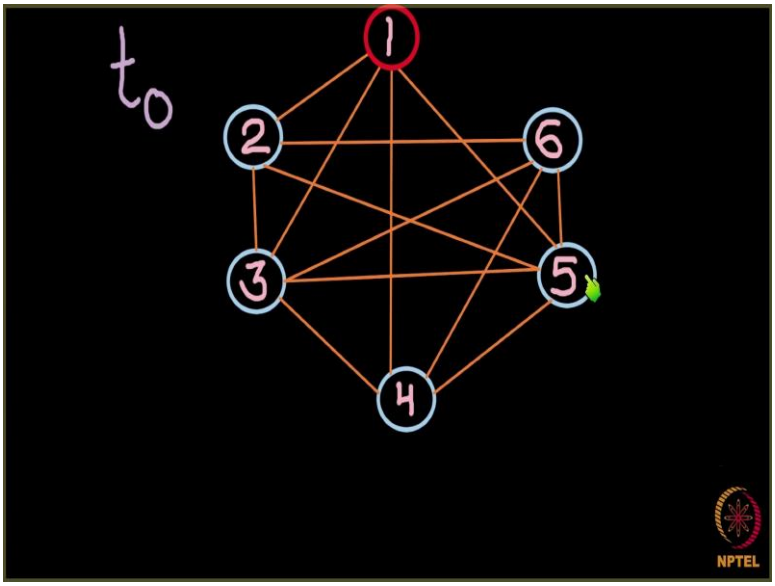


t_0; so, let us talk about day 0. TI is for time; so, you are talking about day 0. At day 0, the lets say the network looks something like this. So, you see there are 6 nodes in this network. 6 people in this network and the blue color is for susceptible. So currently, each of these people is susceptible. None of them has acquired the disease and let us say measles. So, none of these people are suffering from measles.

And these are edges across which your infection can transmit and let us say the probability same across all the edges. Probability of infection is p. So, the pathogen for measles it is having a contagiousness value which is p. Now, let us see what will

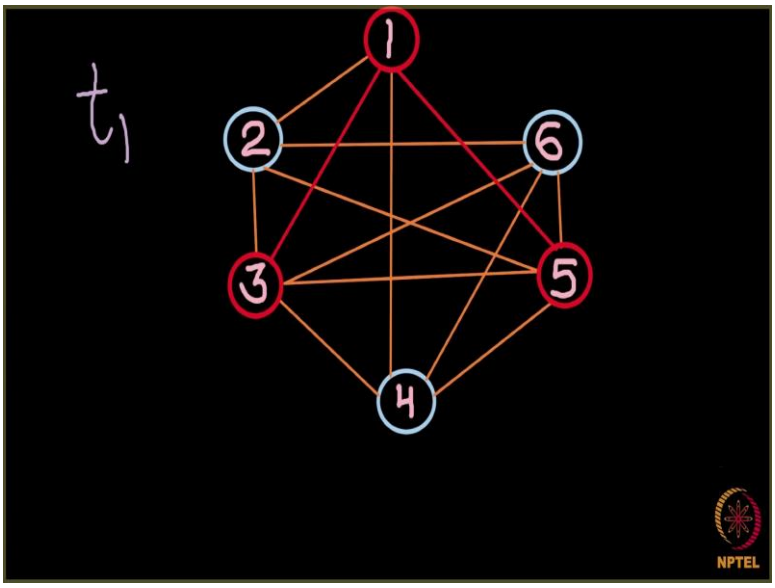
happen? Assume that and yeah one more thing, the value of TI we are going to be taken to 1. So, all though it might not sound realistic, but in the case of measles we cannot say that we remain infected for just one day and after that only you just acquire lifetime immunity. You remain infective for a long period of time, but for just for the sake of simplicity, let us take for the time being TI equals to 1. In the next example, we increase this value and see what happens, but for this example we take the value of TI to be 1. So, here is a new kind of measles, let us say. So, this measles keep you infected just for one day and after this one day you acquire lifetime immunity.

(Refer Slide Time: 02:08)



So, this is a network at day 0 and at day 0 this node 1 gets infected so, this node 1 gets measles the 0th day. What will happen on the 1st day now? Now this node 1 has 4 neighbors 2 and then 3 and then 4 and then 5. So, it has 4 neighbors and it tries each of these neighbors it tries infecting each of these neighbors with the probability of p and let us say, it is able to infect node 3 and node 5.

(Refer Slide Time: 02:38)



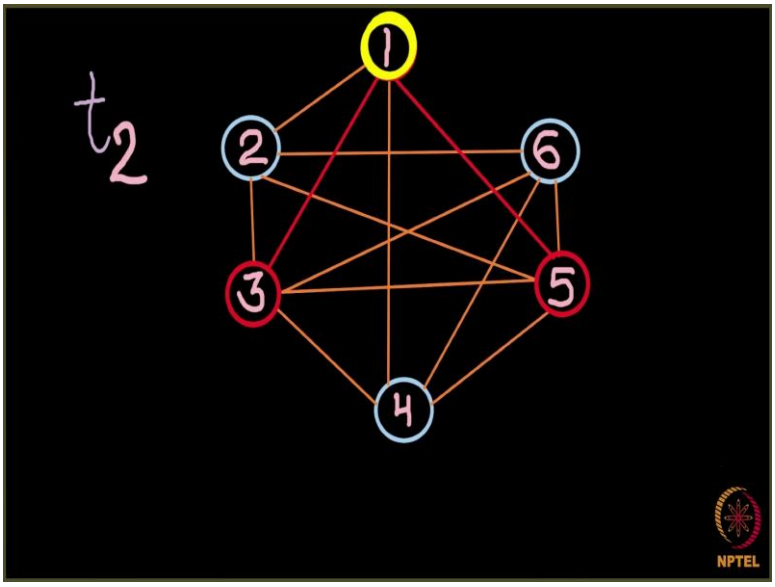
The 1st day, node 1 infects node 3 and node 5. Now you might want to ask, you might ask me a question here. How do I decide which are the nodes which node 1 is infecting? Sorry.

How do I decide it is going to be node 3 and node 5 only? And I will tell you that this is a very random process as we discussed before. So, I am going to toss a coin for each of these edges. So, if the probability of infection is 0.5, I toss up here coin for this edge, for this edge, for this edge and for this edge and here and I get a tail here, I got a head here, I got a tail here and I got a head here and I did it. And you see now, so what will happen. One simulation of this model can go like this. Where node 1 has infected node 3 and node 5, but if you simulated the second time, you again start from this graph and you again toss the coins; you might get some other result. So, it is a random process unlike in the case where we discussed how ideas and behaviors spread on a network, everything there was deterministic.

If you were given the snapshot of a network at a particular time, you would for sure with very certainty you can tell what would happen at the end. But here in this case, there is less certainty. It is more of a random process. So, let us see node 1 here, infected node 3 and node 5. Now, see node 1. Node 1 was infected at day 0 and now it is day 1. So, what is going to happen at the end of day 1? At the end of the day 1, our node 1 it is going to

be recovered. So, node 1 is discovered towards the end of day 1, but it has infected these two nodes in the network, node 3 and node 5.

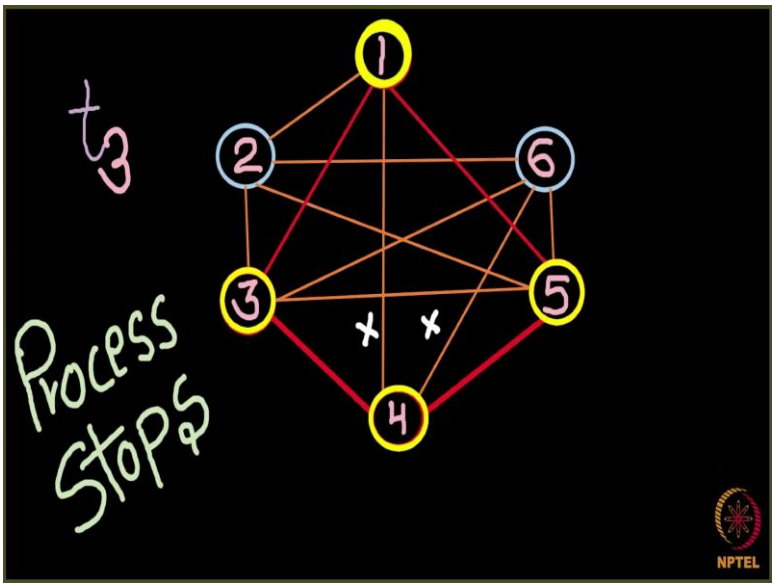
(Refer Slide Time: 04:34)



And now on day 2, node 1 do not get infected anybody, but and nodes 3 and nodes 5; node 3 and node 5 start doing their work and let us say, node 5 end up infected node 4. So, node 3 here is unable to infect any of its neighbors and node 5 here infect node 4. Node 3 here, it's an able to infect node 2 and node 6 and node 4 as well. And you see, node 3 can for sure not infect node 1.

Because node 1 it has recovered from the network. Overall node 3 gets nobody it could infect, but this person node 5 here, infects node 4. What happens next towards the end of the day 2? So, these node 3 and node 5 they were suffering from day 1. They suffered at the end of day 1, right? And now, they have passed one day of infection. So, towards the end of node 2 both of these nodes will become recovered.

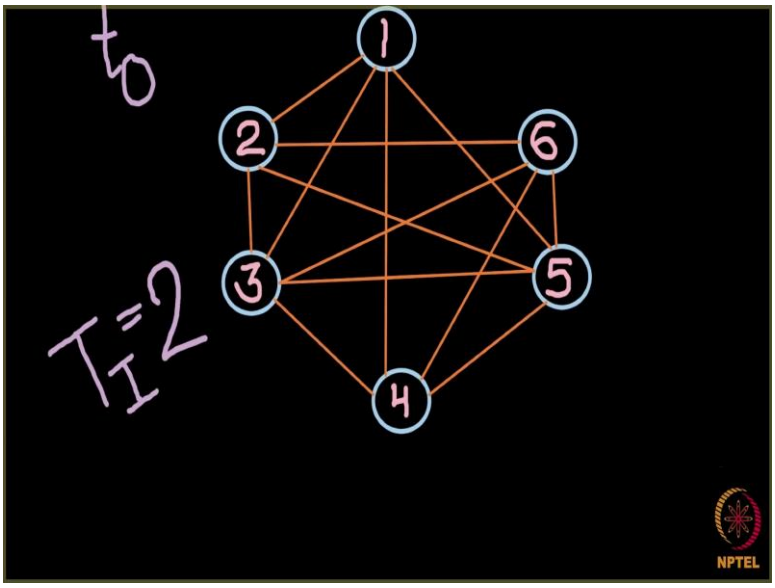
(Refer Slide Time: 05:40)



And on day 3 it is node 4 which goes around, and start infecting the people and let us say, node 4 is unable to infect anybody. Node 4 obviously cannot infect node 3, node 1 and node 5 now because they have recovered. It can infect node 6 but let us say it is unable to infect node 6 and towards the end of day 3, node 4 also gets recovered.

What will happen on day 4 now? On day 4, there are these 6 people out of which 4 have recovered towards a susceptible infection gets over because for this process to go further, we need a node, we need a red node here. We need a node which is infected, but now there is no node here which is infected, no red node here. So, what will happen? The process will stop. So, this was one simulation we did where we took the value of TI that is a infection period to be 1. Every node remains infected for one day.

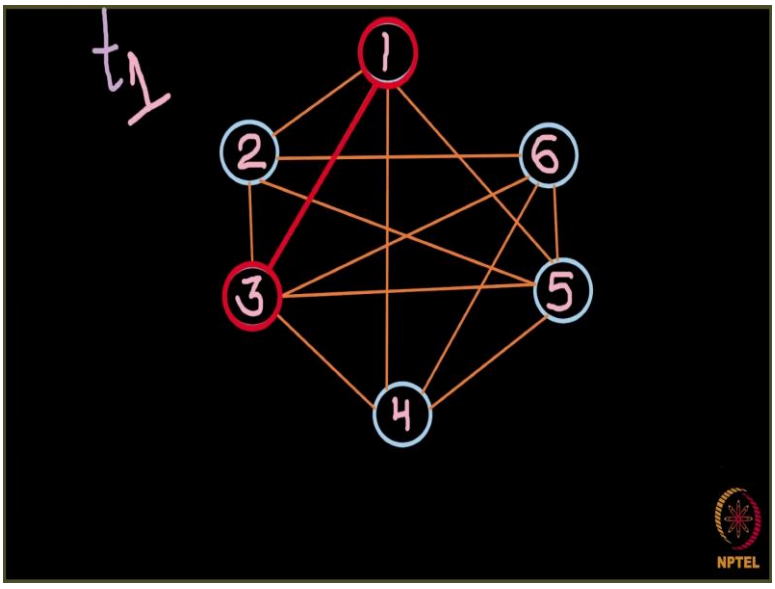
(Refer Slide Time: 06:41)



Let us now look at another example and let us say I have changed this infection period to 2. Now you remain infected with measles for 2 days and after this 2 days, you recover from the disease and again this is node 1 which catches the infection at day 0.

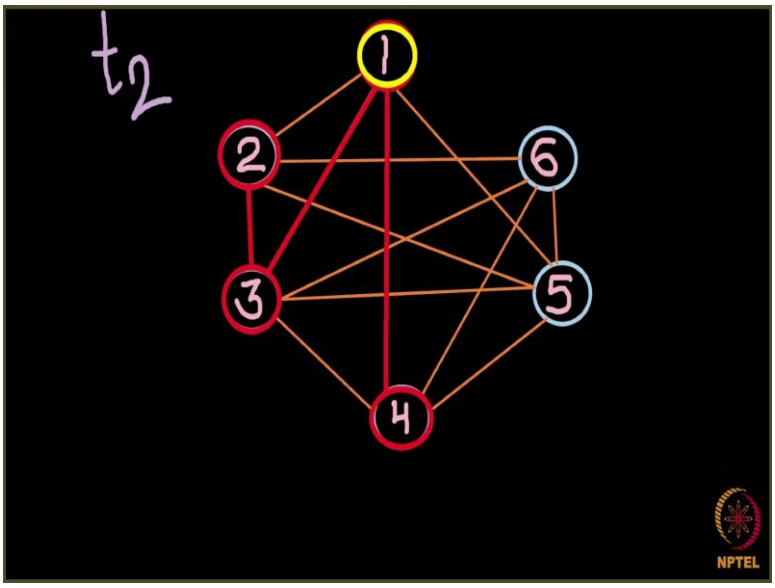
So, now, you see get 2 days. Days 1 and day 1 and day 2 to infects its neighbor and it will become it will recover at the end of day 2.

(Refer Slide Time: 07:11)



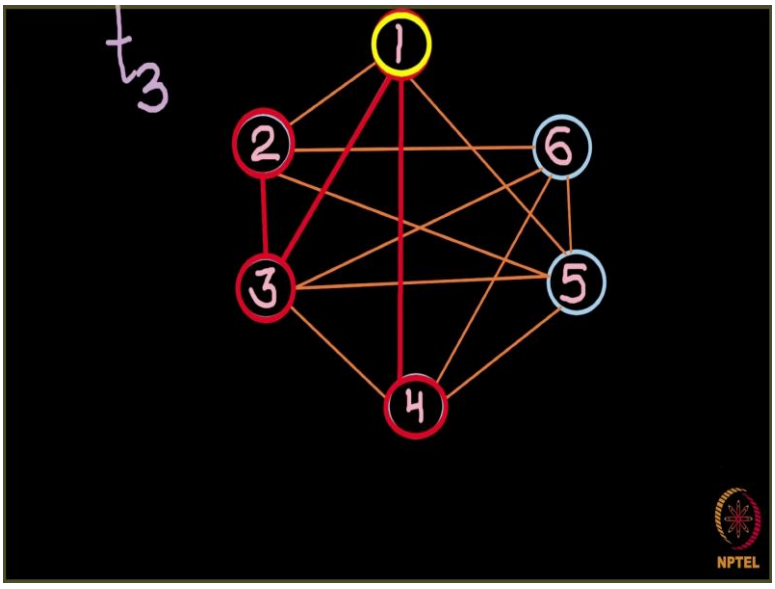
So, node 1 here, let us say day 1 it infects node 3, just node 3.

(Refer Slide Time: 07:18)



And then comes day 2. Now at day 2, both of these nodes will be infecting people. Node 1 as well as node 3. So, let us say day 2, node 3 infected node 2, node 1 is unable to infect anybody at day 2 or let us say rather node 3 infected node 2 and node 1 is able to infect node 4 at day 2. So, node 1 got day 1 to infect people and it is also getting now day 2 to infect people because it remains infected for 2 days, and towards the end of the second day, node 1 recovers.

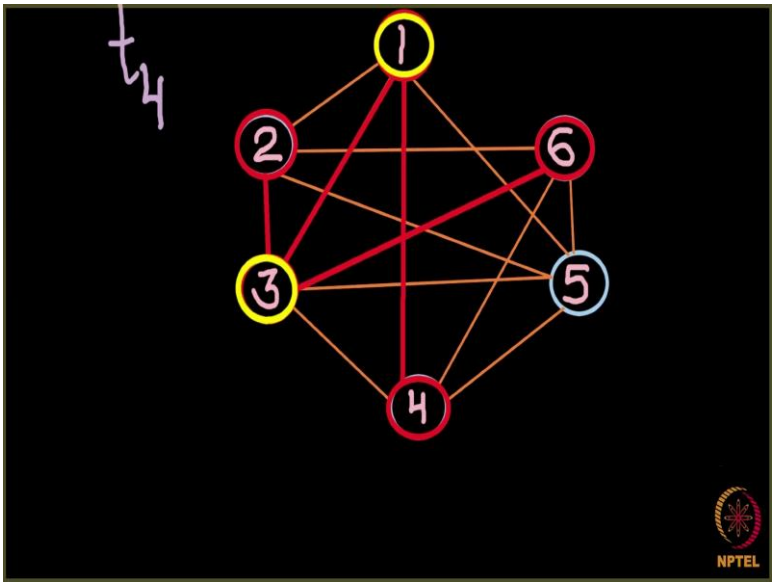
(Refer Slide Time: 07:56)



Now, during the third day, all of these three nodes, 2, 3 and 4 get infect their neighbors and let us say node 3 infects node 6.

If you look back at this network, so let me show you. So, this node 3, it was infected at day 1 and then day 2 it was infected node 2 and again at day 3 infected node 6. And at the end of day 3, what will happen? It will recover.

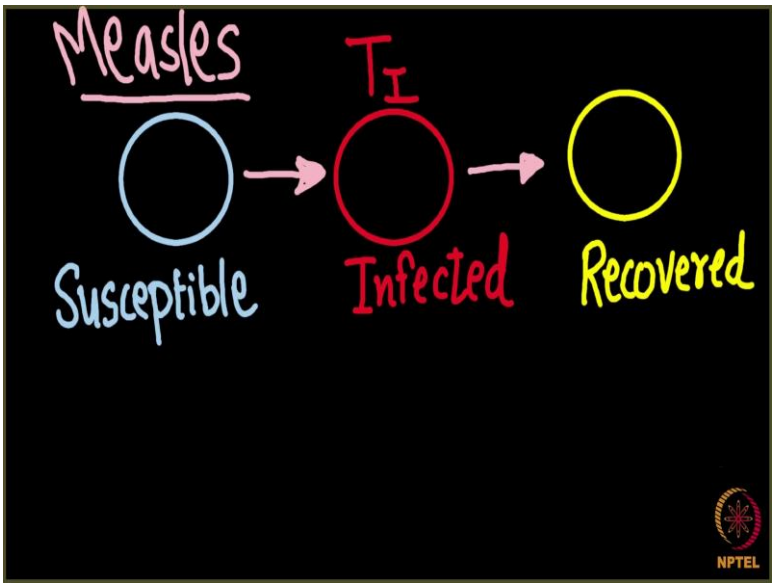
(Refer Slide Time: 08:26)



And now what will happen in day 4? Nodes 2, 4 and 6 get to infect people and let us say node 2 infects node 5 and what will happen towards the end of this day? Nodes 2 and 4 will recover.

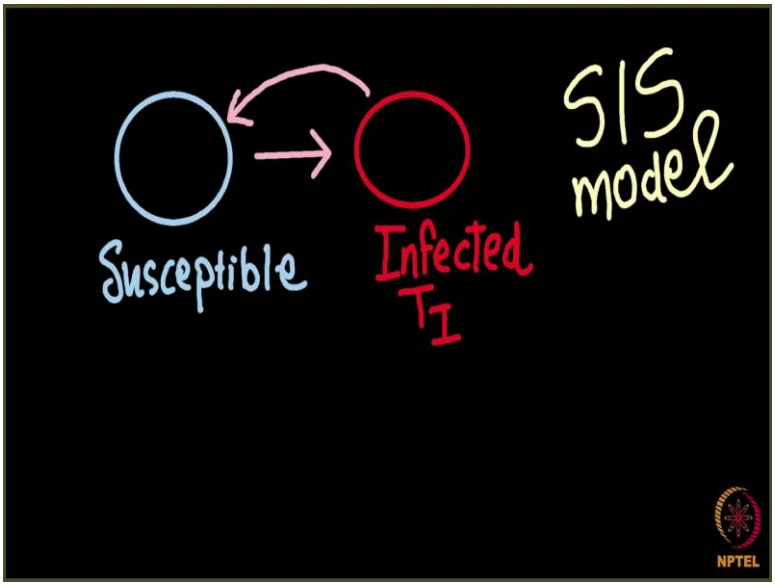
You can easily guess what will happen next, towards day 5. So, towards day 5 also, node 6 and 5 here will be infected and towards day 6 both of these will be recovered; and towards day 6 day 6 both of these will be recovered. So, at the end all the nodes in this network will recover. So, this is how we could have stimulated it for the value TI equals to 2. So, please note that it is one possible simulation. Every time you simulated it, this network will go, and it will be infected in a different way.

(Refer Slide Time: 09:23)



So here what we talked about lets quickly recap. It was a node, so the node can be either susceptible or a node can be infected and it remains infected for TI time during which it infects are other people and finally, the node gets recovered something like in the case of measles. Can I use the same model if I want to simulate the spread of a common flu? What will happen? let us say common cold. So, this infection might sorry. So, this model might not hold good there, right. So, in common cold yes, you become susceptible, you are susceptible and then you become infected after coming in contact with an infected person. Then will you be permanently recovered from common cold? It never happens. What happens is, after recovering from common cold, you enjoy for some days and or let us say some months. So, let us say some years, but after that again that common cold infects you.

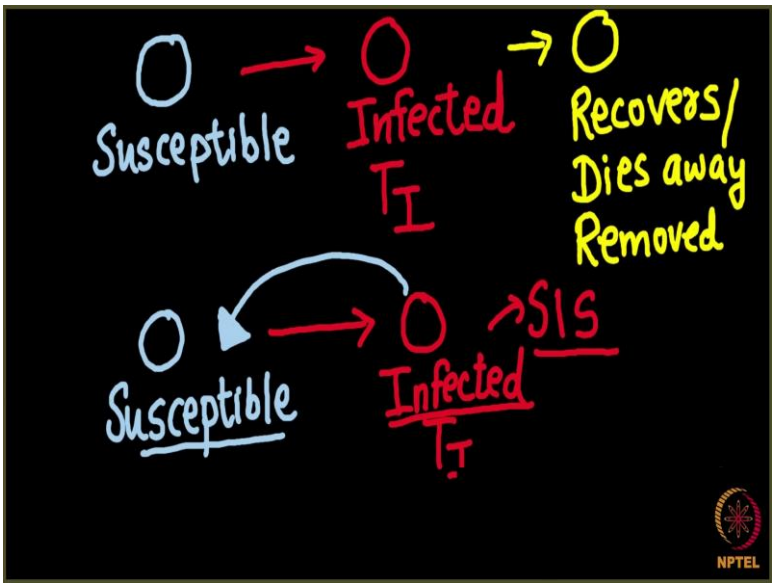
(Refer Slide Time: 10:30)



So, what happens is instead of going to this recovered state you go back to the susceptible state. So, susceptible infected and after remaining infected for TI time period, you go back to the susceptible state and this model is called the SIS model because S refers here for susceptible, I is for infected and S is again for susceptible. So, you become susceptible, you are becoming infected and then you again become susceptible and this is used, this model is used for something like a common cold, which disease which can infect you again and again.

And I am quite sure that you can now create a simulation of SIS model on a network on your own, but still in the next lecture, I will show you how can we simulate a SIS model. How can we simulate an SIS model on a network? So now, we are going to look at 1 simulation of the SIS model. We have looked at the difference between the SIR model and SIS model. Let me quickly recap it.

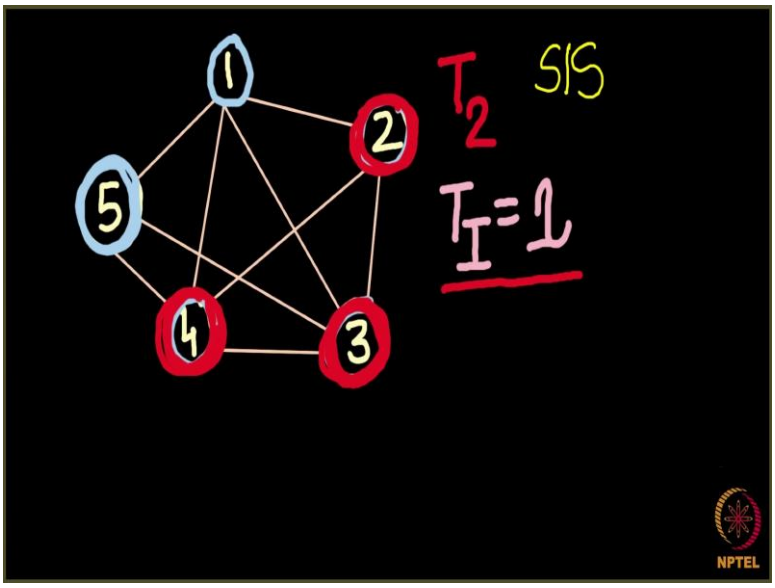
(Refer Slide Time: 11:39)



So, in SIR model here we were having a node and let us say this node is susceptible. That is, it has now got the infection and then we have seen that from susceptible, this node can go and move into an infected state. So, this happens after this node comes in contact with an infected person and this node becomes infected and it remains infected for TI period of time. During which it keeps infecting other people and finally, what happens, this node it either recovers. That is, it develops a lifetime immunity against that, against the disease, or it leaves the network. That is, let us say this node dies away in which case we say that this node has been removed.

So, this was your SIR model and SIS model was a little bit different. It was for a disease which can infect us time and again such as a common flu, cold, common cold. So here again, we have a node which can be initially susceptible and then when it comes in contact with an infected person, this node it becomes infected and it remains infected for TI period of time. And after this TI period of time, what happens is it again goes back to the susceptible state. So, there is no recovered state. After some time, this node becomes susceptible again then it might come in contact with an infected person and again become infected and so on. Now, how will this simulation of such a model look like? So, this is our SIS model and we are going to see how will the simulation of such a model look like. So, let us see.

(Refer Slide Time: 13:47)



So, let us take a network again. So, I take a network here. Let us say there are 5 nodes in the network. 1 and then 2, 3, 4 and 5 and I will put some edges between these nodes. You put some edges, let us say from 1 to 5. From 1 to 2, 1 to 4, 1 to 3 let us say. And then let us say, from 2 to 3, 2 to 4, 3 to 4, 3 to 5, 4 to 5 that is it.

Now, let us say these many it's a network consisting of 5 nodes and let us see how an epidemic or how a contagion will spread here and here we are talking about the SIS spreading model, SIS. Initially, all of these nodes shown in this network they are susceptible. So we will showing susceptibility by blue color. So, let us make all these nodes to be susceptible. So, 1 is susceptible and 2 is susceptible 3, 4 and 5. All these nodes are susceptible.

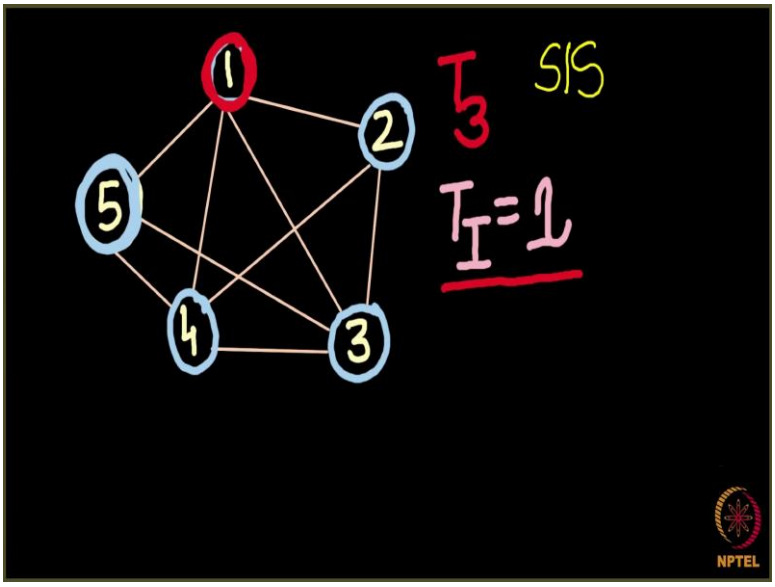
Now, this is T0 and let us say, at T0, an infection counts and infects 1. So, a node 1 is now infecting from common cold and our value of TI, that is, time period during which a node remains infected let it be 1. So, one node remains infected for one day. Now, let us see what will happen during the second day. So, we are modeling what happens during, i am i am sorry, during the first day. So, initially there was a 0th day, now comes the first day. So now first day, this node 1 it will look at all of its neighbors and try to infect some of them.

So, let us say node 1 become successful in infecting this node 2 here and node 4 here. So, during day 1, node 1 infected this node 2 and node 4, and by the end of day 1 what will

happen? Node 1 will recover because at TI it is equal to 1. Towards the end of this day, node 1 will recover. After recovering it will not go back to the recovered state, it will go back to the susceptible state as we have discussed. So, it loses its common cold and come back to the susceptible state; come back to the susceptible state.

First day over; let us now see what happens during the second day. So now is the second day. Second day node 2 and 4, we look at their neighbors and we will try infecting them and let us say they become successful in infecting node 3. So now at day 2, on day 2, node 3 catches the infection and again towards the end of day 2, node 2 and node 4 will go back to the susceptible state. So, node 2 and node 4, now go back to the susceptible states. So, this is node 2 here and node 4 here.

(Refer Slide Time: 17:45)

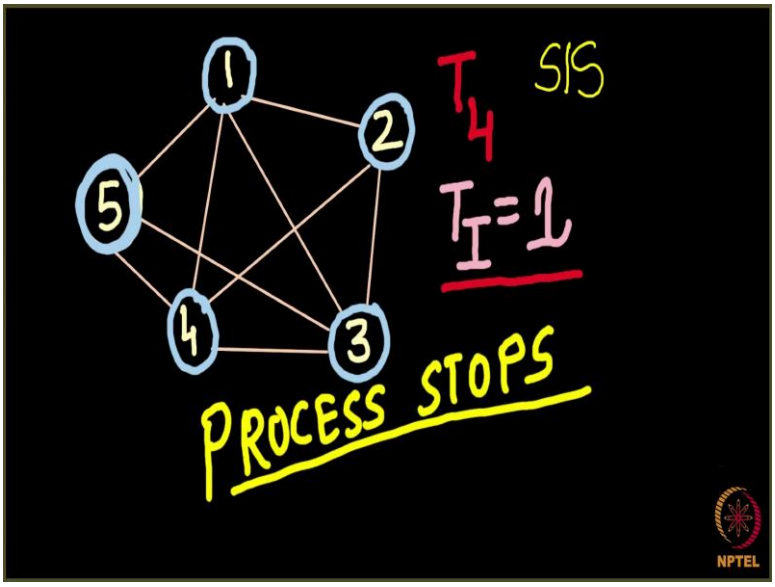


And then, what can happen during the third day let us say; now comes the third day. So, now, in the third day this node 3 is node 3 is infected. So, what node 3 will do?

Let us say it infects node 1. Do you see what happened just now? It infects node 1. So, node 1 as previously also become infected and it is infected certain people it becomes susceptible, but now this node 1 it again gets infected. So, this node 1 it gets infected here and towards the end of day 3 this node 3 now, it comes back to the susceptible state and this process keeps on going; and this process keep on going. So, node 1 may then infect certain nodes and then those nodes will infect certain nodes but suppose at this

iteration what happens is node 1 is unable to infect anybody. So, we are talking about day 4.

(Refer Slide Time: 18:41)



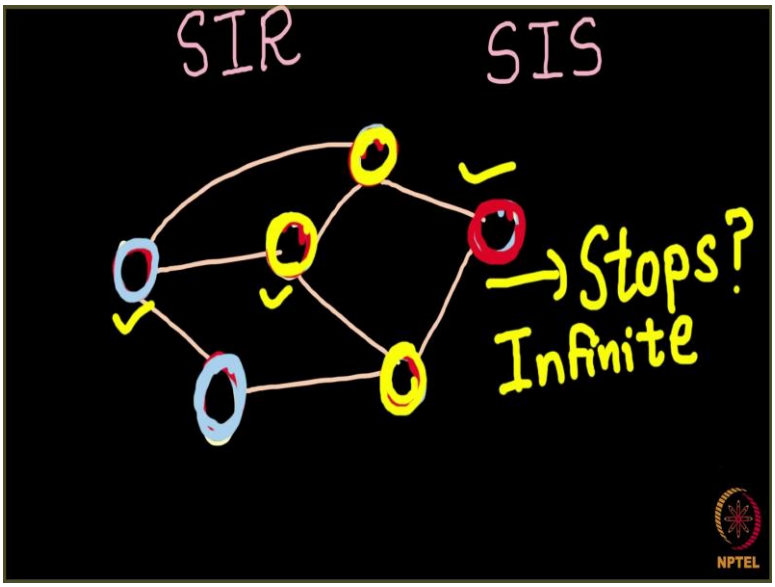
So let us say, at day 4 node 1 is enable to infect anybody and what will happened towards the end of this day; towards the end of day 4?

Node 1 will again go back to the susceptible state. And do you see now what is happened in this network? Everybody now comes back to this susceptible state there is nobody who is infected; and as soon as this happens in this network, there is not even a single node which is infected everybody is susceptible what will happen is the process stops. Now this was the simulation of SIS model on a network.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 135
Comparison between SIR and SIS Spreading Models

(Refer Slide Time: 00:10)



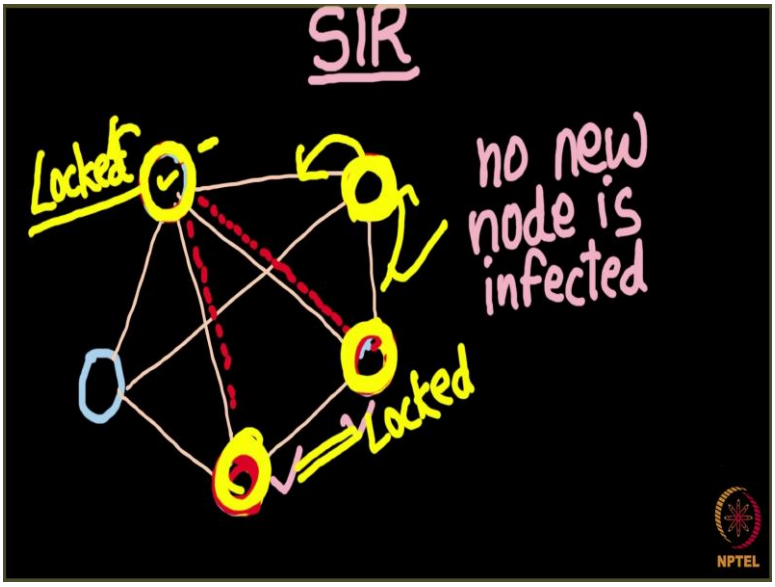
Till now we have talked about two epidemic spreading models, two disease spreading models. One was the SIR model and another was the SIS model. What was happening in both of these models was there were certain nodes in so, we were given a network and there were certain nodes in this network. And these nodes were connecting with the help of some edges, something like this.

And what was happening is, these nodes were changing their colors with time, changing their faces with time. So, initially all of these nodes were susceptible and then, some of the nodes in this network have become infected. And these infected nodes are started infecting the other people and while this was happening some nodes in the network were getting recovered as well and some nodes were turning out from infected, were going from infected state back to the susceptible state so something like this was happening.

These nodes are changing their colors with time and these nodes are changing their colors with time, shifting from one face to another. Now my question to you is this

process where these nodes are changing their colors, somebody is getting infected, and then somebody is getting susceptible, somebody is getting recovered, do you think that this process stops after some time or do this process keeps running for an infinite period of time? What do you think is the answer? You might want to pause the video two minutes and think about it.

(Refer Slide Time: 01:55)



Let us first talk about the SIR model, so let us see what happens in the SIR model. In SIR model initially every node is susceptible. So, I will take a network of five nodes here and initially every node is susceptible, let me put some edges between these nodes. All these are susceptible nodes and we have some edges between these nodes these are the edges.

And then what happens is we need at least one infected person in this network to spread infection. And let us say this is the node here which is infected and this node starts the infection. What is going to happen afterwards is, this node might infect some more nodes so let's say this node passes on the infection to this node and this node here gets infected.

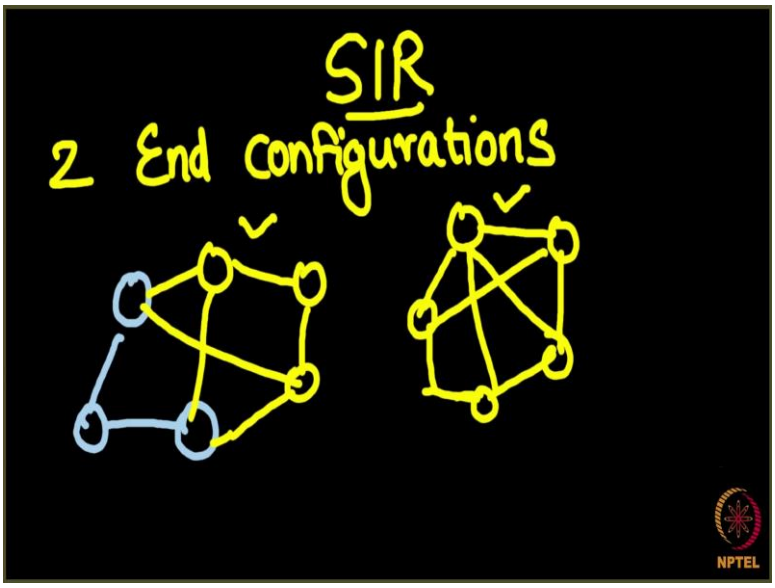
And let's say this node passes over the infection to this node also and this node also gets infected here. But what will happen after certain period of time? After some time this node will go back to the recovered state right or the removed state. And as soon as this node gets goes into the recovered state or removed state, you kind of see that it is locked, it is locked means it is kind of removed from a network. Now we can do nothing with this node it is out of question right.

And then we had these two infected nodes over here, and then these two infected nodes might infect some of the nodes or might not infect some of the nodes and get recovered. So, these nodes are also going to be recovered after some time. So, this is one possible configuration which can occur once the process stops. So, you see here the process will definitely stop in this case. So, let's say that this infection was spreading here. And what has happened there is there comes an iteration where no node is infected. So, this is an iteration where what happens is no new node is infected.

So, we have seen that when no new node is infected the process comes to an end. No new node was infected and with time both of these nodes they also went into the recovered state. Now this network is done, if the process stops nothing is going to happen. But let us say the process does not stop here. So, these nodes were red here these nodes were infected here and the process does not stop here, these nodes keep infecting another nodes even then don't you think that after some period of time this process will come to an end. Because there is a fix supply of nodes after some time both of these nodes are also going to become recovered and as soon as these two nodes become recovered they are also locked and so on.

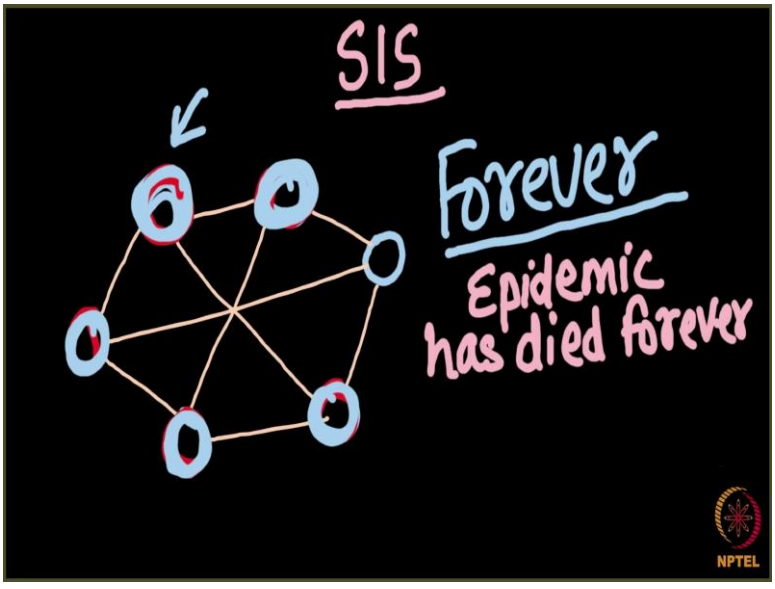
The process keeps going, keeps infecting, and the nodes starts being locked more and more nodes become locked and the time will compare every other node in the network is locked. So, this process will come to an end eventually. So, what can be the ending configuration for this process?

(Refer Slide Time: 05:14)



If we are talking about SIR model there are two end configurations, two end configurations. And the one ending configuration is there are certain nodes in the network which are still susceptible. And then the rest of the nodes in this network are recovered is the one possible configuration there are some edges between them. And second possible configuration is every node in this network has become recovered. And whenever the SIR process runs on any network it will end here or it will end here, it cannot keep running for an indefinite period of time.

(Refer Slide Time: 06:05)

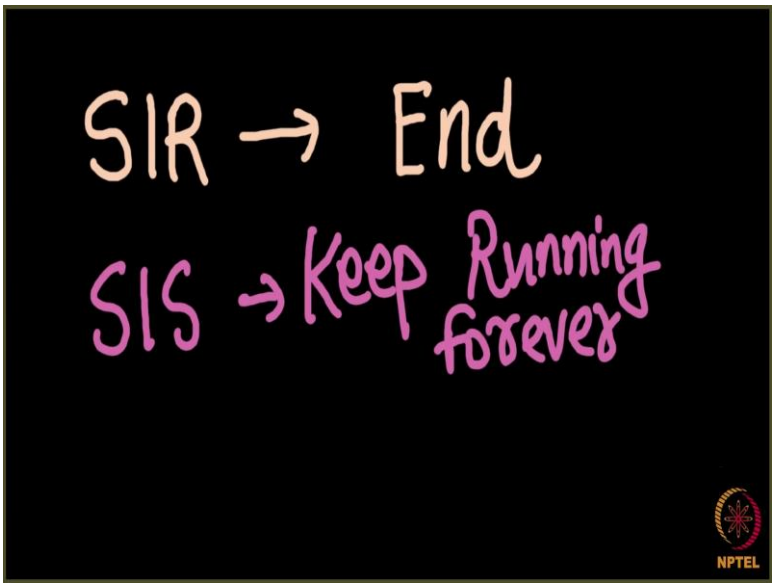


Let us now see what happens in the case of s SIS model. So, here I am talking about SIS model, let us see whether it will come to an end or not. So, there are certain susceptible nodes here I draw some susceptible nodes and these are connected with some edges here we make some edges and then the infection will start from some node. So, let us say this is the node from where the infection starts. And then what will happen with time again this node is going to infect some other nodes. Let us say this node infects these two nodes and what will happen is, this node will now go back to the susceptible state.

So, you see it is not locked here it goes back to the susceptible state and now these two nodes they will infect some more nodes let us say these and these and now these two nodes they go to the susceptible state and then these two again two infected nodes these two infected nodes they can again infect some nodes. So, let us say they infect this node over here. So, this node gets infected and you see it can be a cycle, it can be an indefinite cycle, rather the simplest form of cycle let me tell you.

So, this node over here let us say this node infects this node and it then it becomes susceptible. After some time, this node infects this node so this node is infected and this node becomes susceptible and this process can keep running for forever, this process can keep running forever. So, there is no guaranty that a SIS process will ever stop, but once it stops suppose that these two nodes are also recovered and in the next iteration this node over here it is unable to infect any other node and then it will also turn into susceptible. In this case you see what happen, in this case your epidemic or your contagion has died forever.

(Refer Slide Time: 08:27)



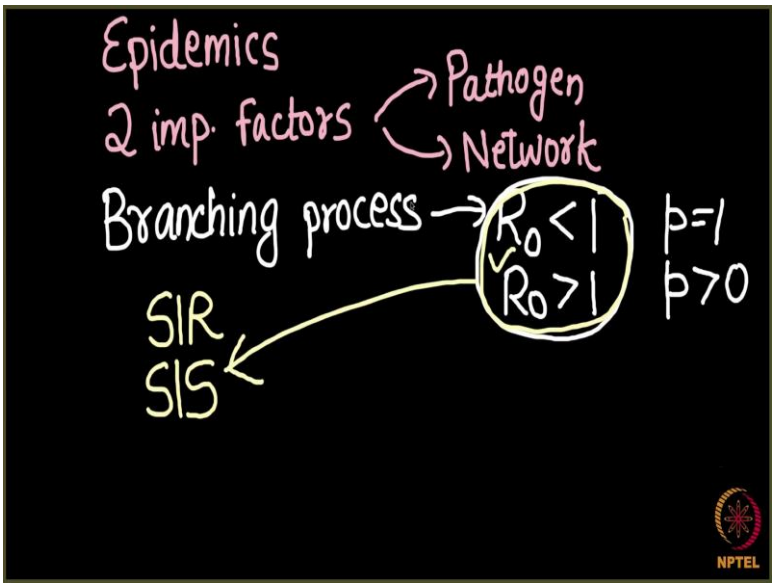
So, what did we see? We see that in the case of in this lecture we saw that, SIR model will definitely come to an end. But if we talk about the SIS model, it can keep running forever, it can keep running forever.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon - 2
Lecture - 136
Basic Reproductive Number Revisited for Complex Networks

Before going further let us quickly recap what all we have done in this chapter.

(Refer Slide Time: 00:12)



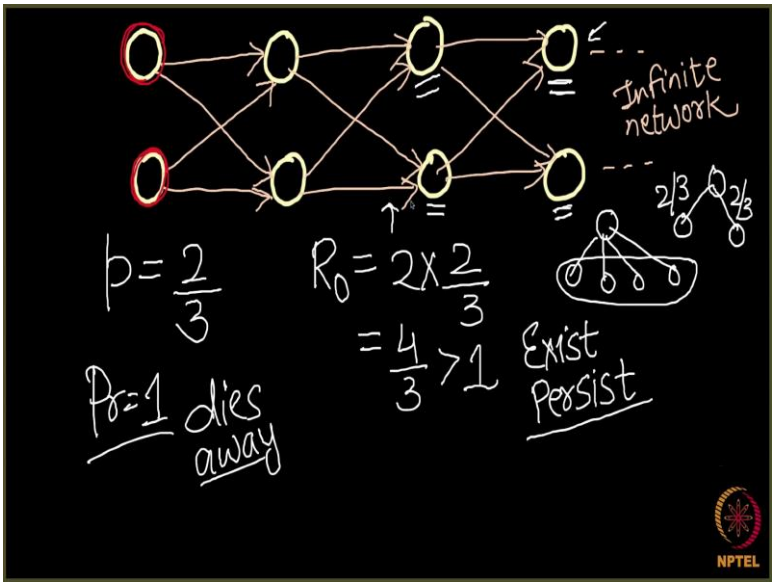
First of all we have looked at what are epidemics and how network scientists can study epidemics. Then we have looked at 2 important factors which are required for the modeling of a contagious disease. And what were those factors? The first one was a pathogen which was spreading and the second one was a network, network on which our contagion is spreading. After that we looked at a very simple process for modeling this spreading of a contagion, which is a branching model, branching process. And, one of the very interesting concept we looked in the branching process was the reproductive number R0.

We have seen that when R0 was less than 1 a disease it spreads sorry a disease it dies away with the probability equals to 1. And, when our R0 it is greater than 1 our disease persists in the network with some positive probability and R0 was responsible was very

helpful in telling us whether a disease is going to be an epidemic or not. And, then we looked at 2 little bit complicated spreading models SIR model and SIS model.

Now, again I will pose a question to you and the question is like here we have seen that there is a reproductive number, which if less than 1 it means that disease for sure dies away. If it is greater than 1 then, then it is quite sure that the not completely sure, but quite sure that the disease will persist in the network. This theory which we did for the branching process, do you think that it will hold here, and the answer can be no, the answer is no. Why this theory does not hold here?

(Refer Slide Time: 02:14)



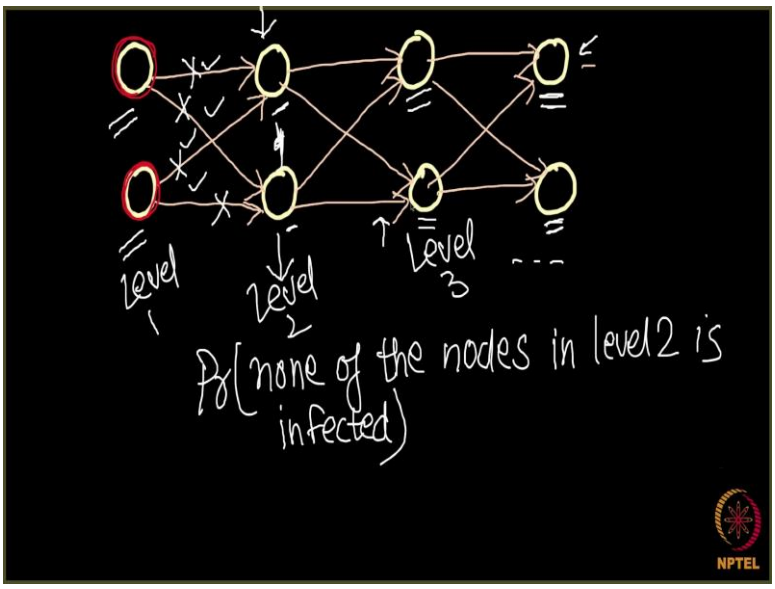
Let us take a small example and try to figure it out. Say we have a network here and let us say network is in this form I will give you a network and we adjust somewhat like this. So, this node is connected to 2 nodes on the next level and this node is connected to both the nodes on the next level. And similarly, this node here, this node here and this and this network goes on like this. So, it is an infinite network. And here I tell you that initially this node and these nodes. So, these 2 nodes are infected initially and then I tell you that the probability of your infection is spreading is 2/3. So, let us try to use the concept of basic reproductive number.

What was a basic reproductive number? What did R0 define? R0 told us the number of secondary infections. So, if a node is here and it has some neighbors here so, on an average how many of the people in this level will be infected, that was what R0 told us.

So, if we look at this network what is happening is every node. So, if we look at this node here this holds for any node in this network, if we look at this node here it is connected to this node and this node. So, 2 nodes in the next level, if I look at this node it is again connected to 2 nodes in the next level.

So, every node is having 2 children and each of these can be infected with the probability of 2 by 3. So, what is my basic reproductive number? It turns out to be 2 into 2 by 3 which is 4 by 3 which is greater than 1, which means that there should be a quite a high probability that my disease you know should exist on this network. There should be a high probability that my disease it should persist in the network, but it is not so, it does not happen like this. Rather, I show you that on this network this disease it for sure with a actually with a probability equals to 1 the disease dies away, which means that our old theory of basic reproductive number is not working here in the case of SIS and SIR model. So, let us see when will this disease die away. So, we will be taking this network.

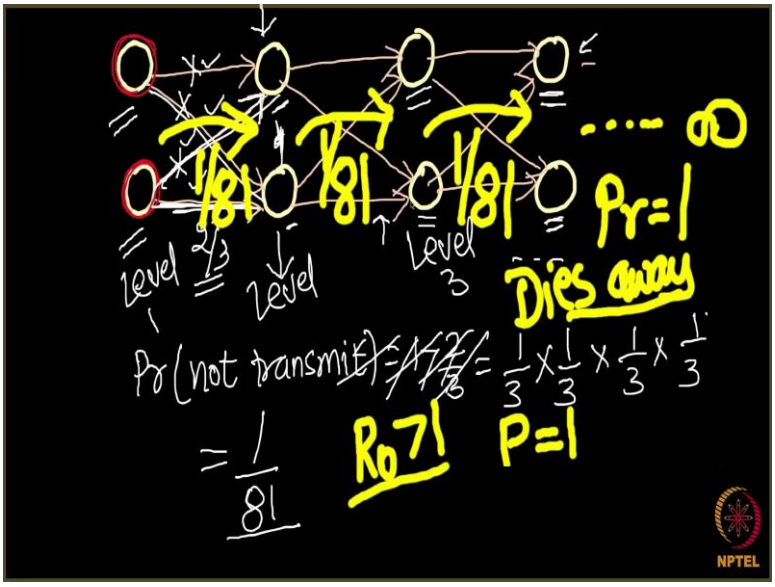
(Refer Slide Time: 05:05)



So, here is the here is our network, initially these 2 nodes are infected, let us look what is the probability that your infection does not come to this level. So, let us say that this is level 1 here, this is level 2, this is level 3 and so on. We are interested in looking at the probability that none of the nodes at level 2 is infected. What is the probability that none of the nodes in level 2 is infected and it is actually simple to calculate. So, when will be a node here infected?

If any of these four links workout then either this node will be infected, or this node will be infected right. So, when we none of these nodes infected when this link also fails, this link also fails, this link also fails and this link also fails, even if one of these links work my infection will come here to the second level. So, none of these links should work. What is the probability that none of these links will work? Let us see.

(Refer Slide Time: 06:27)



So, probability of infection here was 2/3 for every edge right, 2/3 is the probability that my infection will transmit across this edge. So, what is the probability that my infection will not transmit through this edge is nothing, but 1 – 2/3 which is equal to 1/3 right. And, then what is a probability that my disease does not get transmitted from here is again 1/3. So, I am going to multiply this, I am going to multiply this 1/3 and again 1/3 for the third edge and again 1/3 for the fourth edge.

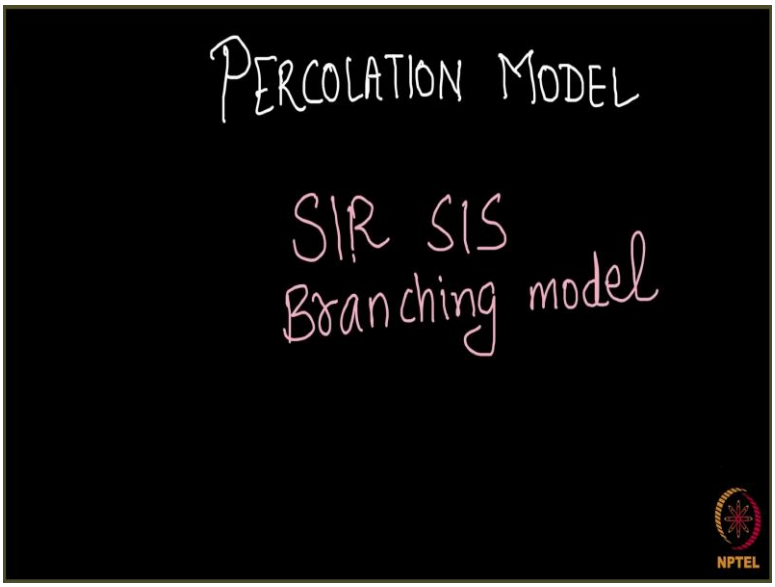
So, it turns out to be 1/81, this 1/81 is the probability that my infection it does not get transmitted from my level 1 to level 2. 1/81 is the probability that my infection does not get transmitted from this level to this level, which is actually a high probability which is on a higher side. When you say there is a high probability that my infection will die away here, even if 1 of the node gets infected here again for going so, from here to here again the disease dies away with the probability 1/81. Again, from here to here the disease dies away with the probability 1/81 and since it is an infinite network with the probability equals to 1, your disease it dies away from the network.

So, even if you see your basic reproductive number was greater than 1, what is happening is counterintuitive, what is happening is with the probability equals to 1 your disease is dying away from this network.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

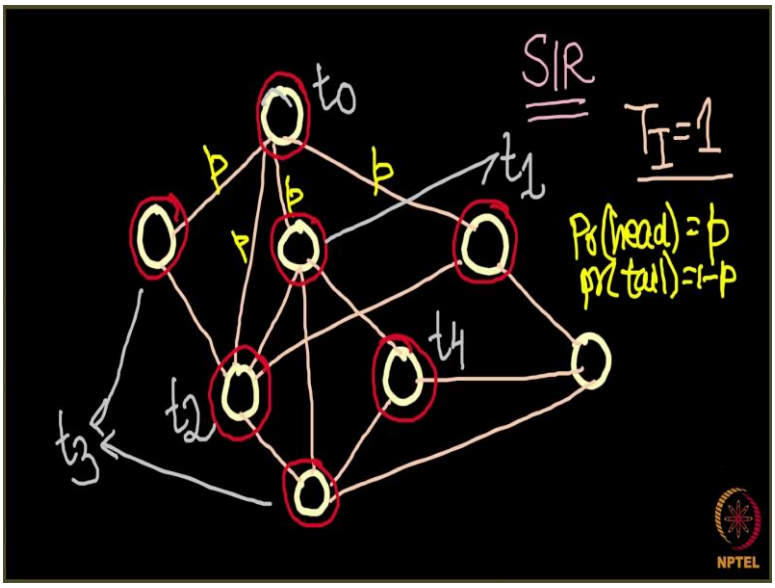
Rich Get Richer Phenomenon – 2
Lecture - 137
Percolation model

(Refer Slide Time: 00:10)



In this lecture we are going to now study a very interesting model which is known as the Percolation model. And this percolation model it is actually not a new model for you know modeling the spreading of your disease is rather it is a new angle, new way of looking at an already existing model. SIR model SIS model and for a matter of an event for your branching model, simple branching model; we are going to look at these models from a different angle. I will tell you what is the different angle.

(Refer Slide Time: 00:49)



So, let us quickly see we will simulate we will quickly simulate SIR model on a network and then we will see how can we look at it in a new manner. So, let us take a network here. So, I will just quickly make a network over here, I will take some nodes and we will take some edges, (Refer Time: 01:28) edges. So, I think this should be enough ok. So, we take this network over here and now we simulate a SIR model on it. And, let us say our $T_I = 1$, that is every node remains infected for 1 time period and after this 1 time period or 1 day that node stops infecting other nodes and let us say the infection starts from this node. So, this is our node which is infected at time 0, the 0 th day this node is infected.

Then now this node has 4 neighbors and it gets a chance to infect each of these neighbors. So, I am sure that you know how this process is occurring. So, each of the edges over here has a probability p of infection. So, what we are going to do is toss a coin for this particular edge and this will be a biased coin, biased coin such that probability of head on that coin will be p and probability of tail on that coin will be 1 minus p right. So, we toss a coin over here and see whether it is a head or tail, if its head we will infect this node else leave it. Similarly, for this edge, for this edge and for this edge and let us say after doing this 4 coin tosses this node over here and this node over here gets infected.

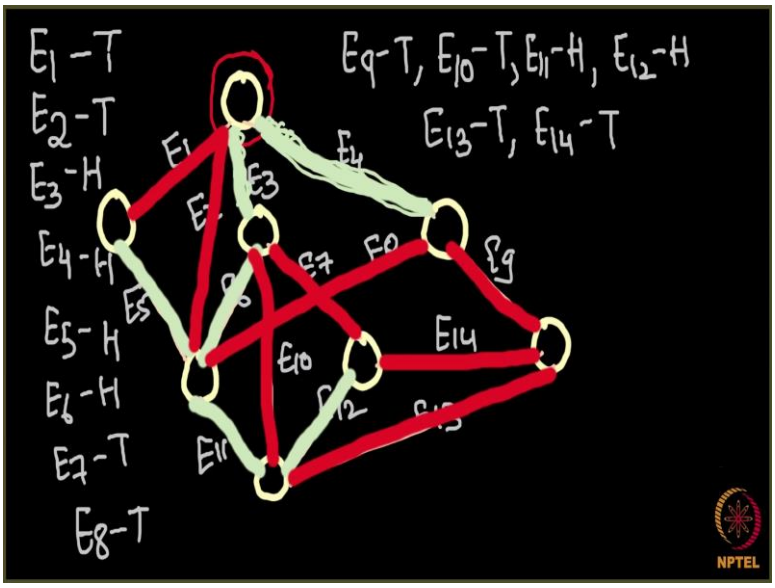
So, both of these nodes they get infected at time t_1. Now, in the next iteration both of this particular node it recovers. and it is no longer able to infect other notes while, these 2 nodes over here they now go ahead and they infect the network. And, let us say that in the next time step that is at time 2 this node over here gets infected so, this node gets infected at time t_2. And, then again both of these nodes recover and then this node over here it sees, and it tries infecting other nodes. And, let us say it ends up infecting this node over here and this node over here.

So, both of these nodes they get infected at time t_3 and then this node has a chance for these 2 nodes now infect further and let us say that in the next iteration this node over here gets infected. So, this happens at time t_4. So, how do we view this process? We are viewing this process as; we are viewing this process as there was this time 0, when this node gets infected day 0. Then it gets a chance to infect its neighbors and that at time 1 at day 1 these 2 nodes get infected and then time 2 these 2 nodes get infected and so on. So, we look at this process in terms of time.

Now, percolation model asks you the question that can you remove this notion of time from here. It will it seems to me to be a very dynamic process everything is running dynamically time 1, time 2, times 3, time 4 can you make static. So, let us say I want to know I was having this initial network over there ok. So, initially my network was like this and then this wait and this node over here, this node was infected. And, let us say now I am interested in knowing when this process will run whether this node over here will get infected or not.

And, then what we did for knowing whether this node will get infected or not we ran the simulation in terms of time. We looked at what will happen at time 0, what will happen that time 1, what will happen at time 2 and so on. And finally, we see that yes this node over here is infected. There is a rather of very new way a novel way to do this a very interesting one, where this process will not run in terms of time. We look at it as a very static process and that is what is called the percolation model. So, how do I make this entire process static is so, I take this network. So, this is my network and I want to see how I can make everything static over here. This is the network; this is the network I am having and no wait 1 second ok.

(Refer Slide Time: 06:30)



This is the network we are having and then we know that at the beginning this node over here is infected. And, we are interested to know whether this node will get infected or not and we know all this is a complete game of tossing a coin only. At time 1 we have tossed a coin for this, whatever gets infected; in the second iteration we toss a coin for that particular node whatever gets infected in third iteration we toss a coin for that particular node and so on.

Now, see what we will do is, first I will just quickly label all of these edges. So, let us say this is E1 E2 E3 E4 E5 E6 E7 E8 E9 10 11 12 13. So, there are 14 edges in this network, what I am going to do is instead of running this process time wise I am now going to flip 14 coins 1 for each edge. So, first I flip a coin here for this edge E1. So, I flip a coin for E1 and let us say I see a tail and then I flip a coin for E2 and I see a tail, E3 let us say I get a head. E4 let us say I again get a head, E5 let us say head, E6 let us say head we are getting so, many heads.

E7 I got a tail, E8 I got a tail and then let me toss it for E9 I get a tail. E10 tail, E11 let us say head, E12 again a head, E13 tail, E14 tail ok. So, see what I am doing is I have flipped a coin for all the edges at the beginning itself, I am not running this process time wise. Now, what do I do in this graph wherever I see a head. So, I imagine this graph in the form of let us say pipes. So, here there are the nodes, and these are the pipes through which let us say some water can flow and actually here some contagion is flowing.

So, these are the pipes through which water flows across these nodes and whenever a head occurs so, I say that a pipe is open and whenever a tail occurs I say that the pipe is closed. So, let me show let us have opened pipes in green color. So, let me open some pipes. So, E 3 is a head and then E 4 is a head, E 4 is a head 1 3, E 3 is a head, E 4 is a head and then E 5 and then E 6, E 7 I am sorry not E 7 E 6 and then we have E 11, where is E 11 E 11 and we have E 12 ok.

So, all these are the open pipes and remaining whenever I get tail I say that these are these are all the closed pipes. So, this is a closed pipe, this is a closed pipe and then closed, closed, closed, closed and these are the closed pipes. So, we can now see these nodes as connected in the form of pipes, where the green pipes are the open and red ones are the closed; now what it has to do with telling us whether some node was infected or not. So, we started our infection over here and we were interested in the question whether towards the end of the process this node over here will get infected or not.

And now, I tell you the answer; the answer says that imagine here is this water flowing from this node and we see that some of the pipes here are closed and some of the pipes here are open. So, I say that if water is able to flow from this node to this node, yes this node will get infected as this node will not get infected. And, you can match it with what we have done previously. So, whichever nodes get got infected are the ones you which you can reach with the help of these open pipes. So, if you want to see whether some node is infected or not let us say I want to know whether this node is infected or not, I will see whether there is a path of open pipes to this node from this node.

So, let us say here and then I can go here, here and here I see that there is a path. So, yes this node will get infected at the end and let us say I want to talk about this node over here. So, I see that there is no path of open pipes from this node over here to this node. So, this node will not be infected towards the end of the process. And, you see we have seen that this entire process of the infection of different nodes in the network in an SIR or let us say SIS model without using any notion of time. So, this is called the percolation model and it is very useful in the analysis, in the theoretical analysis of such processes.

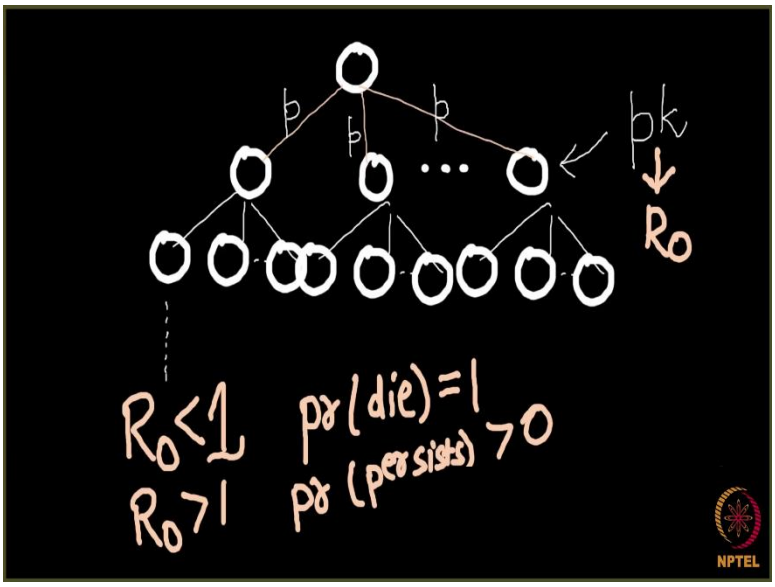
And, I think that by now it would be clear why use the term percolation. We say the term percolation because it is like this water is percolating through this network. And, we can use it in the modeling of the epidemic diseases.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 138
Analysis of basic reproductive number in branching model (The problem statement)

In this lecture we are going to do the Analysis of basic reproductive number.

(Refer Slide Time: 00:18)



So, if you do remember when we talked about a branching model, so branching model where there is a node here and then this node is having k neighbors. So, this node here it is having k neighbors right and then each of these k neighbors, each of these k neighbors are again having some k neighbors. So, something like this, each of these k neighbors are again having k neighbors (Refer Time: 00:52) little bit of overlapping this and so on right.

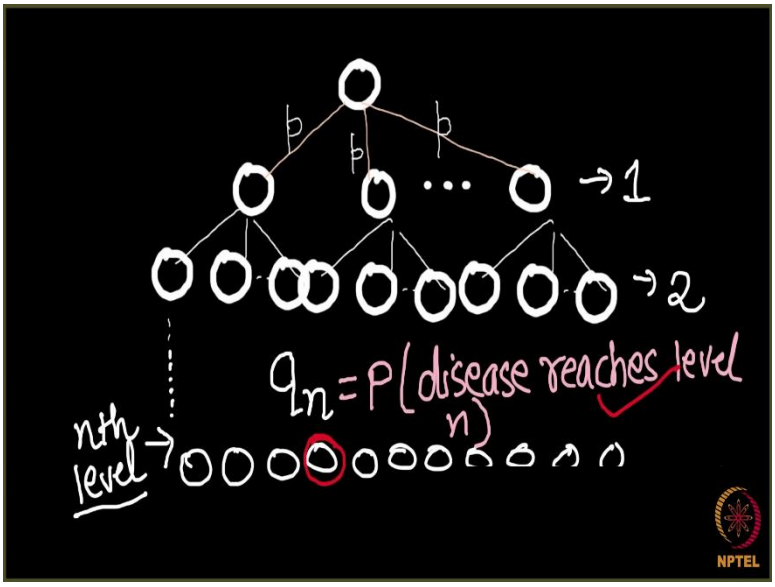
So, this was our basic branching model and then we have looked at the number of expected people who get infected here was given as p into k, where p is the probability that this disease get transmitted across this edge. So, here we were having a number p into k and we call this number as basic reproductive number R0.

And then we made a claim and the claim was we said that if R0 < 1 then the probability that this disease die, the probability that this disease dies equals to 1. That is if R0 < 1 for

sure this disease dies away and if R0 is greater than 1 then with the positive probability the disease persists in the network.

So, the probability that this disease persists in the network is greater than 0, we said this. And we will looked it how intuitively this is true, but we have not looked at a rigorous proof. So, in this lecture we will be doing a rigorous proof for this and the proof is actually very interesting. Let us see how we can do it. So, let us model it a little bit. So, what do we do is, we have this network over here ok.

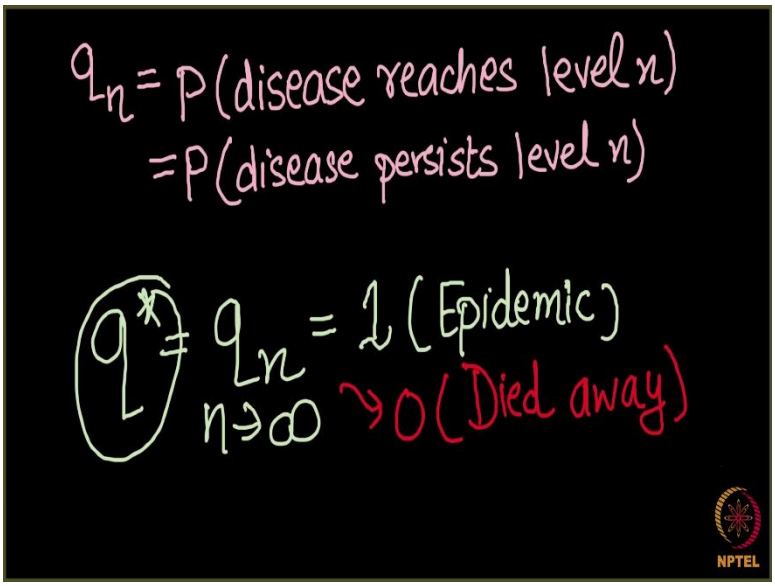
(Refer Slide Time: 02:32)



We have this network here right. First of all, let us see at when do I call this disease to become epidemic. So, let us see this is level 1, this is level 2 and somewhere down here is some level which is the nth level.

So, this is my nth level right and then I take a variable I take a notation let us say q n. And what is q_n? q_n is the probability that are disease it reaches level n. Reaches level n means at least 1 person at this level n should be infected. So, if these are the people at level n say, so at least 1 person in this level should be infected. And if 1 person at this level is infected, I say that my disease reaches level n. So, $q_n$becomes a probability that my disease reaches level n ok.

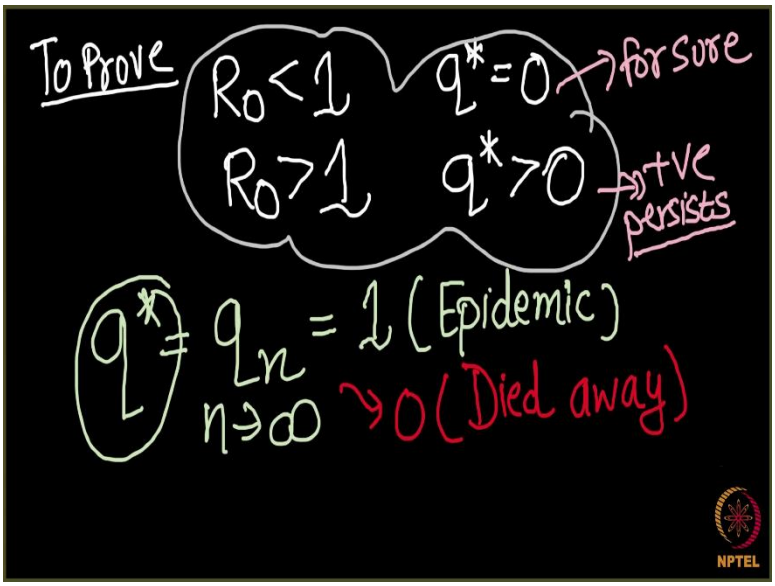
(Refer Slide Time: 03:45)



Let me write it down. So, q_n means the probability that disease reaches level n or I can also say that it is the probability that my disease persists still level n, so far so good. Now let me look at what does this value represent. I take here q* and I define q* as q_n as $n \to \infty$. What does it mean? It means that I keep going down, down, down, down, down and I go to the infinite level and I see there what is the probability that this disease is still persisting at that infinite level, that is my q*.

And you see what if, what does it mean if q* is 1? It means that is even if I go to the infinite level my disease is existing, it means that my disease is epidemic. If my q* is 1, I say that my disease is epidemic and if my q* is 0, I say that my disease has died away ok. So, now what do we have to prove? We have to prove that if our basic reproductive number R 0.

(Refer Slide Time: 05:23)



So, what do we have to prove? We have to prove that if our basic reproductive number R0, if it is less than 1, then we say that the disease for sure dies away, which means that then q* should be 0 right. What is q*? It is 0 when the disease dies away, q* is 0, when the disease dies away. So, when R0 < 1, the probability that your disease exists till your infinite level should be equal to 0; that is your q* should be equal to 0 and second thing when your R0 > 1, then q* should be greater than 0 right.

So, q* if it is greater than 0, it means that if the probability that your disease persists when you go to the infinite level in your network that probability is something positive. So, this is how we write our problem statement rigorously. It is essentially the same thing which we stated before that if R0 < 1, then our disease dies away for sure that is q* equals to 0. If R0 > 1, then with a positive probability our disease it persists in the network. So, we are going to do a proof which is a little bit lengthy.

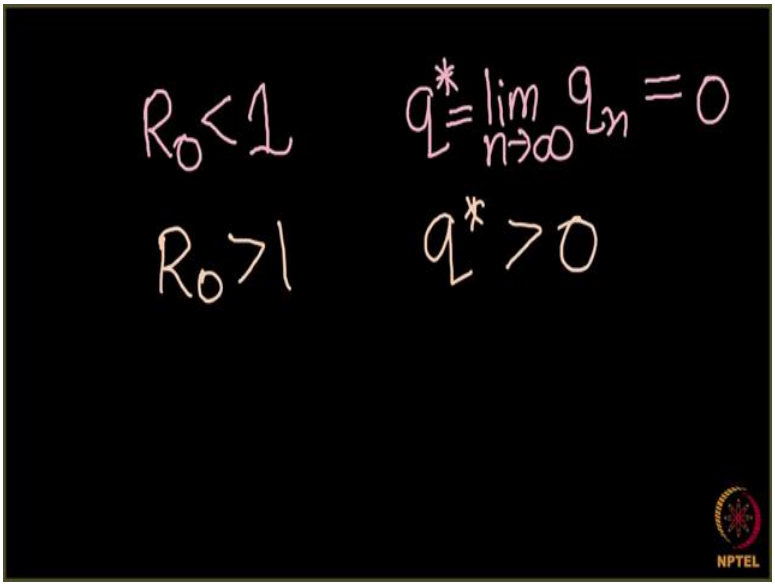
So, at times you may want to look back and forth and see what is happening, although I try my best to keep it as clear as I can ok. This is what we have to prove. And we will continue in the next lecture. So, just look back and see how we have modeled this problem, what is q*, what is q_n and so on. And, we will continue in the next lecture.

Social Network
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon - 2
Lecture - 139
Analyzing basic reproductive number 2

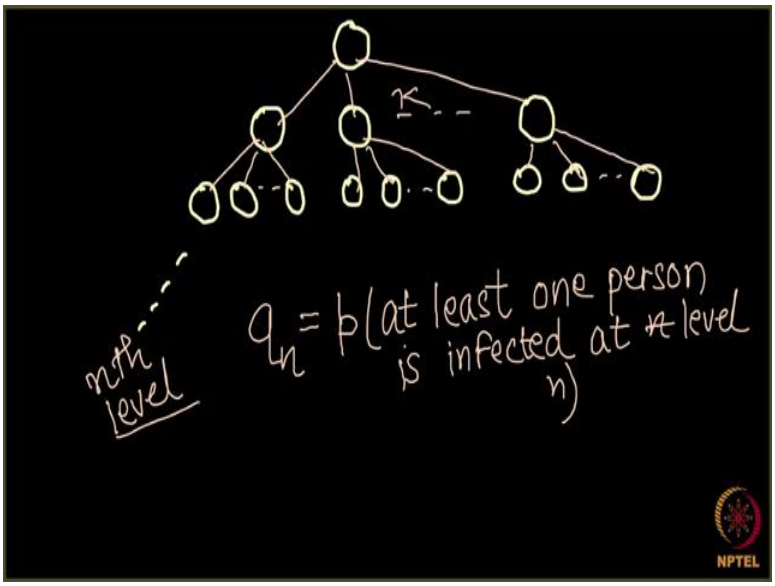
So, we have formulated the problem statement. I will again revise it.

(Refer Slide Time: 00:20)



I keep revising things again and again so that we should not get confused whether we are in proof. At problem statement was simply we have to prove that if the value of R0 basic reproductive number is less than 1, then our q*, q* which is nothing but $\lim_{n\to\infty} q_n = 0$ and if our R0 it is greater than 1 then the same q* > 0. And you remember what is q_n; let us quickly see what is q_n.

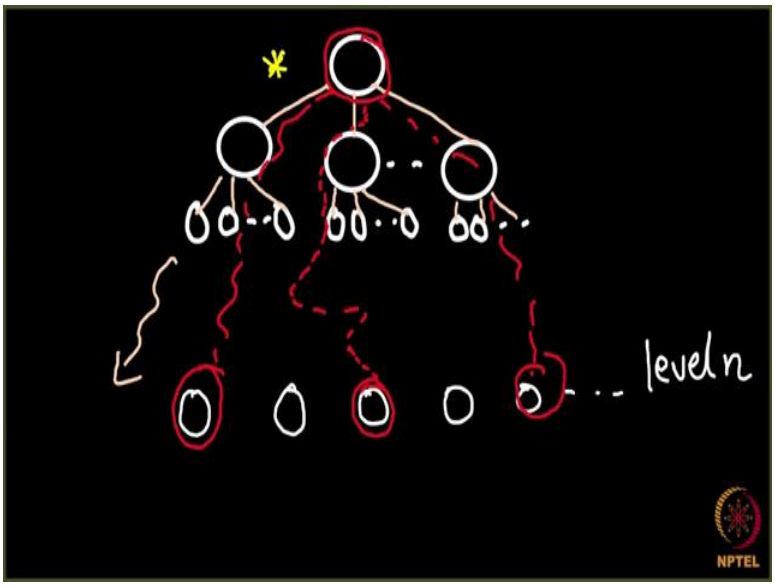
(Refer Slide Time: 00:58)



So, if I have a network over here let us say I have a network here, this node has k neighbors and each of these again have some k neighbors and so on. And it we make which is here and these are the edges k neighbors here. And this has k neighbors here, k neighbors here k neighbors here and so on.

And we got up to the nth level nth level. So, what was q_n ? q_n was the probability that at least at least one person is infected at the nth level. One person is infected at level n. Let us now see how we can find out this value of q_n. So, I make a clearer network over here; so, I need some time.

(Refer Slide Time: 02:02)



And here I can be a little big and here are the edges fine. This is my network right and somewhere down is our nth level. So, let us say this is our level n ok. Now what do we do? I define an event here. Now carefully watch, I define an event here and the event I say that it is a star event, star event.

What do I mean by a star event? We know that one of the node here should be infected right and for this node here to be infected there should be a path right, there should be a path from this node which was initially infected. So, this node here is obviously, infected. For some of the nodded level end to be infected there should be a path from this node to this node to somewhere here to somewhere here some path should be here right.

So, now what does even star means that that path can come from here, that path can come from here, and in fact some node here or that path can come from here, and in fact some node here. But there is at least some path and this path should runs through should go through at least one of these edges either through this either through this or either through this right. So, my star event says that it is the event that your infection reaches this level n through a path which involves this edge.

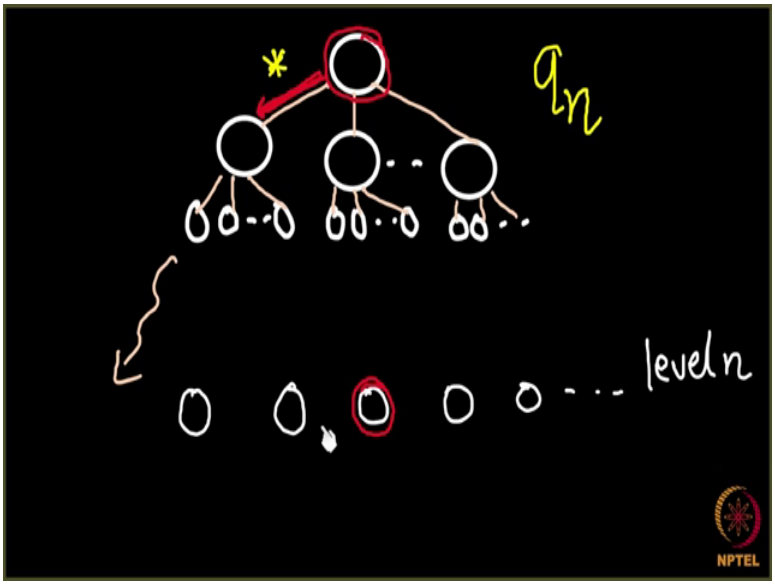
So, infection goes from this edge after this edge I do not know what happens, but it reaches level n, but this edge is for sure is infected. So, star event is the event that your infection reaches level n involving this edge and something happens after this edge any path. So, after this edge infection can go through this edge and have some path, can go through this

edge and have some path can go through this edge and have some path, but for sure it comes from here.

That is my event star ok. What do you think is the probability of this event star? What is the probability that your infection reaches level n through this path, through some of the path here? It is easy how will infection reach here. So, we know that first for infection to reach here may take back all my paths ok.

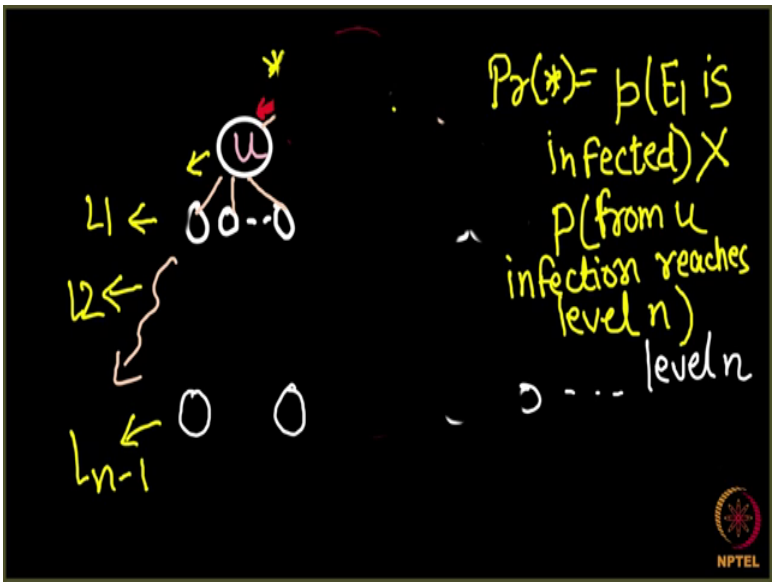
So, for my infection to come through this path first of all this thing should be infected right. This edge should be infected and then something happens afterwards.

(Refer Slide Time: 05:28)



Now, see what was q_n? q_n was the probability that starting from here your infection reaches this level right that is q_n. You start from this level from this node and your infection reaches this right ok.

(Refer Slide Time: 05:52)



Now, tell me can I write the probability of event star as probability that let us call this edges E1 and probability that E1 is infected. The infection passes through E1 multiplied by, what should come after multiplication? After multiplication is the probability that and let us say that this node over here is u. So, into probability that from u infection reaches level n.

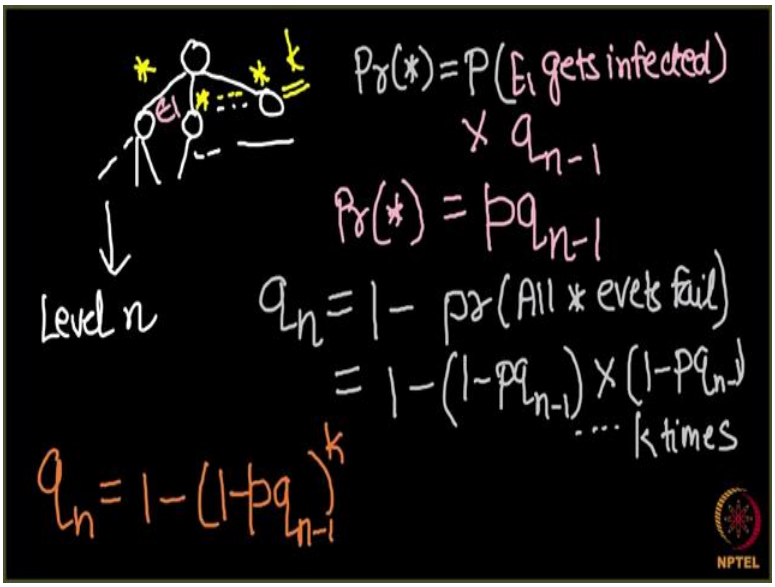
Probability that E1 is infected into probability that from u infection reaches level n. What is this probability that from u infection reaches level n, do you see? If you start from u it is the same as if you as if you are starting from here, just one level has reduced. What is what was q_n? You start from here and your infection should reach nth level. What is q_n - 1? That you start from here and your infection reaches level n - 1.

So, if you look at this node u over here, so assume that this node is not here, all these nodes are not here, I will even remove this, we will take them back wait; we will just rub all these portion. Let us say all this is not here all this is not here; I forget about all these and what I want to know is that you start from u and infection reaches level n.

So, this was here level n, what is this level for u? So, if I start from u, so for u this is level 1, next will be level 2 and this will be level n - 1. So, this thing, probability that your infection starts from u and reaches level n is nothing, but q_n - 1 ok.

So, I will just take everything back right. So, how can I write my probability of event star, what is the probability that E1 is infected? The probability that E1 is infected let us take a new slide and write it there, I will quickly draw all these things here.

(Refer Slide Time: 08:59)



So, we have this node and it has some k neighbors which has some k neighbors again and so on. And here is your level n and then recall what was your event star? Event star was you reach level n through this path and we said that the probability of event star equals to probability that the edge E1 that the edge E1. So, if this is E1 probability that E1 gets infected multiplied by q_n – 1 right. And what is the probability that E1 gets infected is nothing but p.

So, it is pq_{n-1} is the probability of event star so far so good. Now what were we looking at? We start our infection from here and it should reach level n. For reaching level n how many star events are possible, it can go from here, it can go from here, it can go from here. So, there is a star event here, there is a star event here and there is a star event here.

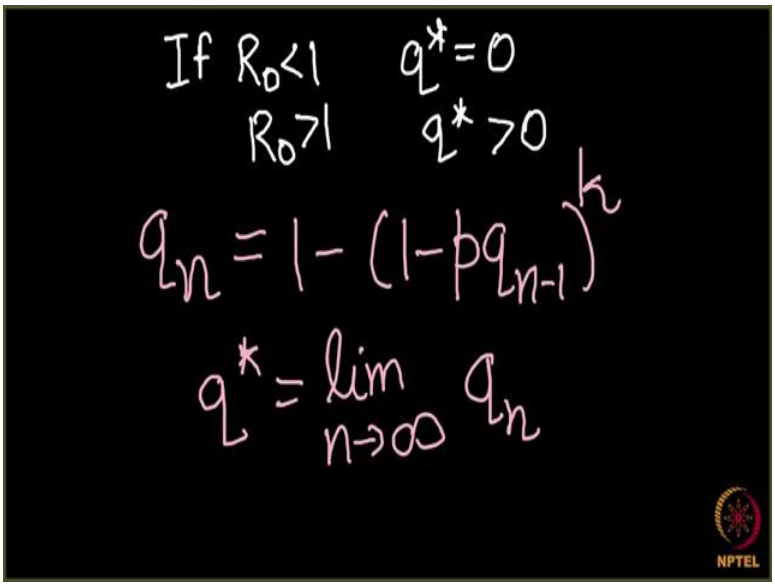
So, there are k star events right and your infection can go through the happening of this star event or the happening of this star event or the happening this star event. If one of these at least one, so even if one of this star event succeed your infection will reach level n right because, what is this star event that your infection reaches level n through this path. So, for your infection node to reach level n all the star event should fail right. So, I can write $q_n . q_n$was my probability that my infection reaches level n.

And for reaching level n it must go through one of the star events. So, can I write q_n as 1 minus probability that all-star events fail? q_n is nothing, but the probability that all the star events fail ok. Now what is the probability that all the star events fail; 1 minus. What is the probability that the first star event fail? So, probability that first star event succeeds is this, probability that first star events will be E1 minus this, which is $1 - (1 - pq_{n-1})$.

What is the probability that second star event fail all of them should fail? So, I put a multiplication here and what is the probability that second star event fail it is a same. So, it is $1 - (1 - pq_{n-1})$ and I keep doing it how many times? How many star events are there that many times, so it is k times right?

So, I can write q_n as $1 - (1 - pq_{n-1})^k$ ok. So, I will quickly put back everything in place whatever we have done till now, we will do it at the end of every small lecture which is this proof is running So, let us quickly see what we have done.

(Refer Slide Time: 12:50)



So, first what was our problem statement? Our problem statement was to prove that if R0 < 1 then q* = 0, if R0 > 1 then q* > 0, this was our problem statement.

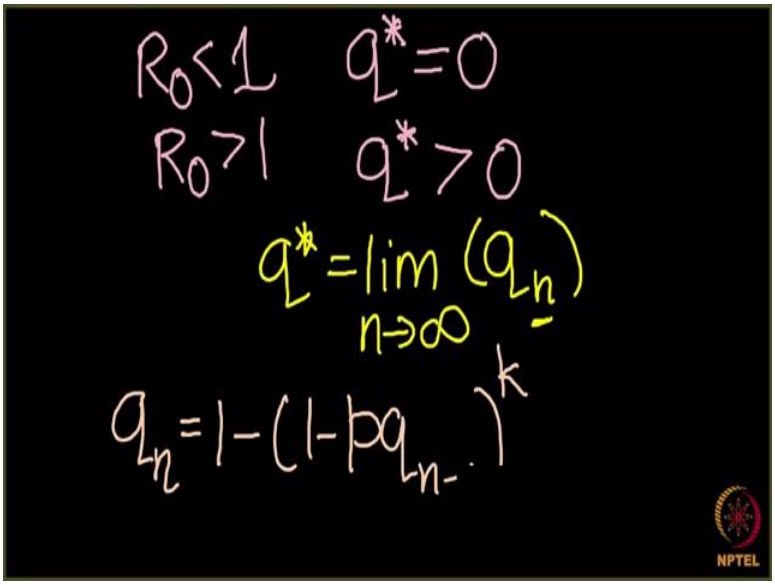
Next in this lecture what we did? In this lecture we derived a formula for q_n and the formula is $1 - (1 - pq_{n-1})^k$. And we also know what is q^* from our first lecture: $q^* = \lim_{n\to\infty} q_n$. So, we will start from here in our next lecture.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 140
Analyzing basic reproductive number – 3

So, now in this lecture we are going to start from the same place where we left of in the last lecture. So, what did we see in the last lecture, first of all our problem statement.

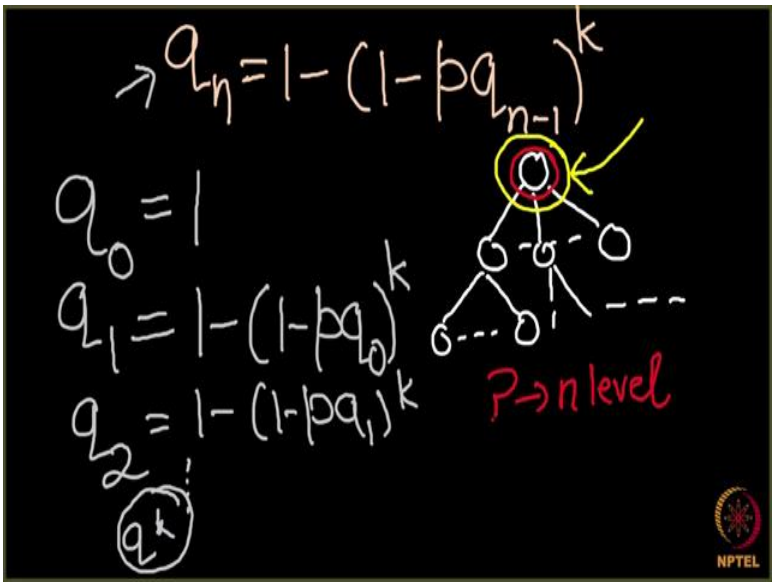
(Refer Slide Time: 00:16)



What is the problem statement? First, if R0 <1 then we have to prove that the value of q* equals to 0 and if R0 is greater than 1 then the value of q* is something which is greater than 0. This thing we have to prove and then we and we know what is q* right.

So, what is q*? q* is nothing but $\lim_{n \to \infty} q_n$ where q_n is the probability that your infection persist till the nth level; that is at least 1 person at the nth level is infected. And in the last lecture we have derived a formula for q_n and the formula was $1 - (1 - pq_{n-1})^k$. What we are going to do in this lecture is now we are going to analyze this formula further.

(Refer Slide Time: 01:16)



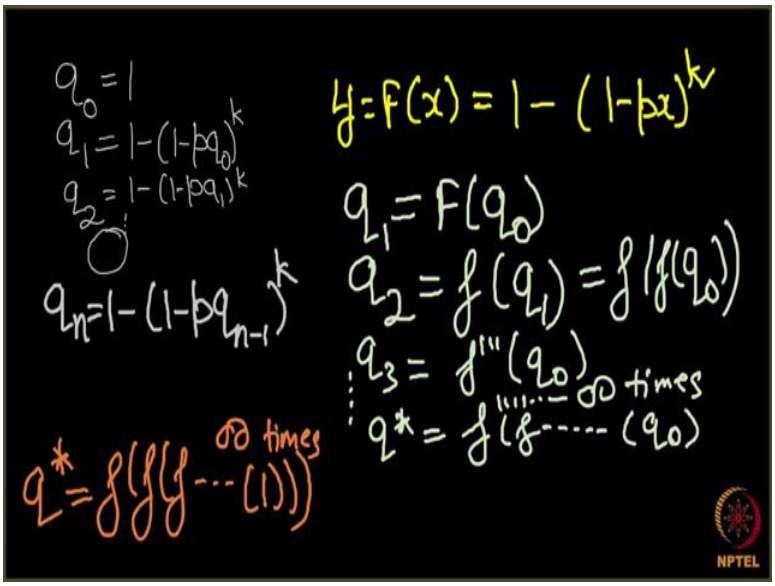
So, let me write this formula here and we will analyzing it $q_n = 1 - (1 - pq_{n-1})^k$ ok. Now what is q_0? Let us look at q_0, q_1, q_2, q_3 and so on. What you think is q_0? You will not get it from this formula. So, q_0 is what, the probability that your infection persists till the 0th level and what is a probability.

So, what was a problem statement? This was a guy who was here having k neighbors and this person was again having k neighbors. This person was again having k neighbors and so on. And your infection started from here and the question was what is the probability that infection reaches here to the nth level was our question.

What is level 0 here? This is level 0 and what is the probability that at least 1 person at this level is infected? It is obviously, 1 right for sure 1 this guy must be infected here. If this guy was not infected here, there was no problem we would be solving right. So, entire problem is because this person at the 0th level is infected it is infected with the probability 1. So, $q_0 = 1$. What is q_1? Finding q_1 is easy. Put the value in this formula what is q_1, $1 - (1 - pq_0)^k$, right.

What is q_2? q_2 is $(1 - pq_1)^k$ and what is our aim? Our aim is to find q* which will come after we will keep repeating this formula. So, we have q_0, from q_0 we can find q_1, from q_1 we can find q_2 and then we have to do this process infinite number of times and finally, we will find the value of q*. We can do it infinite number of times. So, let us see how do we find out this value.

(Refer Slide Time: 03:22)

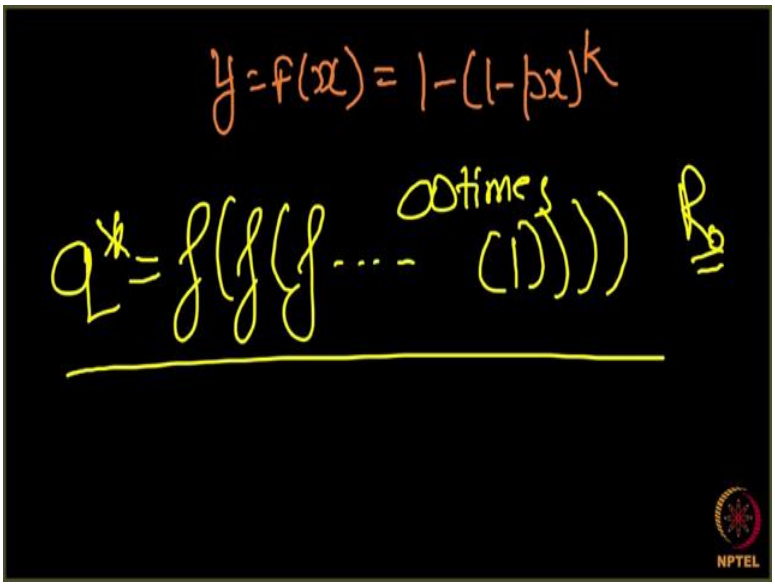


So, now we have all this we know q_0, q_1, q_2 and we know that what is q n, q n is nothing, but $(1 - pq_{n-1})^k$. I try to write it down in the form of a function. It is already in form of a function. If I just take a function y = f(x) we define this function as $1 - (1 - px)^k$.

Now in terms of this function can we see what is q_1, q_0 obviously is 1, what is q_1 according to this function? So, if we see what is q_1 here $(1 - pq_0)^k$. This and this are the same. I have just written this in the form of a function here. So, we can write q_1 as nothing, but 1 minus 1 minus oh sorry. So, we can simply write q_1 as; what $f(q_0)$ right. What is $f(q_0)$? $(1 - pq_0)^k$ which is same as this right. What is q_2 now? q_2 is $f(q_1)$, $(1 - pq_1)^k$ which is nothing, but I put the value of q_1 here. It becomes f''means f of f of so, to $f(f(f(q_0)))$.

Similarly, what is q_3? q_3 is nothing, but $f'''(q_0)$. And similarly, you see what is q* going to be? q* is going to be f dash dash dash dash infinite times means $f(f(f(f(\ldots(q_0))))$. And we also know that the value of q_0 is 1. So, I can write it down as q* = f(f(f(f(…(1)))) right.

(Refer Slide Time: 05:37)



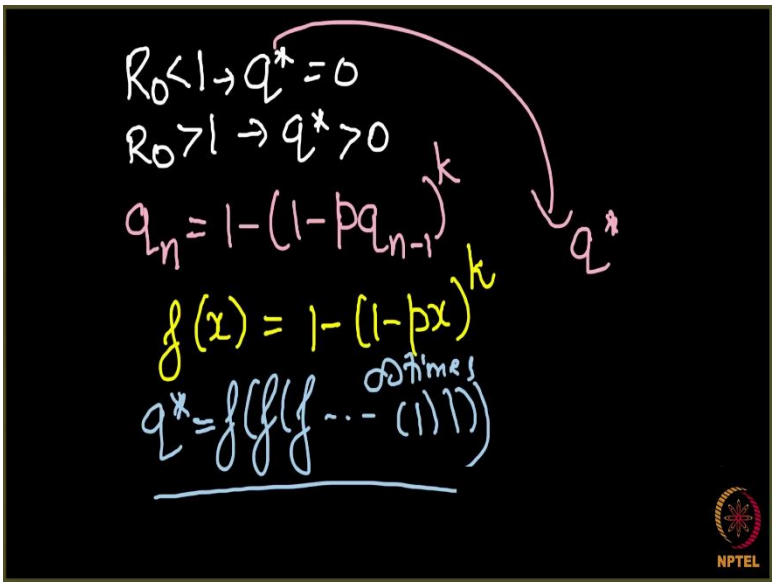
So, what is our overall aim now? Our aim now is we have a function y = f(x) which we define which is $1 - (1 - px)^k$ and I know now the value q* which I have to find is nothing but f(f(f(f(….(1))))).

And how do I find out this value? Once I find out this value my task is done, our aim is to find out the value of q*, obviously given the value of R0 that is to come in the next lecture. Our overall aim is just to find this value of q* and we will be finding it in next lectures. Probably just 1 or 2 lecture more we will be finding what is q*.

Social Networks
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

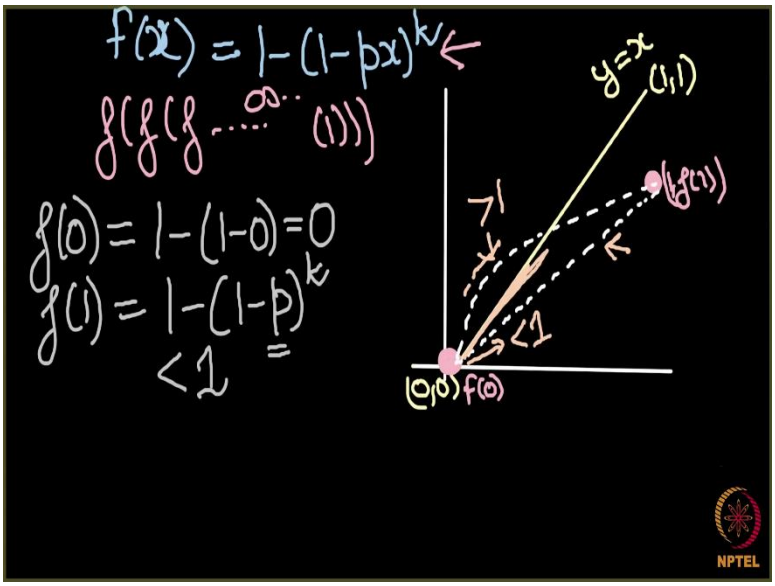
Rich Get Richer Phenomenon - 2
Lecture - 141
Analyzing basic reproductive number – 4

(Refer Slide Time: 00:11)



So, where were we? We will quickly recap from the beginning; our aim to show that if R0 < 1 q* = 0. And, to show that if R0 > 1 then q* > 0, this was our aim. And, then we found out a formula for $q_n = 1 - (1 - pq_{n-1})^k$, but our aim is to find q* right. And, we have defined the function, we have looked at a function and the function were $f(x) = 1 - (1 - px)^k$. And, we have looked at that this q* is nothing, but f(f(f(f(…(1)))) and this is the value which we are after. We need to find out what is this value, let us see how we find this value.

(Refer Slide Time: 01:12)



So, what all we are having $f(x) = 1 - (1 - px)^k$, this is our f(x) and our aim is to find f(f(f ….(1))). How do we find it? First, let us analyze what is this function, how does this function look. Let us see how this function look will. So, let us say, let us try to find f(0). What will f(0) be? It will be (1 – (1 – 0)) right which will be nothing, but 0. So, f(0) is 0 and let us look at what is f(1); $f(1) = 1 - (1 - p)^k$and we know that p is something positive.

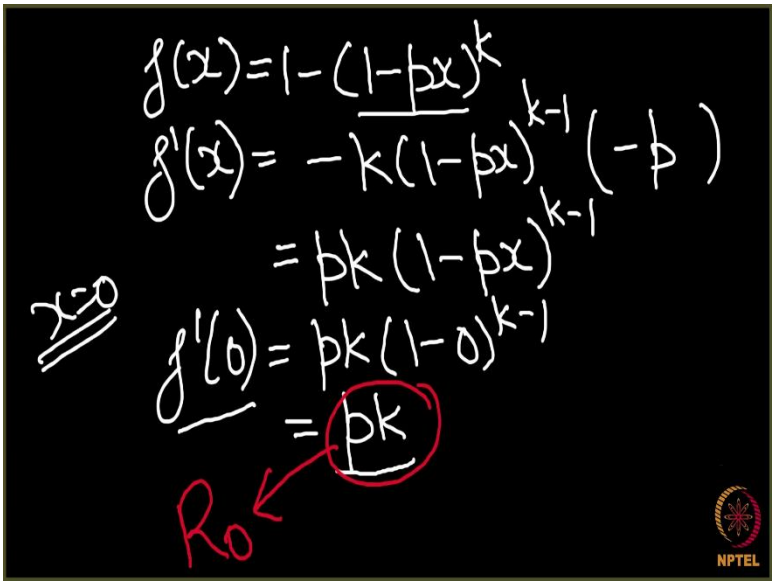
So, this entire value is going to be less than 1. So, f(1) is something which is less than 1 f(0) is 0 and f(1) is something which is less than 1; let us try to see how this plot will look like. So, I draw a plot here let us say y axis and x axis and here it is 0, here it (0, 0) and I first of all I draw a line also here and this sorry and this is a line I say what is this line you all know. So, this line is y equals to x and this point is 1 comma 1 right. And, let us see how a function f(0) look will like, a function f(0) is first of all having a point here for sure because at 0 f(0) is 0.

And we know that f(1) is something less than 1. So, this point is f(0) and f(1) is going to lie somewhere here which is less than 1. So, this point is f(1), (1, f(1)). So, this point is (1, f(1)) ok. We have seen the starting point of this function and the ending point. What happens during middle? What happens here? How does how does this function look like? And, that is also easy to calculate, let us see how we can calculate that. Let us try to find

out the slope of this function at this point, let us try to find out the slope of this function at this point.

So, what is this slope? This slope we know is 1 right, this slope is 1 which is your 45 degree; tan of 45 degree 1. So, if at this point so, now see this function f(x) can either go like this or go like this. When does this function go like this? When this slope of this function here is less than 1 right, if the slope is so, because this is slope equal to 1; so, if this is slope equal to 1, if slope less than 1 then this function will go from downwards of this function y = x. And, when will this function go like? This it will go like this when the slope of this function is greater than 1. So, let us first try to find out the slope of this function. So, what do we do to find out the slope of the function is to differentiate?

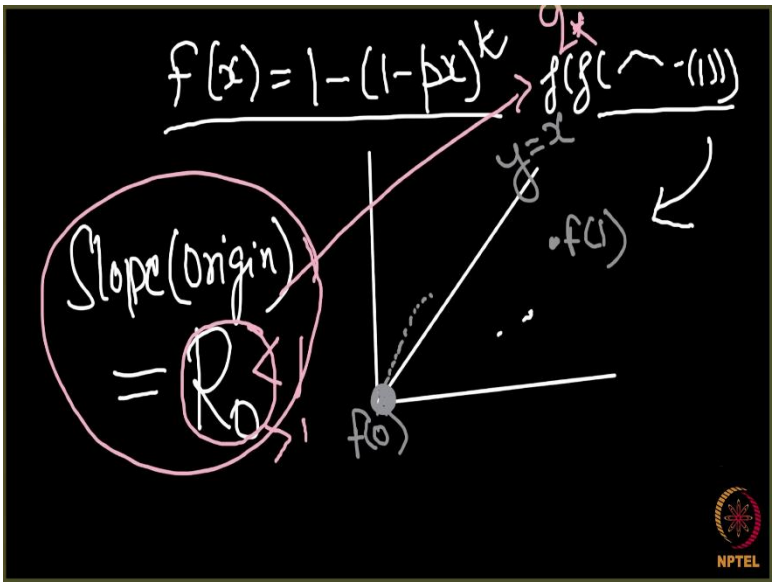
(Refer Slide Time: 04:53)



So, we have the function $f(x) = 1 - (1 - px)^k$, let us differentiate it. So, $f'(x) = -k(1 - px)^{k-1}$ right multiplied by differentiation of (1 – px) which is -p. So, this becomes equal to $pk(1 - px)^{k-1}$ right. So, this is the slope of this function and we are interested to find the slope of this function at x equals to 0 at origin.

What is $f'(0)$? $f'(0) = pk(1 - p * 0)^{k-1}$ which is nothing, but p * k; it is amazing. The slope of this function is p * k. What is p * k? If you remember p * k is nothing, but your basic reproductive number R0, from where we started and where we reached.

(Refer Slide Time: 06:09)



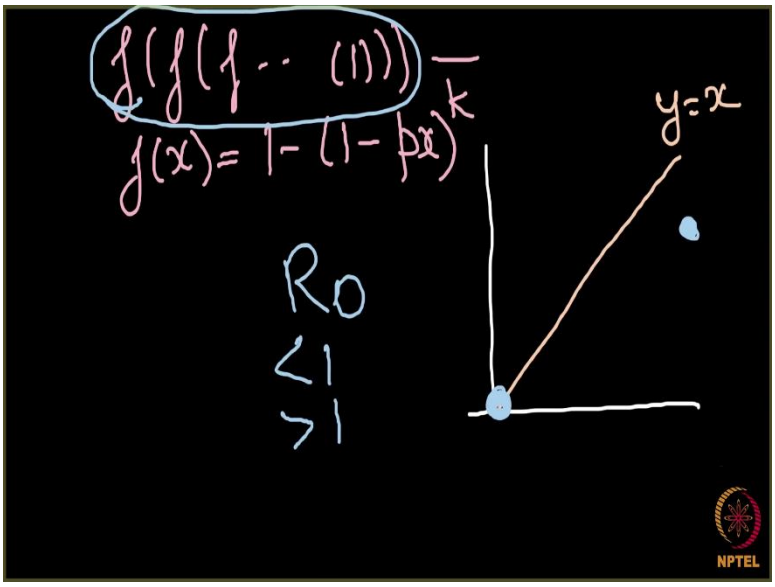
So, you see we started off with this function $f(x) = 1 - (1 - px)^k$ and then we try to plot this function. And, this is the line y, and this is the line y equals to (Refer Time: 06:34) 1 second and this is the line y = x. This is the line y = x and then we saw that here will be our point f(0) and here will be our point f(1). And, then we were interested in finding the slope of this function at this point and it turned out that this slope of this function at origin is nothing, but the value of the basic reproductive number.

So, just remember all these we have this f(x) here and the function look something like this and we wanted to find out f(f(f(f(…...(1)))) and then we tried to look at this function. The function turned out something like this and then we saw something amazing; we saw that the slope of this function at origin is nothing, but R0. So, now we will see for different values of R0, that is R0 can be less than 1 or greater than 1 how does this value which is nothing, but q* turn out to be and that will be the end of this proof which will which will do in the coming up lecture.

Social Network
Prof. S. R. S. Iyengar
Department of Computer Science
Indian Institute of Technology, Ropar

Rich Get Richer Phenomenon – 2
Lecture - 142
Analyzing basic reproductive number – 5

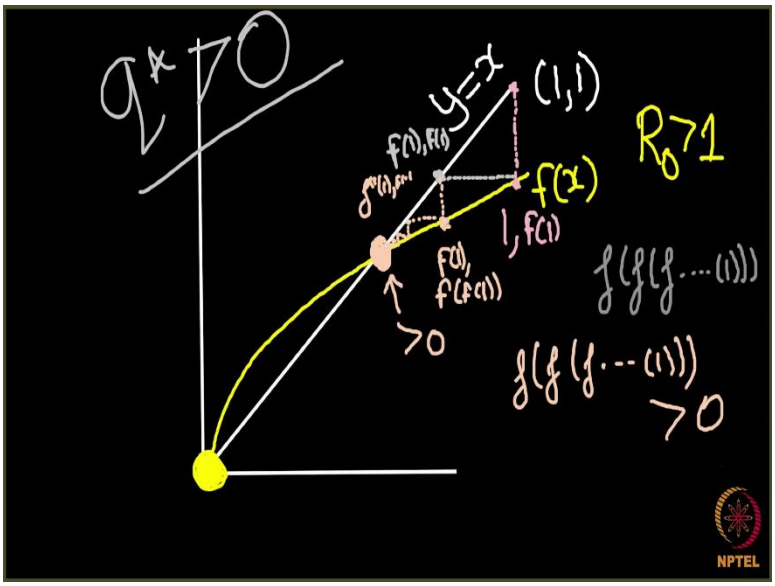
(Refer Slide Time: 00:17)



Now, coming to the final part of this theorem proving; what we were doing, our aim was to find f(f(f(…..(1))), where f(x) was a function which was $(1 – (1 – px)^k)$ and we have looked at this function. So, I will just roughly make it here. So, we have looked at this function, if these are our axis and let us say this is not this let us say, this is our line y = x we saw that the first point of this function was here at origin.

And, second point was something which is less than 1 and then when we saw that the slope of this function at this origin is nothing, but a ray basic reproductive number R0. Now, let us see how we are going to find out this value of q* which is f(f(f(…...(1))), when our R0 < 1 and when our R0 > 1 ok.

(Refer Slide Time: 01:33)



So, let me draw the graph neatly here. This is the graphs and say this is the line, which is y, this is the line y = x and let us say this is the point. What will this point be? (1, 1) and for our function we know that 1 point is here and then 1 point is somewhere here, which is less than 1. And, then let us look at the first case. Let us say the value of R0 > 1, when the value of R0 > 1 we know that our function its slope is going to be greater than 1. So, this line y = x is corresponding to the slope equals to 1. So, slope greater than 1 will go something like this. So, I draw something like this.

So, our function look something like this ok. So, this is our function and I write that this function is my f(x) and you see what is my aim, my aim is to find f(f(f(…..(1))); let us see how do we do it. So, I just make a little bit change to this figure ok, I want some shell points over here yes. This is the point (1, 1), now see something interesting is going to happen. I know that this point over here is (1, 1) right, let us say I draw a line vertically here. I draw a line vertical here, what do you think is this point over here. So, you see the value of x axis remains the same which is 1 and what is the value of y axis over here, this is the function y equals to f(x). So, the value of y over here is f(1) is not it. So, this is this point is (1, f(1)) ok. Now, let me draw a line horizontally over here. When I draw a line horizontally over here, what do you think is this point over here.

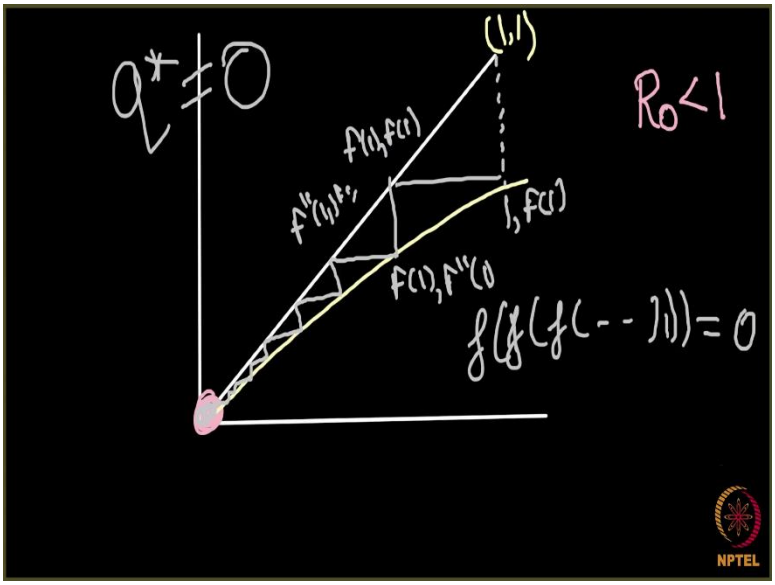
So, this point over here now, you see the value of x changes and what is the value of x becomes; this line is y equals to x. So, the value of x here should becomes x equal to the

value of y and what is the value of y? f(1). So, I get a point here f(1) comma f(1). So, far so good and I think that you are now getting an idea what I am doing. We again draw a line vertically over here and what do you think is now this point over here, this point over here. Again, the value of x axis remains the same which was f(1) and what does the y axis becomes, y here is f(x). So, y axis here is f(f(1)).

So, you see how we are moving from 1 to f(1) to f(f(1)) and then I go horizontally over here, and it will be nothing, but $f''(1)$, $f''(1)$. And, then I can come here, and it will be up triple dash of 1 and when I do keep doing it infinite times, what will happen, when will it converge. And, see what will happen at this point now, at this point both of these curves meet right. So, when I am going to do these process infinite times this process will converge here because, after this I cannot draw horizontally or vertically because, both of the curves have met. So, when I do these infinite times I converge at a point and you see what this point is.

This point is greater than 0, which means that your f(f(f(…...(1))) turns out to be greater than 0 which means that the value of q* in this case is greater than 0. So, we are done with the first part, what happens when R0 > 1. What happens when R0 is less than 1, is entirely the same completely similar.

(Refer Slide Time: 06:26)



So, again here was our curve and here was the line let us say y = x and then 1 point of our plot was here and another point of our plot was here. And, we know here R0 < 1, if R0 <

1 how will my curve look like. It will go something like this, right something like this. And now I can repeat the same procedure. So, we can repeat the same procedure.

This point here is (1, 1) right, now again drop this here and I get, do I get (1, f(1)) and then I go horizontally here, and I get (f(1), f(1)). And, then I come here, and I get (f(1), $f''(1)$) and then I go here I get $(f''(1), f''(1))$ and I keep doing. So, where is this curve going to converge is at 0. So, when you do it infinite times the curve converges at 0. So, in this case f(f(f(f(f(1))))) turns out to be 0, which means that in this case q* = 0.